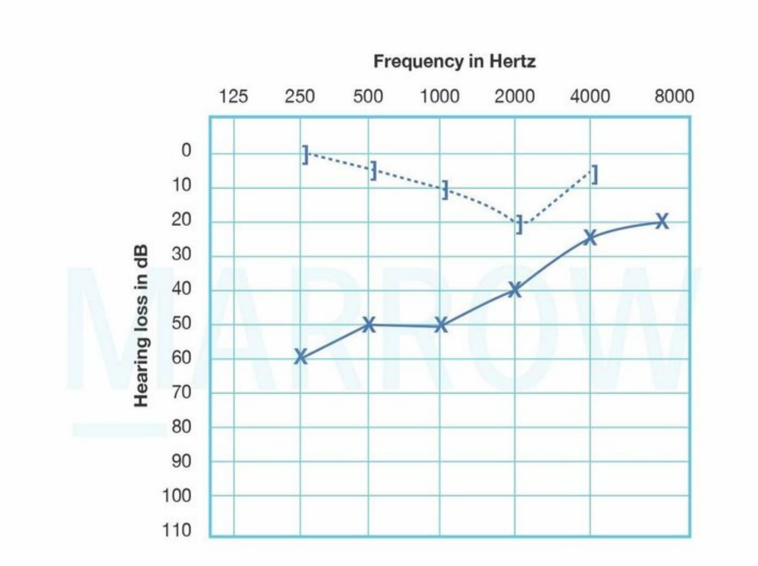
ENT AIIMS 2017

Question 1:

A 25-year-old woman presented with hearing loss that developed shortly after her first pregnancy. Her audiometry report is given below. What is the diagnosis?



a) Ototoxicity

b) Meniere disease

c) Otosclerosis

d) Noise induced hearing loss

Question 2:

Kraissl's lines are

a) Point of maximum tension in a fracture

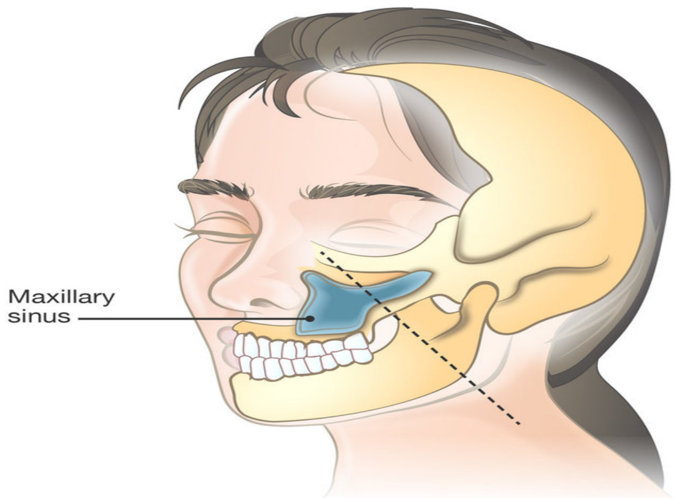
b) Relaxed Tension lines in skin

c) Collagen and elastin lines in stab injury

d) Point of tension in hanging

Question 3:

Identify the following line shown in the below picture:



a) Frankfort line

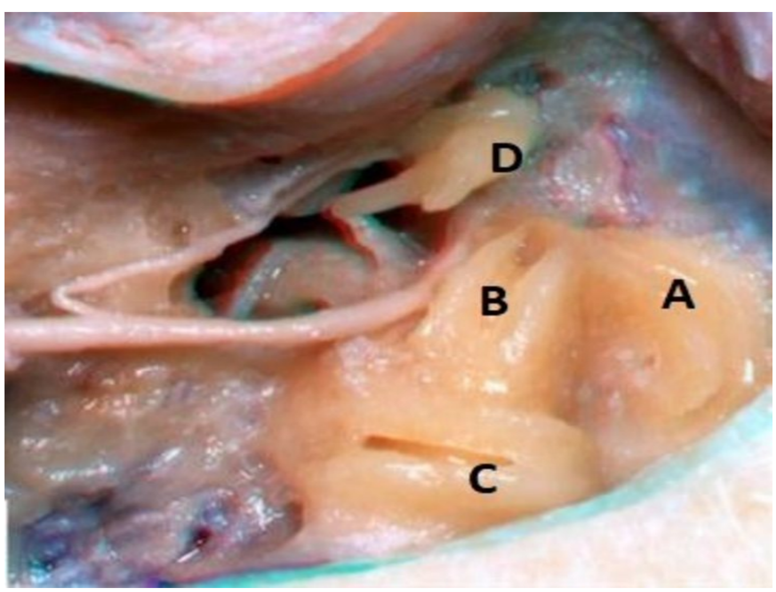
b) Donaldson line

c) Line of Sebileau

d) Ohngren line

Question 4:

The given image is of a patient in whom mastoidectomy is done. Identify the lateral semi-circular canal:



a) A

b) B

c) C

d) D

Question 5:

Which of the following is not a branch of the external carotid artery in Kiesselbach's plexus?

a) Anterior and posterior ethmoidal

b) Sphenopalatine artery

c) Greater palatine artery

d) Septal branch of superior labial

Question 6:

A metal factory worker suffers from noise-induced hearing loss. Which of the following structures will be affected?

a) Crista ampullaris

b) Macula

c) Cupola

d) Outer hair cells

Question 7:

Rhinosporidium seeberi is now classified as a/an _________

a) Aquatic protistan protozoa

b) Fungus

c) Virus

d) Bacteria

Question 8:

Which of the following is not ligated for the control of epistaxis

a) Internal carotid artery

b) Anterior ethmoidal artery

c) Maxillary artery

d) External carotid artery

Question 9:

A 5 year old with gradually progressive hoarseness of voice over three months and respiratory distress for 2 weeks. Diagnosis is

a) Croup

b) Respiratory papillomatosis

c) Vocal nodule

d) Acute epiglottitis

Answer Key

Question No.	Correct Option
1	c
2	b
3	d
4	b
5	a
6	d
7	a
8	a
9	b

Detailed Explanations

Solution to Question 1:

The clinical scenario describes a patient with post-pregnancy hearing loss. The audiogram shows conductive deafness along with Carhart's notch (dip in bone conduction at 2000 Hz). These findings are highly suggestive of otosclerosis.

Ototoxicity, Meniere's disease, and noise-induced hearing loss cause sensory neural hearing loss (SNHL) and not conductive hearing loss.

Otosclerosis is caused by bony overgrowth involving the enchondral layer of the bony otic capsule. It presents with bilateral progressive conductive hearing loss and tinnitus. It has an autosomal dominant inheritance and can worsen with stress and hormonal changes like during pregnancy and menopause.

Stapedial otosclerosis is the most common type of otosclerosis. Here the lesion starts just in front of the oval window in an area called fistula ante fenestrae.

The presence of a reddish hue on the promontory through the tympanic membrane, known as the Schwartze sign, suggests an active focus. PTA shows loss of air conduction with air-bone gap, more for lower frequencies and Carhart's notch may be seen. Tympanometry shows an As type of tympanogram in which there is reduced compliance at or near the ambient air pressure.

Medical management with sodium fluoride is used in the active stage. Definitive treatment is stapedectomy/stapedotomy with the placement of a prosthesis. Carhart notch usually disappears after a successful stapedectomy.

Other options:

Option A: Ototoxicity causes SNHL and shows a high-frequency hearing loss in the SNHL curve.

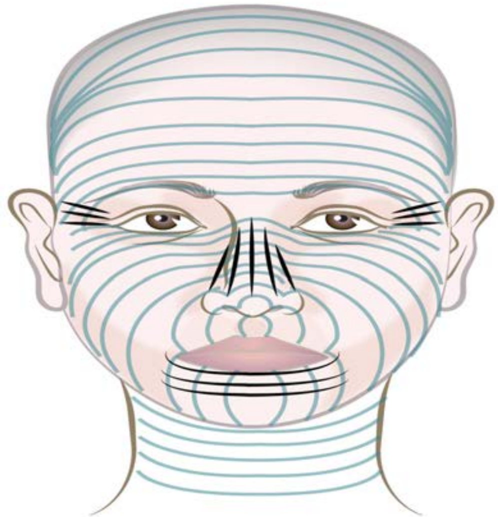
Option B: In Meniere's disease, pure tone audiometry shows an SNHL pattern with a rising curve in the early stages (as lower frequencies are involved first) and as the disease progresses higher frequencies are involved, the curve becomes flat or a falling type.

Option D: Noise-induced hearing loss is SNHL and PTA shows a characteristic acoustic dip at 4000 Hz both for air and bone conduction.

Solution to Question 2:

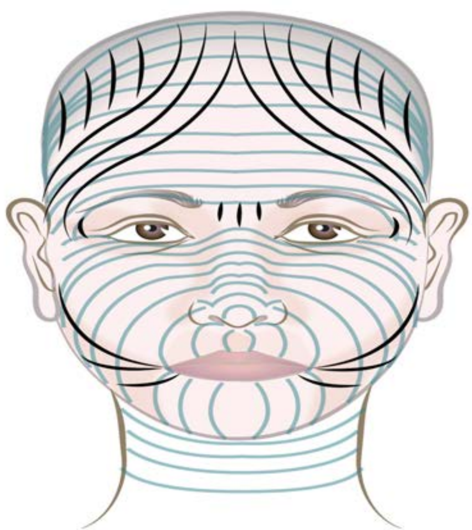
Kraissl's lines are relaxed tension lines in the skin. They lie perpendicular to the long axis of the underlying muscle. They were described in living individuals and reflect underlying muscle action.

Image of Kraissl's lines:



Langer's lines correspond to the natural orientation of collagen fibers in the dermis and are generally parallel to the orientation of the underlying muscle fibers. They were described in cadavers in reference to rigor mortis.

Image of Langer's lines:



Borge's lines are also relaxed skin tension lines observed by pinching the skin in living individuals.

Borge's lines and Kraissl's lines are a better guide for elective incisions in the face and body than Langer's lines.

Linear incisions in the skin tend to have variable degrees of gaping. When a circular skin incision is performed, the skin defect acquires an elliptical configuration. By aligning circular incisions perpendicular to the muscle action, surgeons can minimize tension on the wound edges, reduce the risk of scar contracture, and achieve better cosmetic results.

Difference between Kraissl's line and Langer's line

Kraissl's line	Langer's line
Described in living individual s	Described in cadavers
Lie perpendicular to the long axis of the underlying muscle	Correspond to the natural ori entation of collagen fibers
Reflects the action of the und erlying muscle	They are skin tension vectors observed in the stretched ski n of cadavers exhibiting rigor mortis

Solution to Question 3:

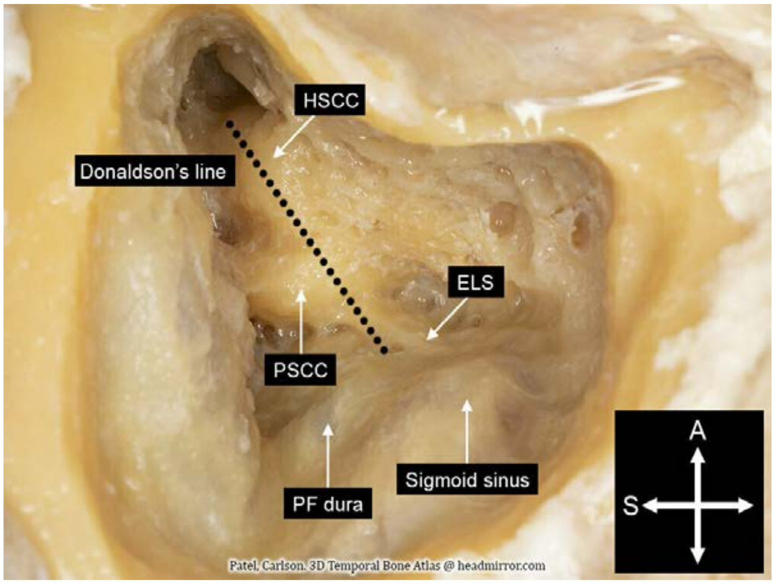
The given image shows Ohngren's line. It is an imaginary line passing through the medial canthus of the eye and the angle of the mandible. It is used for the classification of maxillary carcinoma.

Growth situated above the plane (supra structure) tends to be more aggressive and poorly differentiated and has a poor prognosis. Growth situated below the plane (infrastructure) is more amenable to treatment and has a better prognosis.

Other options:

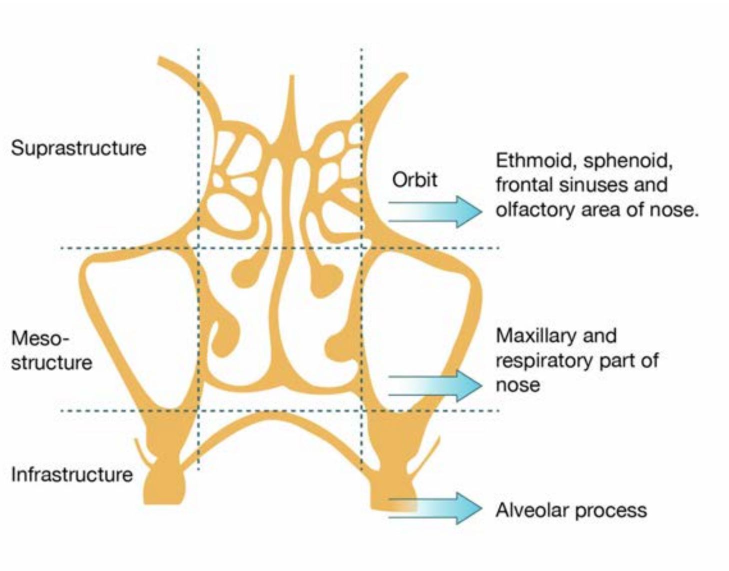
Option A: Proper alignment of the face is determined by the Frankfort plane, a horizontal line bisecting the inferior border of the infraorbital rim and the superior border of the external auditory canal.

Option B: Donaldson's line passes through the horizontal semicircular canal and bisects the posterior canal. It is the landmark for the endolymphatic sac which lies anterior and inferior to it.



ELS: Endolymphatic sac, HSCC: Horizontal semicircular canal, PSCC: Posterior semicircular canal, PF dura: Posterior fossa dura

Option C: Ledermann's classification uses two horizontal lines of Sebileau, one passes through the floor of the orbit and another passes through the floor of the maxillary antra dividing the face into supra structure (ethmoid, sphenoid, frontal sinuses, and the olfactory area of nose), mesostructure (maxillary sinus and respiratory part of nose) and infrastructure (alveolar process). It is used to classify maxillary carcinoma.



Solution to Question 4:

The structure marked as 'B' in the given image represents the lateral (horizontal) semi-circular canal. The tympanic (horizontal) segment of the facial nerve lies below the lateral horizontal semicircular canal.

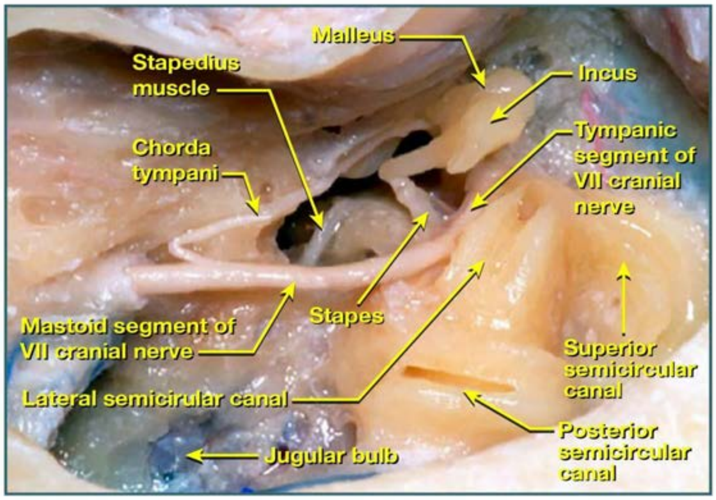
Other markings represent:

A: Superior semicircular canal (SSC)

C: Posterior semicircular canal (PSC)

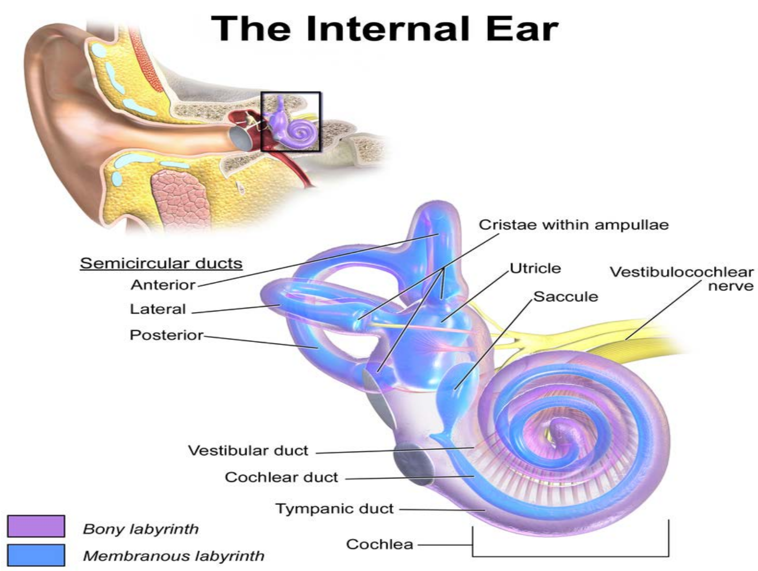
D: Incus

The semicircular canals are positioned in the superior part of the mastoid region. These canals are situated adjacent to the middle ear, with the lateral semicircular canal being the nearest to the middle ear. The superior and posterior canals share a common crescent-shaped space in between. The one that lies in close proximity to the lateral semicircular canal is the posterior semicircular canal.



The semicircular canals are part of the bony labyrinth. They house the semicircular ducts which are the end organs for angular momentum. They are 3 semicircular canals, lateral (horizontal), superior (anterior), and posterior. They lie in planes at right angles to one another.

Stimulation of semicircular canals produces nystagmus. The nystagmus is rotatory from the superior canal, horizontal from the horizontal canal, and vertical from the posterior canal.



Solution to Question 5:

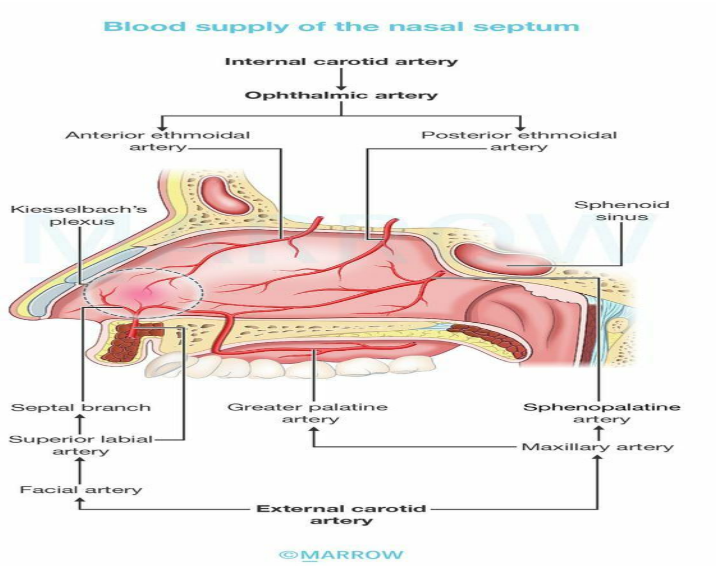
Anterior and posterior ethmoidal arteries are branches of the ophthalmic artery which in turn is a branch of the internal carotid artery.

Kiesselbach’s plexus is an arterial anastomosis at the anterior inferior part of the nasal septum (Little's area). It includes the anterior ethmoidal artery, septal branch of the superior labial artery, septal branch of the sphenopalatine artery, and septal branch of the greater palatine artery. It is the usual site of epistaxis in children and young adults.

The blood supply of the nasal septum can be divided into two main systems, the internal carotid system, and the external carotid system.

• Internal carotid system: Branches of the ophthalmic artery, anterior ethmoidal artery, and posterior ethmoidal artery.

• External carotid system: Sphenopalatine artery (branch of the maxillary artery) gives nasopalatine and posterior medial nasal branches, septal branch of the greater palatine artery (branch of the maxillary artery), and septal branch of the superior labial artery (branch of the facial artery).



Solution to Question 6:

Noise-induced hearing loss (NIHL) damages the outer hair cells, starting in the basal turn of the cochlea. Outer hair cells are affected before the inner hair cells.

NIHL occurs after chronic exposure to less intense sounds. The damage caused by noise trauma depends on the frequency of noise (frequency of 2000 to 3000 Hz causes more damage) intensity, duration of the noise, and preexisting ear disease.

The audiogram in noise-induced hearing loss shows a typical notch at 4000Hz both for air and bone conduction known as acoustic dip/boiler's notch. Patients at this stage complain of high-pitched tinnitus and difficulty hearing in noisy surroundings. Pre-employment and annual audiograms for persons working at places where noise is above 85 dB (A) help in the early detection of NIHL.

The human ear can hear frequencies from about 20 to 20,000 Hz. Below 20 Hz, the vibrations are infra-audible, and above 20,000 Hz, they are ultra-sonic. Pure tone audiometry can detect only up to 8000 Hz.

Extended high-frequency audiometry is recommended for early detection of hearing loss, as it can identify hearing impairments before the frequencies necessary for speech comprehension are affected. Distortion product otoacoustic emissions is a more sensitive tool than conventional audiometry, particularly in children under 5 years old for early detection of hearing loss.

A noise of 90 dB(A) SPL (acceptable noise levels, sound pressure level), 8 hours a day for 5 days per week is the maximum safe limit as recommended by the Ministry of Labour. As per WHO, 85 dB(A) is considered the highest safe exposure level up to a maximum of 8 hours.

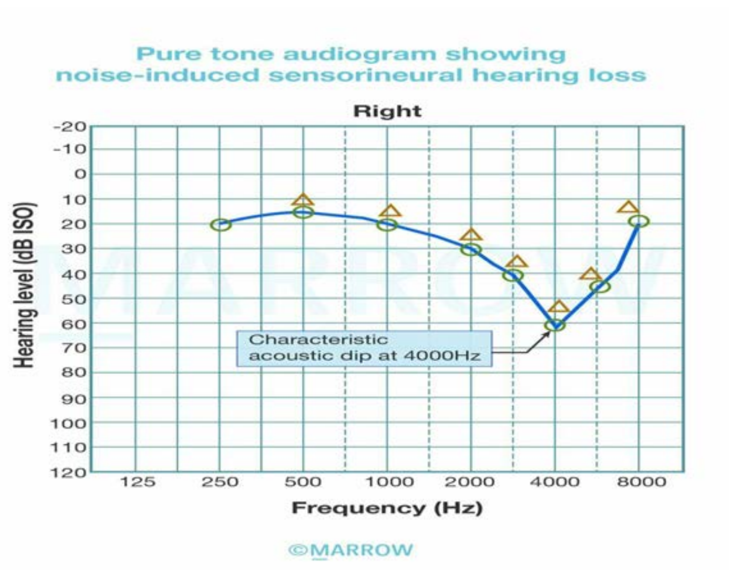
The silence zone extends to a distance of 100 meters around the premises of hospitals, nursing homes, educational institutions, and courts.

Permissible limits of noise as per the noise pollution (Regulation and Control) rules:

Noise levels are measured in dB(A), which prioritizes middle-range frequencies for assessing noise-induced hearing loss while giving less weightage to low and high frequencies.

The image below shows an acoustic dip at 4000Hz:

Zone/Area	Day(6 am to 10 pm) limits in dB(A)	Night(10 pm to 6 am) limits in dB(A)
Industrial	75	70
Commercial	65	55
Residential	55	45
Silence	50	40



Other options:

Option A: Crista ampullaris is a thickened ridge of neuroepithelium in the ampullated end of semicircular ducts.

Option B: Macula is the sensory epithelium of the utricle

Option C: Cupula is a gelatinous mass extending from the surface of the crista (peripheral receptor located in the ampullated end of semicircular ducts) to the ceiling of the ampulla, into which project the cilia of sensory hair cells.

Solution to Question 7:

Rhinosporidium seeberi is not a classic fungus, but rather the first known human pathogen from the DRIPs (Dermocystidium, Rosette agent, Icthyophonus, and Posorospermum) clade, a novel clade of aquatic protistan parasites (Ichthyosporea).

It causes a granulomatous disease called rhinosporidiosis. It is endemic in south India. Patients typically have a history of bathing in water bodies shared by animals and present with recurrent epistaxis, nasal obstruction by a red vascular mass (mulberry mass), and hypertrophic turbinates.

For diagnosis, a nasal smear can be taken to demonstrate sporangia.

Treatment is surgical excision of the mass with a diathermy knife and cauterization of base. Dapsone can also be used postoperatively to reduce the incidence of recurrence.

The image below shows a polypoidal mass of rhinosporidiosis:

Solution to Question 8:

The internal carotid artery is not ligated in epistaxis control due to a high percentage of cerebral complications and high mortality.

The first step in treating epistaxis is to pinch the nose with the thumb and index finger for about 5 minutes. The patient should sit leaning slightly forward over a basin to spit out blood and breathe through the mouth (Hippocratic or Trotter’s method).

If the bleeding point is localized, cauterization can be done with electrocautery. In cases of uncontrolled bleeding or when the bleeding is posterior into the throat, anterior or posterior nasal packing may be required, respectively. If bleeding continues despite these measures, endoscopic procedures or vessel ligation can be considered.

The sphenopalatine artery is ligated first through a trans-nasal endoscope (Transnasal endoscopic sphenopalatine artery ligation). However, if the bleeding persists, maxillary artery ligation is done. If bleeding is still not controlled, external carotid artery ligation is done. It is important to note that if the bleeding site is located superior to the level of the middle turbinate, the anterior and posterior ethmoidal arteries should be ligated.

Solution to Question 9:

The given clinical scenario of a child presenting with chronic hoarseness (3 months) of voice and respiratory distress for 2 weeks suggests the diagnosis of respiratory papillomatosis (juvenile papillomatosis).

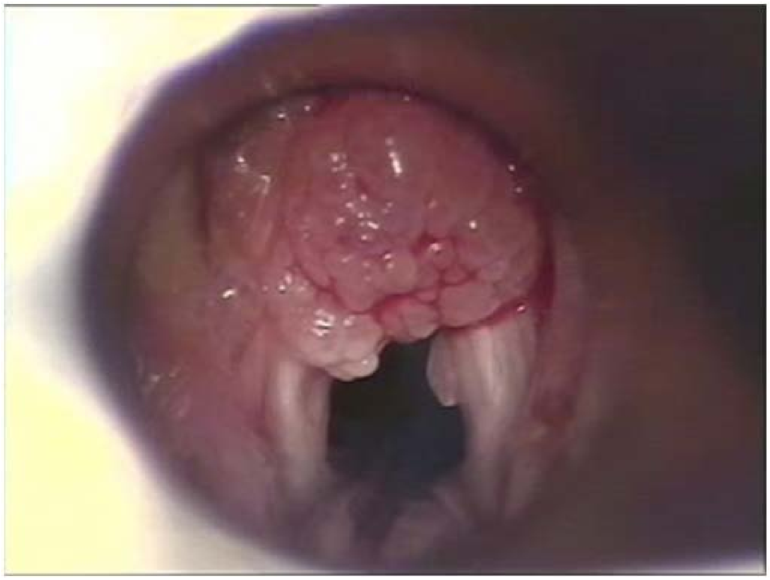
Croup (Option A) and acute epiglottitis (Option D), which cause acute upper respiratory tract obstruction, are ruled out due to the chronic nature of the symptoms. In croup, the upper airway obstruction becomes apparent after 1 to 3 days of low-grade fever, rhinorrhea, and pharyngitis. On the other hand, in epiglottitis, a child experiences respiratory distress within hours of developing a fever.

Vocal nodules (Option C) are not typically associated with respiratory distress.

Respiratory papillomatosis is the most common respiratory tract neoplasm in children. They are simple warts (benign tumors) caused by human papillomavirus types 6 and 11 commonly seen in children aged 3 to 5 years.

The course involves remissions and exacerbations of recurrent papillomas, mostly affecting the glottic area (vocal cords), and supraglottis. The patient presents with progressively worsening hoarseness, sleep-disordered breathing, dyspnea, stridor, and severe airway obstruction if left untreated.

Diagnosis confirmed by direct laryngoscopy, which shows grape-like projections (shown below), and biopsy. Treatment consists of micro laryngoscopy and CO_2 laser excision. Antivirals like interferon alpha2a, ribavirin, cidofovir, and acyclovir are used as adjunct therapies.



Other options:

Option A: Croup (laryngotracheobronchitis) is caused by the parainfluenza virus. The patient initially presents with an upper respiratory tract infection (rhinorrhea, pharyngitis, cough, low-grade fever) followed by a characteristic barking cough, hoarseness, and inspiratory stridor.

Option C: Vocal nodule (singer's or screamer's nodule) occurs as a result of chronic vocal abuse. They appear symmetrically at the junction of the anterior one-third and posterior two third of the vocal cord. The patient presents with chronic hoarseness and voice fatigue.

The image below shows symmetric vocal nodules:

Option D: Acute epiglottitis is caused by streptococcus pyogenes, pneumococci, Hemophilus influenzae, and staphylococcus aureus. It is characterized by an acute rapidly progressive course of high-grade fever, dyspnea, and stridor.

ENT AIIMS 2018

Question 1:

While performing a caloric test, what type of nystagmus would you expect on cold water irrigation?

a) Horizontal nystagmus towards the same side

b) Vertical nystagmus towards the same side

c) Horizontal nystagmus towards the opposite side

d) Vertical nystagmus towards the opposite side

Question 2:

In which of the following conditions is Gelle's test negative?

a) CSOM with cholesteatoma

b) Senile deafness

c) Meniere's disease

d) Otosclerosis

Question 3:

What is the definitive treatment of intractable vertigo due to Meniere’s disease?

a) Acetazolamide

b) Labyrinthectomy

c) Vestibular neurectomy

d) Intratympanic gentamicin

Question 4:

A patient comes with an ulcer on the tongue with a duration of 3 months. On examination, there is a 3.5 cm x 2.5 cm lesion with induration of 8 mm. There was no lymphadenopathy. What is the stage of this lesion according to 8th AJCC guidelines?

a) Stage I

b) Stage II

c) Stage III

d) Stage IV

Question 5:

Which of the following features will not be manifested in an infection of the post-styloid compartment of the parapharyngeal space?

a) Trismus

b) Vocal cord palsy

c) Parotid swelling

d) Horner's syndrome

Question 6:

A patient presents with oral cavity findings are seen in the image below. He is a smoker. What would you choose as your next step in the management of this patient?

a) Give corticosteroids

b) Cessation of smoking and do biopsy

c) Lifestyle modifications

d) Injection vitamin B12

Question 7:

A teenager has a mass in his nasopharynx that has minimally spread into the sphenoid sinus with no lateral extension. Which stage of nasopharyngeal angiofibroma could this be?

- **a)** Stage IA
- **b)** Stage IB
- **c)** Stage IIA
- **d)** Stage IIB

Question 8:

A patient complains that he feels like the room is spinning when he gets up from lying down or turning his head. He has no history of loss of consciousness. Which of the following could be the probable diagnosis?

- **a)** Benign paroxysmal positional vertigo
- **b)** Meniere's disease
- **c)** Labyrinthitis
- **d)** Syncope

Answer Key

Question No.	Correct Option
1	c
2	d
3	b
4	b
5	a
6	b
7	b
8	a

Detailed Explanations

Solution to Question 1:

In caloric testing, cold water irrigation produces horizontal nystagmus towards the opposite side of irrigation. The direction of nystagmus can be remembered by the mnemonic, COWS (cold–opposite, warm–same).

The mechanism for the nystagmus is the creation of a temperature gradient by the cold water, which pulls the cupula away from the utricle, causing nystagmus towards the oppsite side. As the horizontal semicircular canal is closer to the outer ear canal, most of the response comes from it and the nystagmus is horizontal.

The Fitzgerald–Hallpike test (bithermal caloric test) is done to assess this phenomenon. The patient lies in a supine position with head tilted forward at a 30° angle, ensuring the horizontal canal becomes vertical. Alternatingly, water at temperatures of 30°C and 44°C (7° below and above normal body temperature, respectively) is irrigated into the ears for a duration of 40 seconds each. The examiner closely observes the patient's eyes for the emergence of nystagmus until its endpoint is reached. The time taken from the start of irrigation to the endpoint of nystagmus is recorded and plotted on a calorigram.

If no nystagmus is evoked from either ear, the test is repeated with water at a temperature of 20°C for a duration of 4 minutes before labeling the labyrinth dead. A 5-minute interval should be allowed between testing each ear to ensure accurate results.

Solution to Question 2:

Gelle’s test is negative when the ossicular chain is fixed as in cases of otosclerosis.

Gelle’s test is a test of bone conduction and examines the effect of increased air pressure in the ear canal on hearing. In normal individuals, due to the increased intralabrynthine pressure applied by the Seigel's speculum, there is immobility of the basilar membrane, as the tympanic membrane and the ossicles are pushed inwards (medially). This results in decreased hearing. But, in diseased individuals, where the ossicular chain and the tympanic membrane are fixed or disconnected, no change in hearing is observed.

Gelle’s test is positive in normal persons and in those with sensorineural hearing loss.

Note: This test was commonly used for the diagnosis of otosclerosis, but has since been replaced by tympanometry.

Solution to Question 3:

The definitive treatment of intractable vertigo due to Ménière’s disease is labyrinthectomy.

Labyrinthectomy involves uniform destruction of hearing and vestibular function. Ideal candidates have no functional hearing and have failed more conservative treatments, such as gentamicin injection. Despite this morbidity, labyrinthectomy has the highest rate of vertigo control (>95%), more than vestibular neurectomy (78-90%), and is considered the single definitive treatment.

The step-by-step management approach for Ménière's disease consists of dietary modifications and vestibular suppressants as the initial measures, followed by intratympanic gentamicin ablation, and finally, surgical intervention.

Primarily, the management of Menier's disease involves a sodium restricted diet and diuretics like acetazolamide (Option A). They are more useful for short-term control.

Gentamicin (Option D) is a selective vestibulotoxic aminoglycoside antibiotic that is preferentially taken up by type I hair cells. It is opted for only if dietary modifications and vestibular suppressants have failed to control the symptoms. If used judiciously, hearing is preserved.

The options for surgical management include - Endolymphatic sac surgery, vestibular neurectomy, and labyrinthectomy. Endolymphatic sac surgery is currently declining in popularity due to lack of evidence of vertigo control.

Vestibular neurectomy (Option C) refers to selective ablation of the vestibular nerve. Even though it is considered a hearing sparing surgery and carries comparable rates of vertigo control as labyrinthectomy, it is an intradural procedure and carries a greater risk of complications. Hearing loss will still progress as it would have in the natural course of illness.

Solution to Question 4:

The maximum diameter of the ulcer is less than 2-4 cm and the depth of invasion is less than 10mm which implies T2; There is no lymph node involvement (N0) and no metastasis (M0). A T2N0M0 tumor is classified as stage II according to the 8th AJCC guidelines.

8th AJCC guidelines for the staging of oral tumors:

The changes in the 8th AJCC update include:

• Depth of invasion (DOI) and extranodal extension have been included as independent prognostic factors.

• A new category of tumor added - HPV positive, p16 positive

• In the past, the lip was considered a primary site for oral cancers; now - it is divided into mucosal and cutaneous lip, mucosal lip is included in the oral cavity.

The gold standard investigation for oral cancer includes an incisional biopsy. An MRI is the imaging modality of choice for assessing primary extent. Surgical management with chemotherapy and adjuvant radiotherapy is the mainstay of management.

Staging	Tumor Size	Lymph Node	Metastasis
I	T1 (<2 cm, depth <5 mm)	N0 (No regional Lymph Nodes)	M0
II	T2 (2 - 4 cm, depth <10 mm)	N0	M0
III	T3 (>4 cm, depth>10 mm)	N0	M0
III	T1, 2, 3	N1	M0

Staging	Tumor Size	Lymph Node	Metastasis
IVA	T4a (cortical bone of mandible or maxillary sinus or skin of face)	N0,1	M0
IVA	T1, 2, 3, 4a	N2	M0
IVB	Any T	N3	M0
IVB	T4b (masticator space, pterygoid plates, skull base or encases carotid vessels)	Any N	M0
IVC	Any T	Any N	M1

Solution to Question 5:

Trismus is not a feature of infection of the post-styloid compartment of the parapharyngeal space. Trismus is seen in pre-styloid abscess due to the involvement of medial pterygoid muscle.

Parapharyngeal space is divided into two compartments by the styloid process and its attachments. Clinical features depend on the compartment involved.

In a post-styloid abscess, there is a bulge of the pharynx behind the posterior pillar, accompanied by paralysis of cranial nerves IX, X (resulting in vocal cord paralysis), XI, XII, and the sympathetic chain, which can manifest as Horner's syndrome. Swelling in the parotid region may also be present. Notably, trismus, the inability to fully open the mouth, is usually absent in this type of abscess.

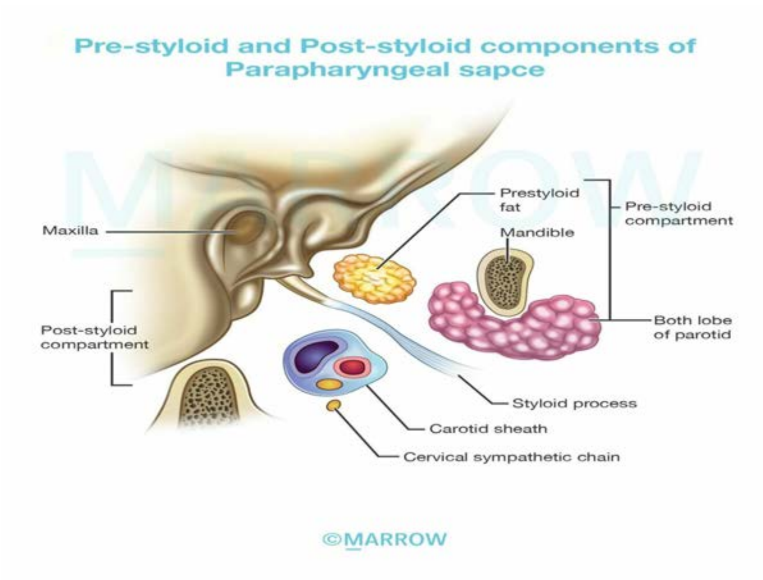
A pre-styloid abscess exhibits distinct features including the prolapse of the tonsil and tonsillar fossa, along with trismus caused by the spasm of the medial pterygoid muscle. Marked odynophagia is common. Additionally, an external swelling can be observed behind the angle of the jaw.

Fever, sore throat, torticollis (caused by spasm of the prevertebral muscles), and signs of systemic toxicity are common features.

Diagnosis can be established using a contrast-enhanced CT scan (CECT). If there is suspicion of thrombosis of the internal jugular vein or an aneurysm of the internal carotid artery, magnetic resonance arteriography (MRA) can be useful.

Patients require intravenous antibiotic therapy with amoxicillin-clavulanic acid, carbapenem antibiotics plus drugs that target anaerobic organisms, such as clindamycin or metronidazole. A parapharyngeal abscess is drained by an external horizontal incision made on the skin a few centimeters below the angle of the mandible.

The image attached below shows the components of both spaces:



Solution to Question 6:

The patient, who is a chronic smoker, presenting with a wrinkled white patch on the ventral surface of his tongue is likely to have leukoplakia. Confirmation of this diagnosis will require a biopsy of the lesion. Leukoplakia is a premalignant lesion, with a malignant potential of 1 - 17.5%, hence it is imperative that the patient is advised on cessation of smoking.

Leukoplakia is a premalignant condition of the oral cavity characterized by whitish patches or plaques that cannot be rubbed off. They appear as small, well-circumscribed lesions and can extensively involve a large area of the oral mucosa.

Variants of leukoplakia are:

• Homogenous variant: Smooth white patch. Less asociation with malignancy.

• Non-homogenous variants:

• Speckled leukoplakia is whitish lesions on an erythematous base. They have the highest risk of malignant transformation.

• Proliferative verrucous leukoplakia is rare and has a high risk for malignant transformation.

• Erosive (erythroleukoplakia) is when it is interspersed with erythroplakia and has erosions and fissures. This also has malignant potential.

Management includes smoking cessation and reduction in alcohol consumption. A biopsy should be done. A CO_2 laser is often used for ablation.

A photographic record of the lesion is useful for long-term follow-up in cases of conservative treatment.

Given below is a list of premalignant lesions of the oral cavity with their risk of malignancy:

High-risk lesions	Medium risk lesions	Low-risk lesions
Erythroplakia(homogenous/speckled)Seen asred velvetylesionsSpeckledtype is the most aggressiveThe treatment of choice issurgicalablation	Oral submucous fibrosisHypersensitivereactionto betel nut product.Other risk factors are - tobacco, chili, vitamin deficiencyClinical features - trismus, ankyloglossia, soreness, and burning sensation in the mouth.Management includes antioxidants, intralesional triamcinolone, and surgical excision.	Oral lichen planusReticulate fine lace-like pattern.Management istopicalcorticosteroids.
Proliferative verrucous leucoplakiaAwhite patchylesion thatcannot be rubbed off.Speckled leucoplakia has an erythematous baseThe treatment of choice is excision	Syphilitic glossitisAssociated with tertiary syphilis	Discoid lupus erythematosus
Chronic hyperplastic candidiasisWhite patchesthatcan be rubbed offwith ahyperaemicborder.Management includes prolonged treatment withtopicalantifungalsorsystemicantifungaltherapy. If no improvement or if there are any signs of dysplasia,surgical excision is done.		Dyskeratosis Congenita

Solution to Question 7:

The mass has spread into the sphenoid sinus with no lateral extension, so it's Stage 1B according to the Radkowski classification.

Multiple classification systems exist for the staging of juvenile nasal angiofibroma, of which the most commonly used ones are the Fisch and Radkowski classification systems:

Staging of juvenile nasopharyngeal angiofibroma (Radkowski, et al):

Fisch staging system of juvenile nasopharyngeal angiofibroma:

Juvenile nasopharyngeal angiofibroma (JNA) is a highly vascular and locally aggressive tumor seen almost exclusively in adolescent males, associated with significant morbidity and occasional mortality.

Note: The Fisch system is considered to be the more robust and practical of the two.

Stag e	Spread
IA	Limited to the nose and/or na sopharyngeal vault
IB	Extension into ≥ 1 sinus
IIA	Minimal extension into the p terygopalatine fossa
IIB	Full occupation of pterygopal atine fossa with or without er osion of orbital bones
IIC	Infratemporal fossa with or without cheek or posterior to pterygoid plates
IIIA	Erosion of skull base-minima l intracranial
IIIB	Erosion of skull base-extensi ve intracranial with or witho ut cavernous sinus

Type	Details
1	Tumourlimited to the nasoph aryngeal cavity; bonedestruct ion negligibleor limited to th e sphenopalatine foramen
2	Tumourinvading the pterygo palatine fossaor the maxillary, ethmoid, or sphenoidsinus w ith bone destruction
3	A:Tumour invading theinfrat emporal fossaororbital region withoutintracranial involvem ent
B:Tumour invading theinfrat emporal fossaororbital region withintracranialextradural(pa rasellar) involvement	
4	A:Intracranialintraduraltumo rwithoutinfiltration of the ca vernous sinus, pituitary fossa, or optic chiasm
B:Intracranialintraduraltumo rwithinfiltration of the caver nous sinus, pituitary fossa, or optic chiasm	

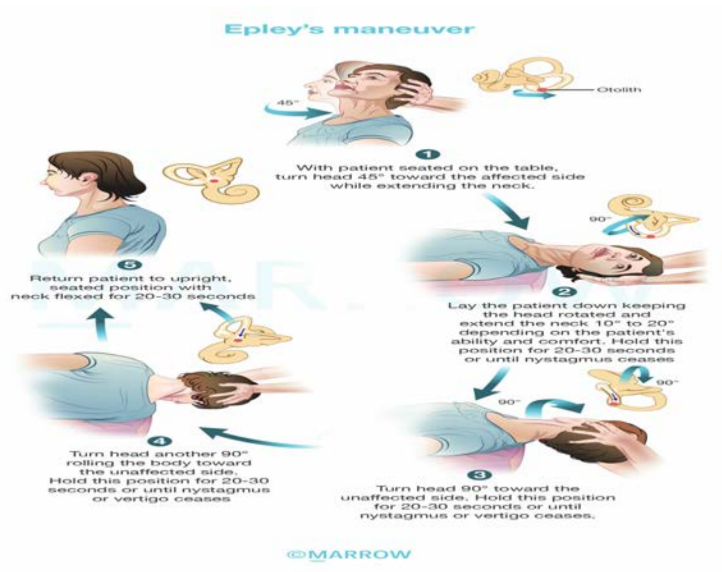
Solution to Question 8:

The given clinical scenario, with complaints of dizziness associated with postural change and no history of loss of consciousness is suggestive of benign paroxysmal positional vertigo (BPPV).

BPPV is one of the common causes of recurrent vertigo, and the episodes are brief (<1min) and are always provoked by changes in head position. The attacks are caused by free-floating calcium carbonate crystals usually dislodged from the posterior canal.

A positive Dix-Hallpike maneuver confirms the diagnosis of BPPV. In this maneuver, the patient begins sitting up with their head turned 45 degrees toward the ear to be tested. Then the patient is made to lie down quickly with their head past the end of the bed. Then, the patient's neck is extended 20 degrees below the horizontal. The clinician then watches the patient's eyes for torsional and up-beating nystagmus, which should persist for no more than one minute. This would indicate a positive test.

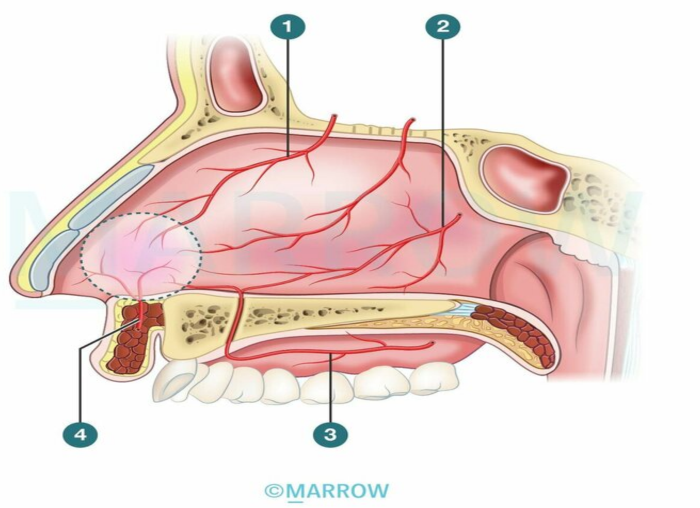
Epley's maneuver is performed to reposition the crystals into the macula from the semicircular canals. This is only a temporary cure and there is a high rate of recurrence.



ENT AIIMS 2019 (May & Nov)

Question 1:

Which of the following marked structures is a branch of the internal carotid artery?



a) 1

b) 2

c) 3

d) 4

Question 2:

Which of the following is not an etiological factor for head and neck malignancies?

a) EBV infection

b) HPV infection

c) Betel nut

d) Vitamin A deficiency

Question 3:

Which of the following structures are not removed in radical neck dissection?

a) Tail of parotid

b) Sublingual gland

c) Submandibular gland

d) Level 2b lymph nodes

Question 4:

Which of the following features is not associated with superior semicircular canal dehiscence syndrome?

a) Gaze dependent nystagmus

b) Positive Fistula test

c) Sensorineural hearing loss

d) Positive Tulio phenomenon

Question 5:

A patient undergoes surgery of the lateral part of the base of the skull. Postoperatively, the patient presents with aspiration but does not have a change in voice. Which of the following is the reason?

a) Superior laryngeal nerve injury

b) Glossopharyngeal nerve injury

c) Recurrent laryngeal nerve injury

d) High vagal injury

Answer Key

Question No.	Correct Option
1	a
2	d
3	b
4	c
5	a

Detailed Explanations

Solution to Question 1:

The structure marked 1 is the anterior ethmoidal artery, which is a branch of the ophthalmic division of the internal carotid artery.

Other structures marked in the diagram are as follows:

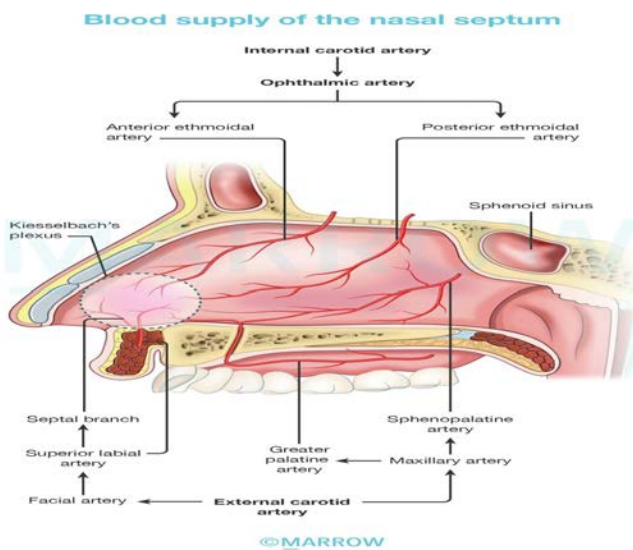
2 – Sphenopalatine artery

3 – Greater palatine artery

4 – Superior labial artery

2, 3 and 4 are derived from branches of the external carotid artery.

The nasal septum is supplied by branches of both internal and external carotid arteries.



Solution to Question 2:

Vitamin A deficiency is not a risk factor for head and neck malignancies. Vitamin A deficiency is a risk factor for esophageal cancer.

The head and neck cancers include cancers of the oral cavity, pharynx, larynx, paranasal sinuses, nasal cavity, and salivary glands. Squamous cell cancer is the most common head and neck malignancy, often linked to tobacco and alcohol use. Non-squamous malignancies include thyroid cancer, salivary gland cancer, and sarcomas.

The risk factors for head and neck malignancies are listed below:

Oral cavity cancer:

• Tobacco, smokeless tobacco like Bidi, Chutta, Paan (oral tobacco mixed with betel leaf, areca nut, and salted lime), Khaini, and marijuana

• Alcohol, poor oral hygiene, and intake of red meat and salted meat

• Infections with human papillomavirus, and HIV

• Genetic conditions like Li-Fraumeni, ataxia telangiectasia, Bloom syndrome, and Fanconi's anemia.

Oropharyngeal cancer:

• Smoking and alcohol

• HPV 16 and 18

Cancer of the larynx and hypopharynx:

• Tobacco and alcohol

• Nickel, mustard gas, and asbestos exposure

• Long-term exposure to sulphuric acid and hydrochloric acid in battery plant workers.

• Postcricoid carcinoma is associated with previous radiation and sideropenic dysphagia

• HPV infection and immunosuppression

Nasopharyngeal cancer:

• Chinese population

• Epstein Barr virus infection

• Nitrosamines in salted fish and meat

• Vitamin C deficiency

Cancers of the nasal cavity and paranasal sinus:

• Tobacco

• Occupational exposure to dust from wood, textiles, and leather. Nickel and chromium dust and mustard gas exposure.

• HPV 16 and 18

Thyroid cancer:

• Females

• Previous radiation

• Inherited conditions like Gardner syndrome, familial polyposis, and Cowden syndrome. Medullary thyroid cancer associated with MEN-2 syndrome

Salivary gland carcinoma:

• Previous radiation

• Exposure to nickel alloy dust, and silica dust

• Smoking and heavy alcohol consumption were linked to a higher risk of salivary gland cancer in men but not in women

• Early menarche and nulliparity

Solution to Question 3:

Sublingual gland is not removed in radical neck dissection (RND). Submandibular gland is removed in RND.

RND involves the removal of:

• All lymph nodes from Levels I through to V (submental, submandibular, upper, middle, and lower jugular, and posterior triangle regions) along with its fibrofatty tissue.

• The spinal accessory nerve

• Internal jugular vein

• Sternocleidomastoid muscle

• Tail of parotid

• Submandibular gland

• Omohyoid muscle

Solution to Question 4:

Sensorineural hearing loss is not seen in superior semicircular canal dehiscence syndrome (SSCDS). They may experience mild to moderate hearing conductive loss.

Conductive hearing loss is due to the abnormal transmission and dissipation of acoustic energy through the defect (third window), leading to reduced conduction to the cochlea for lower frequencies. This can manifest as a low-frequency air-bone gap at multiple frequencies. There is no defect in the cochlea or the auditory pathway.

Superior semicircular canal dehiscence (SSCD) is a condition in which bone between the apex of the superior semicircular canal and the middle cranial fossa is absent. This causes endolymph in the labyrinthine system to move with sound or pressure, causing an activation of the vestibular system.

Patients may have vestibular symptoms like vertigo triggered by loud sounds (Tullio phenomenon)or pressure such as coughing and can also experience hyperacusis for bone-conducted sounds. They may perceive the sounds of their own eye movements or footsteps in the affected ear.

The patients may exhibit nystagmus when exposed to loud sounds or during the Valsalva maneuver or pressure in the ear canal (positive fistula test). The direction of the nystagmus aligns with the affected superior semicircular canal, and some patients may tilt their heads in the plane of the affected superior semicircular canal in response to loud sounds.

Noncontrast HRCT of the temporal bone can demonstrate the defect. The click-evoked VEMP (vestibular evoked myogenic potential) response and vestibuloocular reflex are pronounced in SSCD patients, in which the evoked eye movements align with the affected superior canal.

For mild symptoms, avoidance of triggers is recommended. In case of severe disabling symptoms, surgical resurfacing of the superior semicircular canal or plugging of the superior semicircular canal may be attempted.

Solution to Question 5:

In the given clinical scenario, the patient experiences aspiration but does not exhibit any change in voice following base of skull surgery. This suggests an injury to the superior laryngeal nerve (SLN).

In SLN injury, only the cricothyroid muscle is involved which is used only in high-pitch vocalization, normal speech will not be affected. SLN also provides sensory supply to the mucosa above vocal cords, hence its lesion would lead to increased susceptibility to aspiration.

Other options:

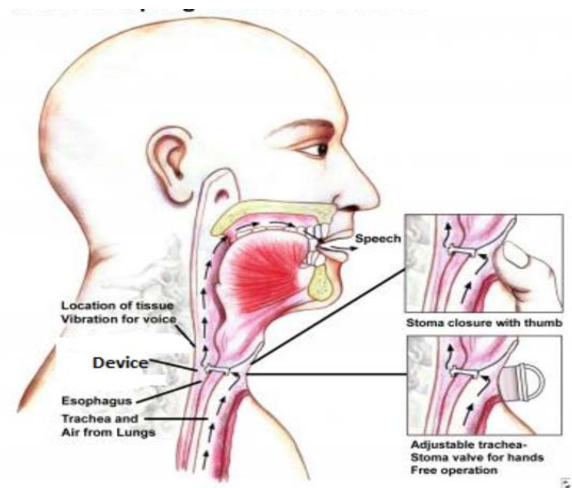
Option B: The glossopharyngeal nerve does not provide sensory or motor supply to the larynx.

Option C and D: The patient also has normal vocalization which rules out recurrent laryngeal nerve (RLN) and high vagal injury. RLN supplies all the intrinsic muscles of the larynx except cricothyroid, therefore an RLN lesion would lead to disordered vocalization. A high vagal injury would mean that both the SLN and RLN functions are affected, therefore that can also be ruled out.

ENT AIIMS 2020 (May and Nov INI-CET)

Question 1:

A patient underwent total laryngectomy for laryngeal cancer. Identify the modality of voice production in this patient shown in the image given below.



a) Electro-larynx

b) Chicago implant

c) Tracheoesophageal prosthesis

d) Esophageal speech

Question 2:

Anterior most air cells of anterior ethmoidal sinus are known as?

a) Bulla ethmoidalis

b) Onodi cell

c) Agger nasi

d) Haller cell

Question 3:

A patient complaining of vertigo without hearing loss has consulted an ENT surgeon. The surgeon performs a diagnostic maneuver and cautiously performs a therapeutic maneuver. What is the diagnostic maneuver?

a) Hampton's maneuver

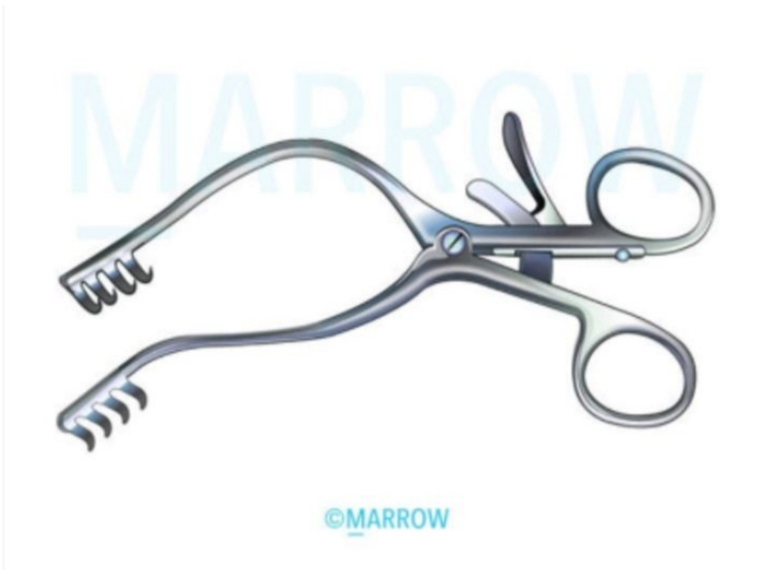
b) Dix-Hallpike maneuver

c) Epley's maneuver

d) Simon's maneuver

Question 4:

Identify the retractor shown in the below image?



a) Joll's retractor

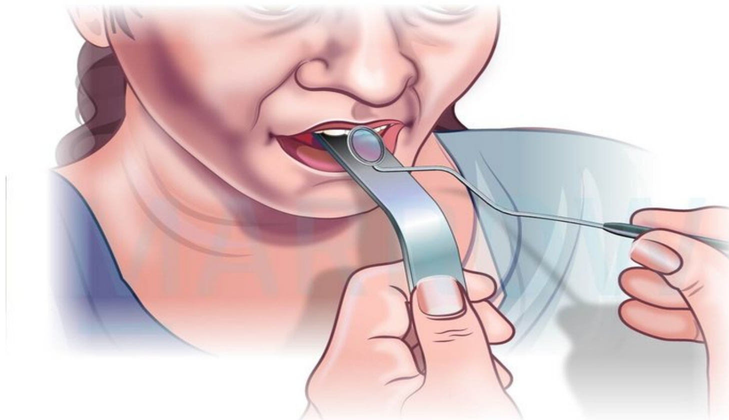
b) Perkins retractor

c) Mastoid retractor

d) Langenbeck retractor

Question 5:

A senior resident in the hospital is performing the following clinical examination as shown in the image below. Which of the following structures will not be visible to him?



a) Adenoids

b) Arytenoids

c) Torus tubarius

d) Upper surface of soft palate

Question 6:

Which of the following is not a branch of the external carotid artery in Kiesselbach's plexus?

a) Anterior and posterior ethmoidal

b) Sphenopalatine artery

c) Greater palatine artery

d) Septal branch of superior labial

Question 7:

The radiograph shown below is done for better assessment of the frontal sinus. What is the common name of this view?

a) Water’s view

b) Caldwell view

c) Pierre's view

d) Towne’s view

Question 8:

Which of the following is not seen in a case of otitis media?

a) Bezold abscess

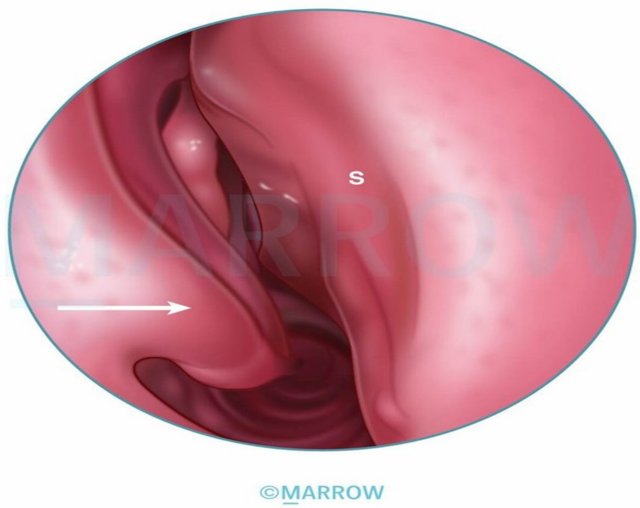
b) Griesinger's sign

c) Battle sign

d) Delta sign

Question 9:

Given below is the endoscopic view of the right nasal cavity. Identify the marked structure?



a) Inferior Turbinate

b) Middle turbinate

c) Superior Turbinate

d) Uncinate process

Question 10:

Which of the following are the objective tests for the hearing?

a) 1, 2, and 4

b) 1,2, and 3

c) 2,3, and 4

d) 1,2,3, and 4

Question 11:

Identify the correct sequence of the auditory pathways: A.Superior Olivary nucleus B. Cochlear Nucleus C. Inferior Colliculi D. Lateral lemniscus E. Medial geniculate ganglion

a) ABCDE

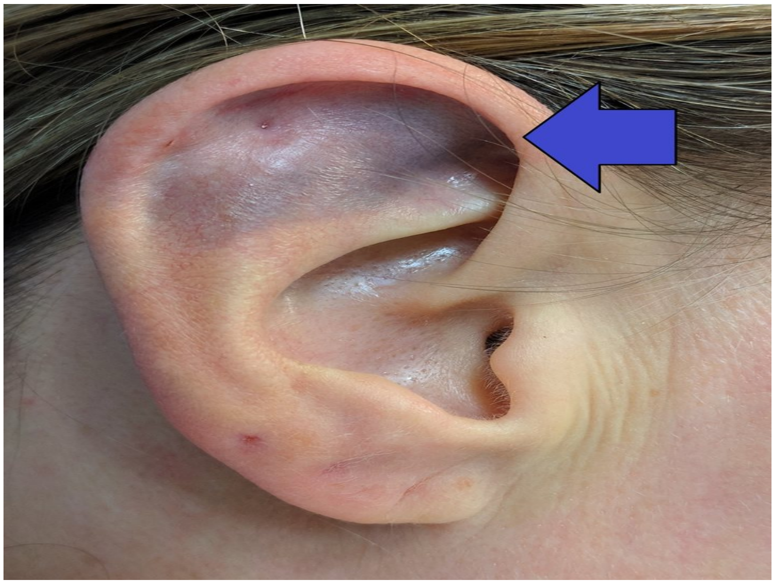
b) BADCE

c) BACDE

d) BAEDC

Question 12:

A 55-year-old female came to the hospital with a history of trauma to the ear. On examination, there is swelling, tenderness, and discoloration of the ear. Which of the following is false regarding the given image?



a) All cases should receive prophylactic antibiotic

b) May lead to Pugilistic ear

c) Caused by accumulation of blood in perichondrial space

d) Resolves spontaneously

Answer Key

Question No.	Correct Option
1	c
2	c
3	b
4	c
5	b
6	a
7	b
8	c
9	a

10	a
11	b
12	d

Detailed Explanations

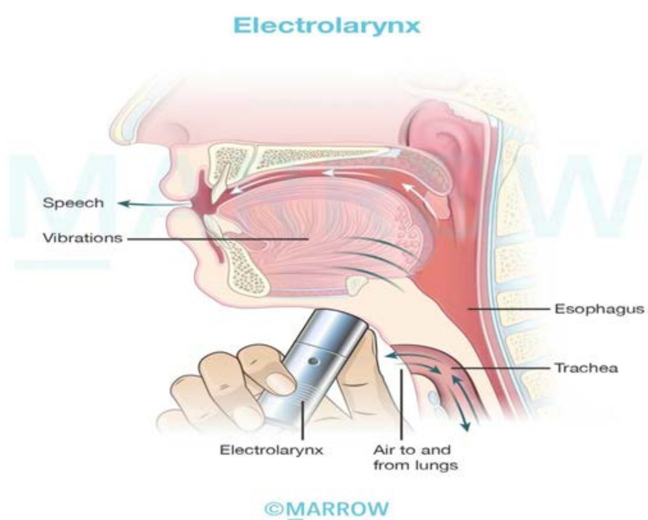
Solution to Question 1:

The modality of voice production shown in the given image is a tracheoesophageal voice prosthesis.

In a tracheoesophageal prosthesis, the exhaled air from the trachea is diverted to the esophagus through a device that connects the trachea and esophagus. It has a one-way valve, which prevents food from entering into the trachea. A variety of tracheoesophageal prostheses are available including the provox, Blom-Singer, and Panje devices.

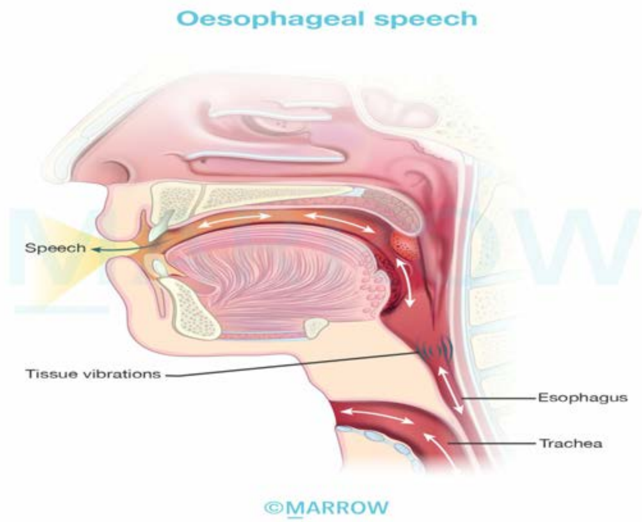
Other options:

Option A: Electrolarynx is a vibrating device that produces a low-pitched sound in the hypopharynx. It is then articulated by the palate, tongue, and teeth.



Option B: Chicago implant is a hearing aid.

Option D: In oesophageal speech, the patient is trained to swallow air and slowly release it into the pharynx. 6-10 words can be spoken in a rough manner before re-swallowing the air.



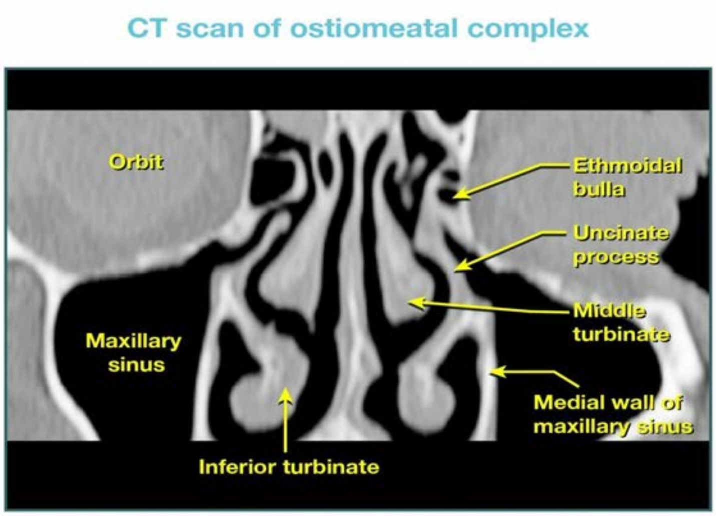
Solution to Question 2:

Agger nasi is the anterior-most air cells of anterior ethmoid air cells. When enlarged it may cause mechanical obstruction to frontal sinus drainage.

Other options:

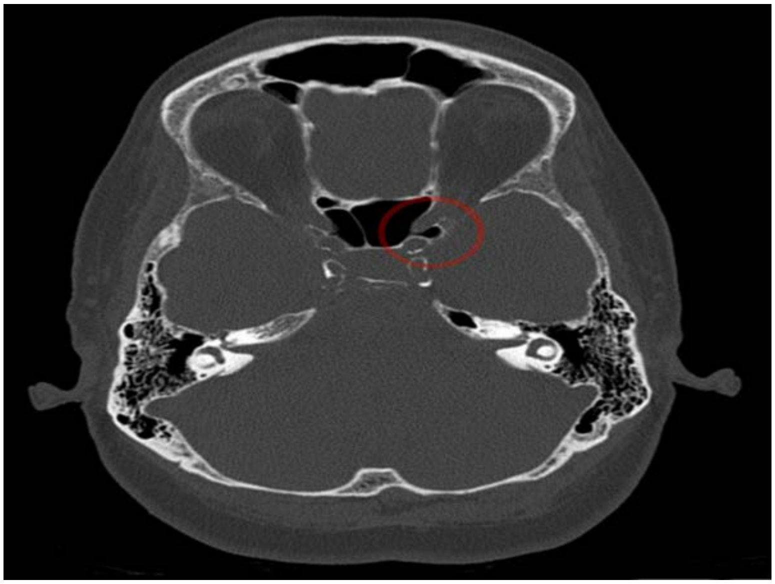
Option A: Bulla ethmoidalis is the largest air cell in the ethmoidal air sinus. It is situated behind the uncinate process and forms the posterior boundary of hiatus semilunaris through which the maxillary, frontal, and anterior ethmoidal sinuses drain.

The image below shows the ethmoidal bulla:



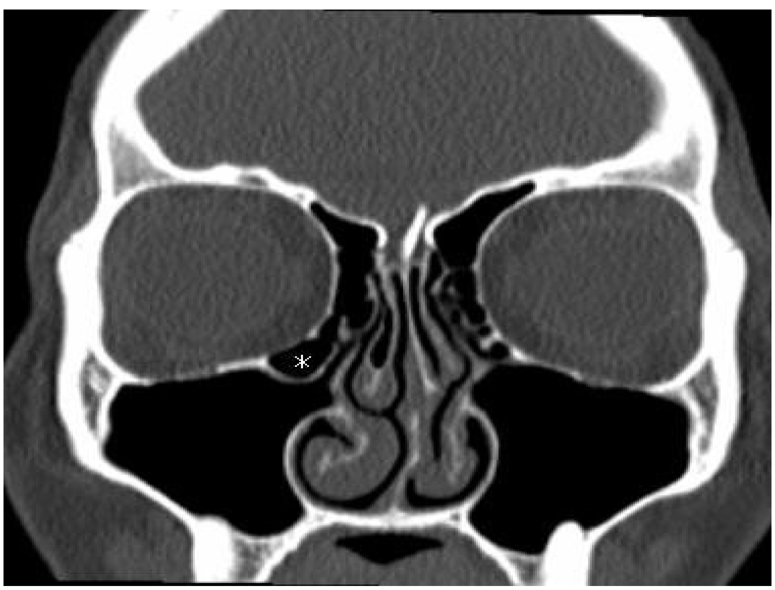
Option B: Onodi cell is formed from posterior ethmoidal air cell. The optic nerve lies in its lateral wall and can be injured while operating in the cells.

The image below shows Onodi air cell:



Option D: Haller cells are formed from either anterior or posterior ethmoid air cells. They present in relation to the floor of the orbit and roof of the maxillary sinus, enlargement may prevent drainage of the maxillary sinus.

The image below shows Haller cell:



Solution to Question 3:

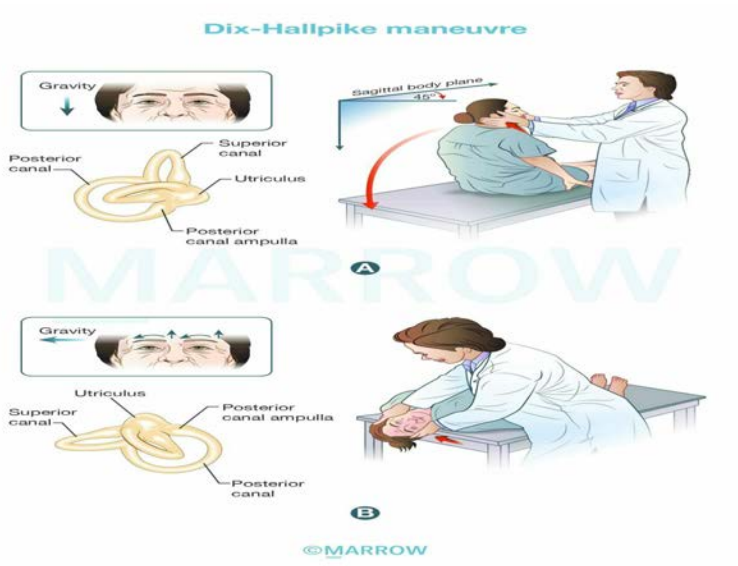
The given clinical scenario of a patient with vertigo without hearing loss is suggestive of benign paroxysmal positional vertigo (BPPV). The diagnostic maneuver used in BPPV is the Dix-Hallpike maneuver, the gold standard for diagnosing BPPV. The therapeutic maneuver is Epley's maneuver.

BPPV is characterized by vertigo, recurrent brief (<1 minute) episodes triggered by specific head movements, and is not associated with hearing loss or other neurological symptoms.

It is thought to be caused by the dislodgement of crystals of calcium carbonate from the degenerating macula of the utricle. When this crystal settles on the cupula of the posterior canal, it causes displacement of the cupula and vertigo.

A patient presenting with vertigo should be evaluated for ruling out the central lesions (like a tumor in the fourth ventricle, cerebellum, temporal lobe, or multiple sclerosis) before confirming the diagnosis of BPPV. The Dix-Hallpike maneuver helps to differentiate a peripheral from a central lesion.

Dix-Hallpike maneuver: The maneuver starts with the patient in a sitting position. Their head is turned 45 degrees towards the ear to be tested. Then the examiner quickly brings the patient to a supine position with their head 30 degrees below the horizontal. The 45-degree head rotation is maintained during the procedure. Now the examiner observes the patient for nystagmus.



The features of nystagmus differentiate between BPPV and central lesions:

The therapeutic maneuver is the Epley maneuver which displaces the otolith out of the posterior canal. This ends the sensation of vertigo with positional changes relieving the symptoms.

Nystagmus	BPPV	Central lesions
Latency	2-20 seconds	No latency
Duration	<1 minute	>1 minute
Direction	fixed, towards the lower-placed ear	changing
The result on repetitions of the test	nystagmus is not seen (fatiguable)	nystagmus persists (nonfatiguable)
Symptoms during the test	severe vertigo	Absent/mild

Solution to Question 4:

The instrument shown in the image is Mollison's mastoid retractor. It is used in mastoidectomy to retract soft tissues after incision and elevation of flaps. It is self-retaining and hemostatic.

Jansen's self-retaining mastoid retractor (shown below) is also used in mastoidectomy and is similar to Mollison's retractor.

Option A: Joll retractor (shown below) is a self-retaining retractor used in thyroid surgery.

Option D: Langenbeck retractor is useful for holding open wounds, as in an open appendicectomy and they are often used in pairs (one in each of the assistant’s hands).

Solution to Question 5:

The senior resident would not be visualizing the arytenoids as he is performing a posterior rhinoscopy indicated by the post-nasal mirror that he is holding.

The posterior rhinoscopy mirror (St. Clair Thompson mirror) has two bends in the shaft and is bayonet shaped.

The structures seen in the posterior rhinoscopy are:

• Adenoids

• Posterior ends of nasal turbinates

• Opening of the Eustachian tube

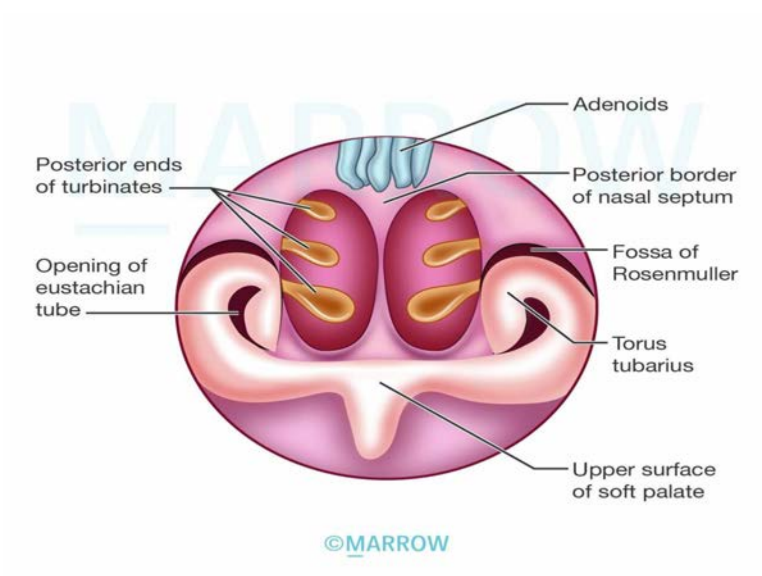
• The posterior border of the nasal septum

• Fossa of Rosenmuller

• Torus tubarius

• The upper surface of the soft palate

The images below show the posterior rhinoscopy mirror and structures seen in the posterior rhinoscopy:



Indirect laryngoscopy is done using a laryngeal mirror. It is straight without any bends with a mirror at the end.

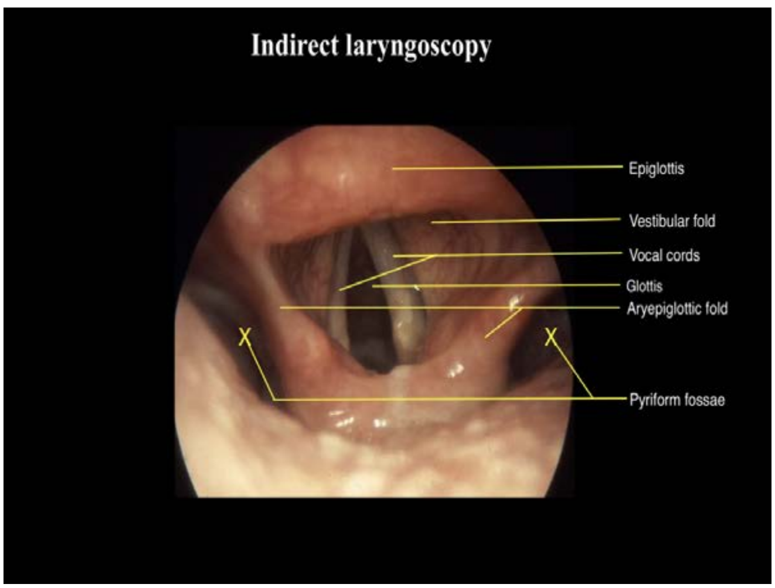
Structures seen during indirect laryngoscopy (in order):

• Oropharynx: Base of the tongue, lingual tonsils, valleculae, medial and lateral glossoepiglottic folds.

• Laryngopharynx: Both piriform fossae, post cricoid region, posterior wall of laryngopharynx.

• Larynx: Epiglottis, aryepiglottic folds, arytenoids, cuneiform and corniculate cartilages, glossoepiglottic folds, pharyngoepiglottic folds, aryepiglottic fold, true and false vocal cords, anterior commissure, posterior commissure, subglottis and rings of the trachea.

The images below show the laryngeal mirror and structures seen during indirect laryngoscopy:



Solution to Question 6:

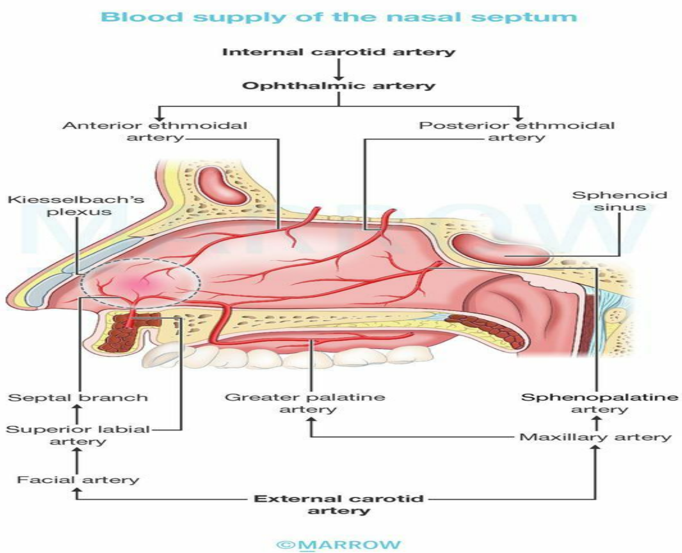
Anterior and posterior ethmoidal arteries are branches of the ophthalmic artery which in turn is a branch of the internal carotid artery.

Kiesselbach’s plexus is an arterial anastomosis at the anterior inferior part of the nasal septum (Little's area). It includes the anterior ethmoidal artery, septal branch of the superior labial artery, septal branch of the sphenopalatine artery, and septal branch of the greater palatine artery. It is the usual site of epistaxis in children and young adults.

The blood supply of the nasal septum can be divided into two main systems, the internal carotid system, and the external carotid system.

• Internal carotid system: Branches of the ophthalmic artery, anterior ethmoidal artery, and posterior ethmoidal artery.

• External carotid system: Sphenopalatine artery (branch of the maxillary artery) gives nasopalatine and posterior medial nasal branches, septal branch of the greater palatine artery (branch of the maxillary artery), and septal branch of the superior labial artery (branch of the facial artery).



Solution to Question 7:

The frontal sinus is best seen in Caldwell's view (occipitofrontal view or nose-forehead position).

The structures seen in Caldwell's view include:

• Frontal sinuses (seen best)

• Ethmoid sinuses

• Maxillary sinuses

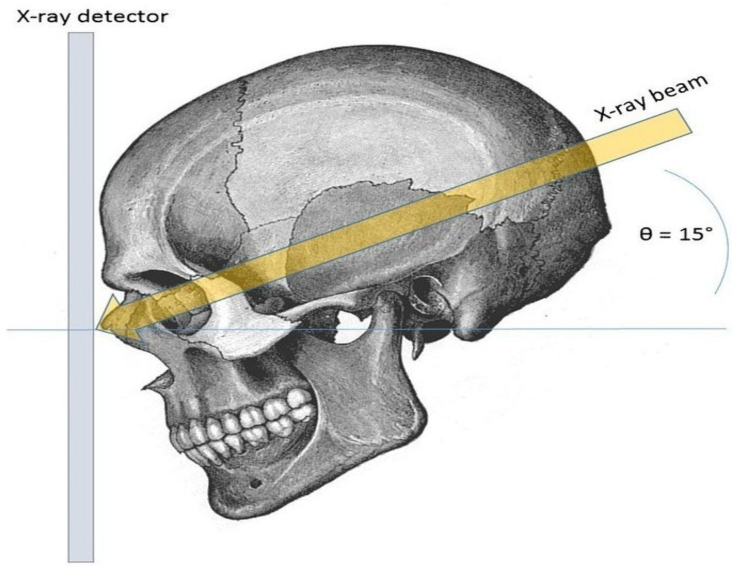
• Frontal process of the zygoma and the zygomatic process of the frontal bone

• Superior margin of the orbit and lamina papyracea

• Superior orbital fissure

• Foramen rotundum (inferolateral to superior orbital fissure)

The nose and forehead are touching the X-ray film and the X-ray beam is projected 15 degrees caudally in Caldwell's view.



Other options:

Option A: Water's view (occipitomental view or nose-chin position) is a radiographic view where an X-ray beam is angled at 45 degrees to the orbitomeatal line. Structures seen are the maxillary sinus (best seen), sphenoid sinus (if the film is taken with an open mouth), frontal sinus, zygoma, zygomatic arch, nasal bone, and infratemporal fossa.

Option C: Pierre's view is the occipitomental with an open mouth. It helps in visualizing the sphenoid sinus as well, in addition to the sinuses that can be seen in water's view.

Option D: Towne's view is an angled anteroposterior radiograph of the skull. Structures seen are the petrous part of the pyramids, arcuate eminence, superior semicircular canal, mastoid antrum, internal auditory canal, tympanic cavity, cochlea, and external auditory canal. This view is taken for acoustic neuroma and apical petrositis.

Solution to Question 8:

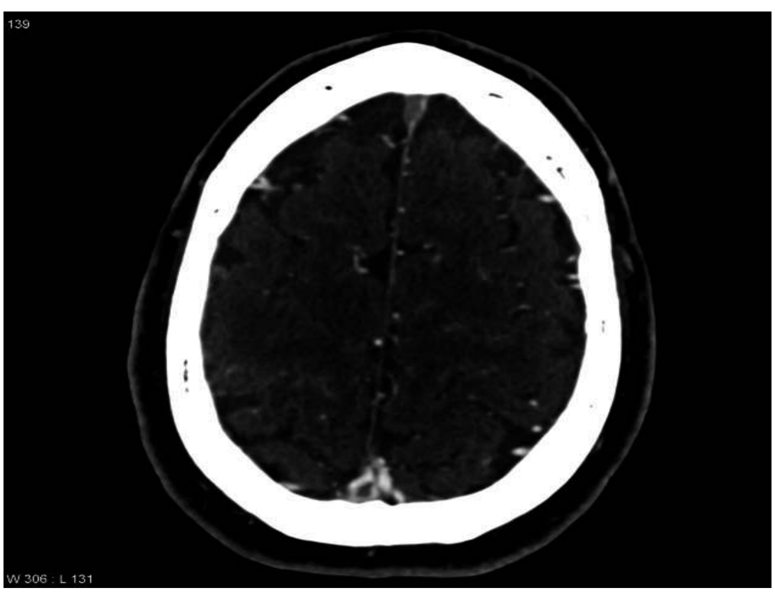
Battle sign refers to retro auricular or mastoid ecchymosis that typically results from a fracture of the middle cranial fossa following head trauma. It is prominent when there is a fracture of the temporal bone. Battle sign is not seen in the case of otitis media.

Option A: Bezold abscess is a complication of acute otomastoiditis where the infection/pus breaks through the thin medial side of the tip of the mastoid and presents as a swelling in the upper part of the neck.

Option B: Griesinger sign refers to edema of the postauricular soft tissues overlying the mastoid process as a result of thrombosis of the mastoid emissary vein seen in sigmoid sinus thrombosis. It is a complication of acute otomastoiditis.

Option D: Delta sign is a contrast-enhanced CT finding seen in sagittal sinus thrombosis, which can occur as a complication of acute mastoiditis or chronic suppurative otitis media. It is a triangular area with rim enhancement and the central low-density area is seen in the posterior cranial fossa on axial cuts. MR venography is useful to assess the progression or resolution of a thrombus.

The image below depicts a filling defect (thrombus) outlined by contrast medium in the sagittal sinus.



Solution to Question 9:

The marked structure is the lowermost projection in the lateral wall of the nose, the inferior turbinate. The middle turbinate (Option B) is seen as superior to it.

The lateral wall of the nose contains three or sometimes four projections of variable size namely superior, middle, and inferior turbinates along with a corresponding meatus under each turbinate.

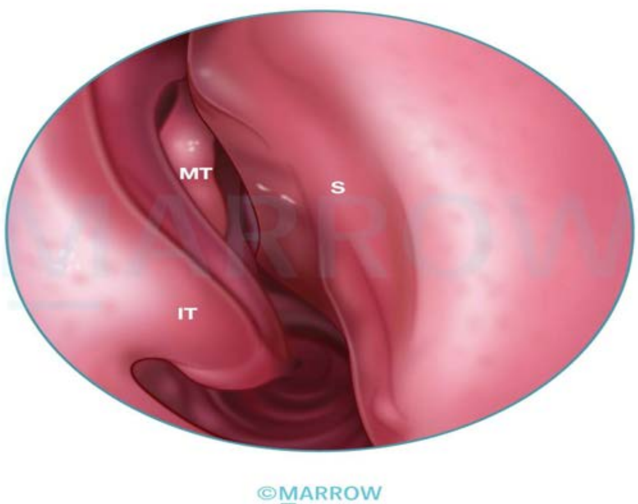
The inferior turbinate is the largest turbinate and occupies the lower third of the lateral nasal wall. It is a separate bone. Middle and superior turbinates (Option C) are ethmoturbinal, part of the ethmoid bone.

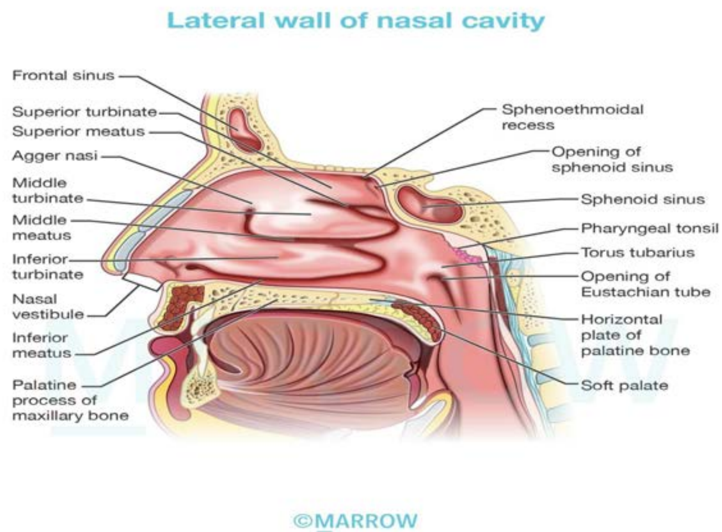
Other important structures which could be seen in the above image are:

• IT: Inferior turbinate

• S: Septum,

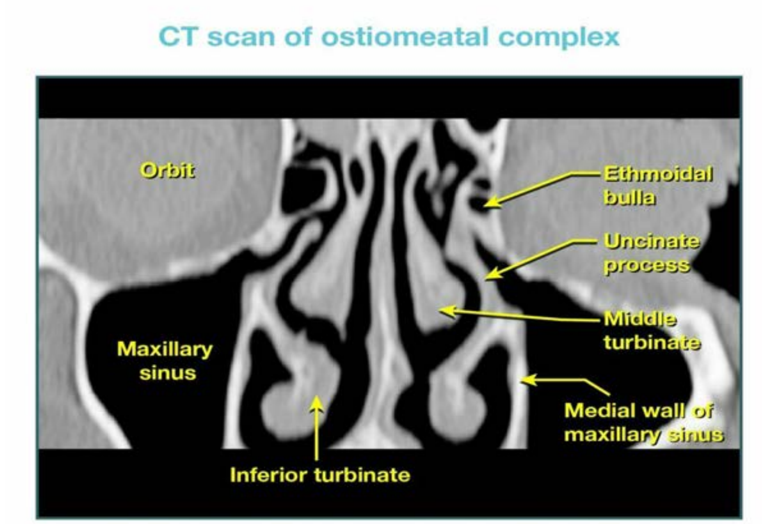
• MT: middle turbinate





Uncinate process (Option D) is a hook-like structure running parallel to the anterior border of bulla ethmoidalis. The posteroinferior end of the uncinate process is attached to the inferior turbinate.

The image below shows a CT image of the uncinate process.

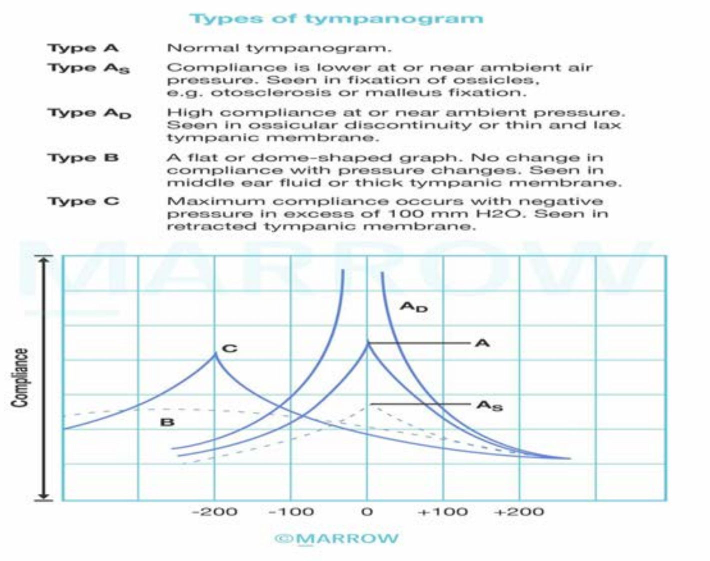


Solution to Question 10:

BERA, OAE, and tympanometry are the objective tests for hearing. PTA (pure tone audiometry) is a subjective test for hearing as it relies on patient responses to pure tone stimuli. Other examples of subjective tests include tuning fork tests and speech audiometry.

Tympanometry is an objective test of the middle ear and eustachian tube functions (conductive compartment of hearing). Here, the compliance of the TM, i.e., how much it moves to the pressure differentials applied from the external ear is measured.

The image below shows the types of tympanogram:

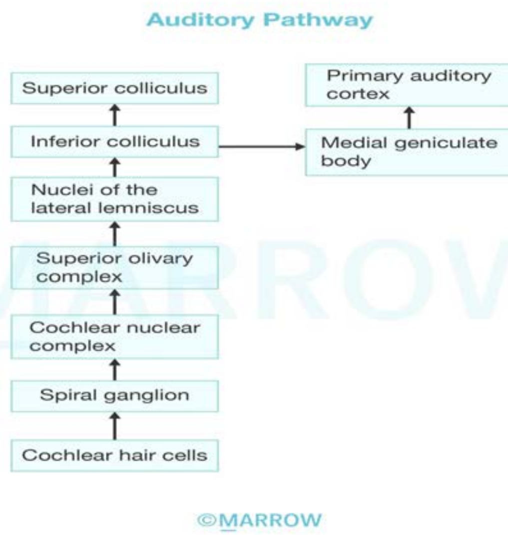


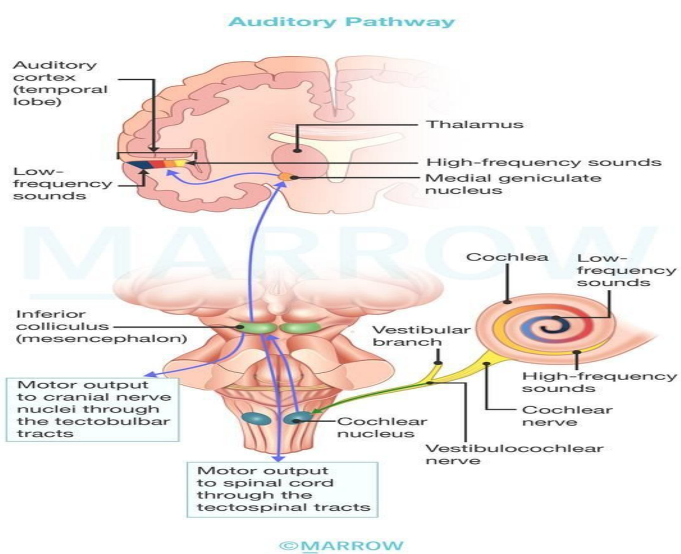
Solution to Question 11:

Cochlear nucleus (B), superior olivary nucleus (A), lateral lemniscus (D), inferior colliculi (C), and medial geniculate ganglion (E) represent the correct sequence of the auditory pathway.

The correct sequence of structures involved in the auditory pathway from the periphery to the center is

- Vestibulocochlear nerve
- Cochlear Nucleus
- Superior olivary nucleus
- Lateral lemniscus
- Inferior colliculus
- Medial geniculate body
- Auditory cortex





Solution to Question 12:

The given clinical history of trauma to the ear along with swelling, tenderness, and discoloration of the pinna is highly suggestive of hematoma of the auricle (perichondrial hematoma). It does not resolve spontaneously.

It is a collection of blood between the auricular cartilage and its perichondrium known as perichondrial space (Option C). It is commonly seen in boxers, wrestlers, and rugby players due to blunt trauma and can result in a deformity called cauliflower ear, wrestler's ear, or pugilistic ear (Option B). Severe perichondritis may occur if the hematoma gets infected.

All cases should receive prophylactic antibiotics (Option A). Aspiration of the hematoma under strict aseptic precautions and a pressure dressing is done to prevent reaccumulation. Aspiration may need to be repeated. When aspiration fails, incision and drainage should be done.

ENT NEET 2018

Question 1:

Which of the following is an indication of high tracheostomy?

a) Vocal cord palsy

b) Tracheomalacia

c) Foreign body obstruction

d) Suspicion of Laryngeal carcinoma planned for surgery later

Question 2:

Which of the following nerves is responsible for referred otalgia from tonsillitis?

a) Glossopharyngeal nerve

b) Facial nerve

c) Trigeminal nerve

d) Vagus nerve

Question 3:

Water's view is used to best visualize which of the following sinuses?

a) Maxillary sinus

b) Ethmoidal sinus

c) Frontal sinus

d) Sphenoid sinus

Question 4:

The following instrument is used for:

a) Septoplasty

b) Myringoplasty

c) Myringotomy

d) Adenoidectomy

Question 5:

The vertical fracture of the nasal septum is known as _______.

a) Chevallet fracture

b) Citelli fracture

c) Tripod fracture

d) Jarjaway fracture

Question 6:

Where is the electrode placed in cochlear implant procedures?

a) Round window

b) Oval window

c) Scala vestibuli

d) Scala tympani

Answer Key

Question No.	Correct Option
1	d
2	a
3	a
4	c
5	a
6	d

Detailed Explanations

Solution to Question 1:

Laryngeal carcinoma is an indication of high tracheostomy. It is done at the level of the 1st tracheal ring and an opening is made above the isthmus of the thyroid gland.

The disadvantage of a high tracheostomy is subglottic stenosis at the level of the cricoid and perichondritis of the cricoid. However, in laryngeal carcinoma this would be a temporary measure as the definitive management is total laryngectomy.

Tracheostomy involves making an opening and placing a stoma in the anterior wall of the trachea.

Functions:

• Provides an alternative pathway for breathing by bypassing the obstruction.

• Improves alveolar ventilation by reducing dead space by 30-50% and reducing airway resistance

• Uses a cuff to protect the airways from secretions and blood

• Allows suction of secretions in the setting of the impaired cough reflex.

• Allows positive pressure respiration

• Allows administration of anesthesia

Indications:

• Respiratory obstruction due to

• Infections (laryngo-tracheo-bronchitis, epiglottitis)

• Trauma due to external injury, mandibular or maxillofacial fractures

• Neoplasms

• Foreign bodies or edema

• Bilateral abductor paralysis

• Congenital anomalies

• Retained secretions

• Inability to cough due to coma, spasm, or paralysis of respiratory muscles

• Aspiration of pharyngeal secretions

• Painful cough

• Respiratory insufficiency due to emphysema, chronic bronchitis

Solution to Question 2:

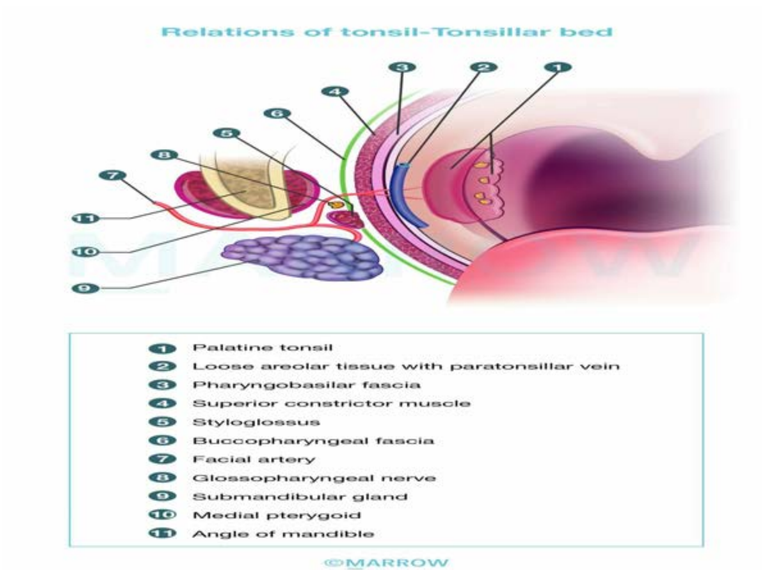
The tympanic branch of the glossopharyngeal nerve (cranial nerve IX) is responsible for referred otalgia from tonsillitis.

The tonsils and tonsillar fossa are supplied by the glossopharyngeal nerve. The glossopharyngeal nerve and the styloid process lie in relation to the lower part of the tonsillar fossa. Therefore, any irritation or pain can be referred to the ear along the tympanic branch of the glossopharyngeal (Jacobson's) nerve.

Causes of referred otalgia via the glossopharyngeal nerve include acute pharyngitis, tonsillitis, peritonsillar abscess, post-tonsillectomy, foreign bodies embedded in the base of the tongue or tonsillar area and an elongated styloid process (Eagle's syndrome).

The ear receives extensive sensory innervation via four cranial nerves (V, VII, IX, and X) and two spinal segments (C2 and C3).

The image below shows the anatomic relations of the tonsil and tonsillar bed:



Solution to Question 3:

Water’s view is used to best visualize the maxillary sinus. Frontal and ethmoidal sinuses can also be visualized. Sphenoid sinus can be seen in Water's view with open mouth.

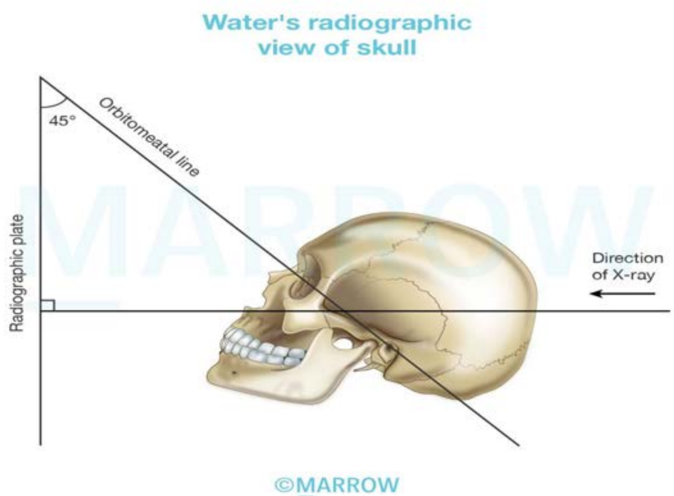
It is also known as the occipitomental view. The X-ray beam is angled at 37°-45° to the orbitomeatal line.

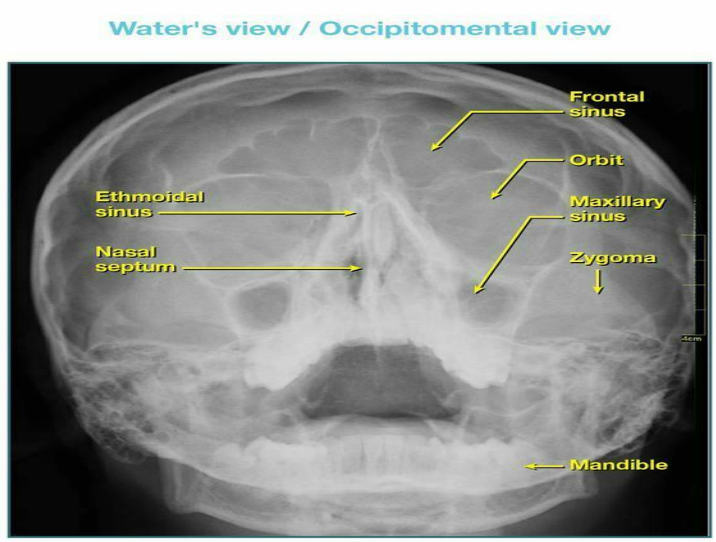
The recommended imaging views for different sinuses are as follows:

• Ethmoidal sinus: Rheese's view

• Frontal sinus: Caldwell's view

• Sphenoid sinus: Hirtz view





Solution to Question 4:

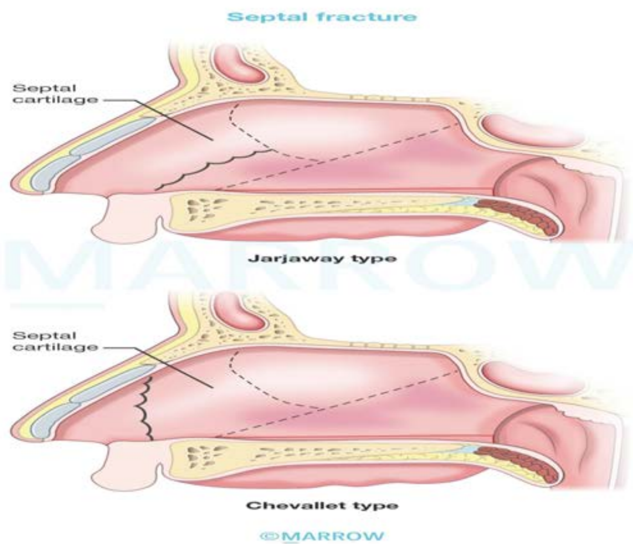
The given instrument with an arrowed blade and angled shank is a myringotome. It is used for myringotomy.

Solution to Question 5:

A Chevallet fracture usually results from a blow to the nose from below. This fracture line runs vertically. It extends from the anterior nasal spine upwards to the junction of the bony and cartilaginous dorsum of the nose.

Injuries to the nasal septum can result from trauma to the nose from the front, side, or below. The septum may get buckled on itself, fracture vertically and horizontally, or get crushed

A Jarjaway fracture (option D) results from a blow to the nose from the front. The fracture line runs horizontally backward. It begins just above the anterior nasal spine and extends just above the junction of septal cartilage with the vomer.



Early recognition and treatment of septal injuries are crucial, including drainage of hematomas.

Dislocated or fractured septal fragments should be repositioned and supported between mucoperichondrial flaps using mattress sutures and nasal packing. Nasal pyramid fractures often coincide with septal fractures and should be treated simultaneously.

Neglecting injuries to the septum can lead to deviations in the cartilaginous nose, as well as asymmetry in the nasal tip, columella, or nostrils.

Other options:

Option B: Citelli fracture is a term that does not exist. Citelli's abscess is an extratemporal complication of otitis media, occurring behind the mastoid, more towards the occipital bone. Citelli's angle refers to the sino-dural angle between the sigmoid sinus and middle cranial fossa dural plate. It is an important landmark in mastoidectomy.

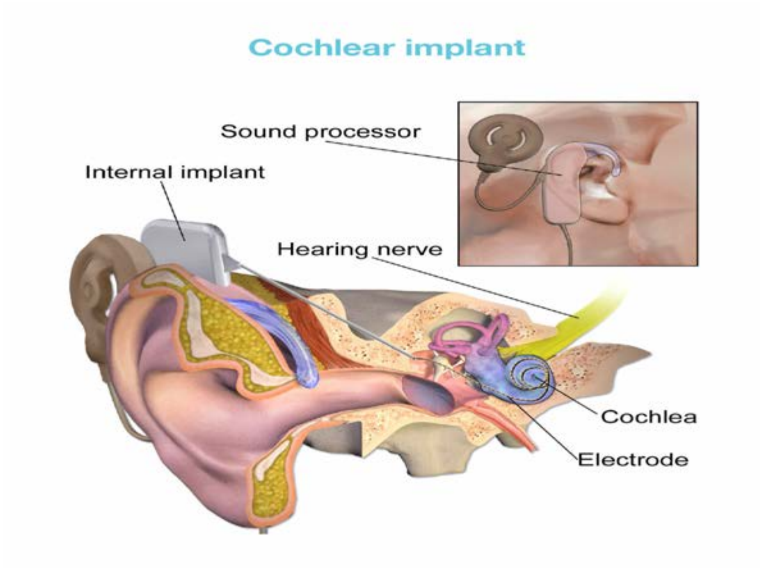
Option C: A tripod fracture refers to a fracture of the zygomatic complex. There are three commonly occurring disruptions - frontozygomatic, infraorbital rim and the zygomaticomaxillary buttress.

Solution to Question 6:

The electrode of a cochlear implant is placed in the scala tympani. It allows for the electrodes to be in close proximity to the ganglion cells and their dendrites which are located in the modiolus and bony spiral lamina of the cochlea.

A cochlear implant is an electronic device for severe to profound SNHL and who cannot benefit from a hearing aid. It converts sound to an electrical signal (transduction) and delivers this to the auditory nerve. Prerequisites include an intact cochlear nerve and higher auditory pathway.

In the auditory brainstem implant, the site of the implant is the lateral recess of the fourth ventricle. It is used when the cochlear nerve is not intact.



ENT NEET 2019

Question 1:

Which of the following is not a feature of tubercular otitis media?

- **a)** Ear ache
- **b)** Multiple perforations
- **c)** Pale granulation
- **d)** Foul smelling ear discharge

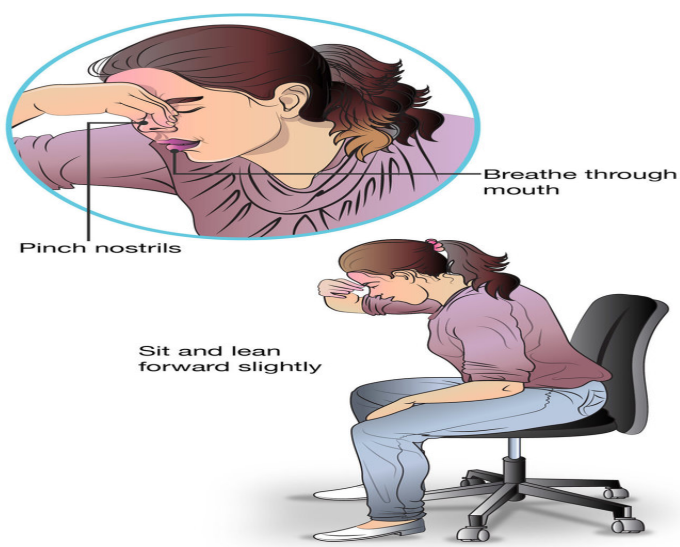
Question 2:

Which among the following statements is true about keratosis obturans?

- **a)** Failure of migration of desquamated epithelium along posterior meatal wall
- **b)** Widening of meatus and facial nerve palsy might be seen
- **c)** Associated bronchiectasis and sinusitis
- **d)** All of the above

Question 3:

What is the maneuver depicted in the image given below?



a) Epley's maneuvre

b) Trotter's method

c) McGovern's technique

d) Valsalva maneuvre

Question 4:

Pott's puffy tumor is

a) Subperiosteal abscess of frontal bone

b) Subperiosteal abscess of ethmoid bone

c) Mucocele of frontal bone

d) Mucocele of ethmoid bone

Question 5:

Identify the lesion of vocal cord in the image given below :

a) Reinke's edema

b) Malignancy

c) Tracheomalacia

d) Laryngeal papilloma

Question 6:

Inspiratory stridor is found in what kind of lesions:

- **a)** Supraglottic
- **b)** Subglottic
- **c)** Tracheal
- **d)** Bronchus

Answer Key

Question No.	Correct Option
1	a
2	d
3	b
4	a
5	d
6	a

Detailed Explanations

Solution to Question 1:

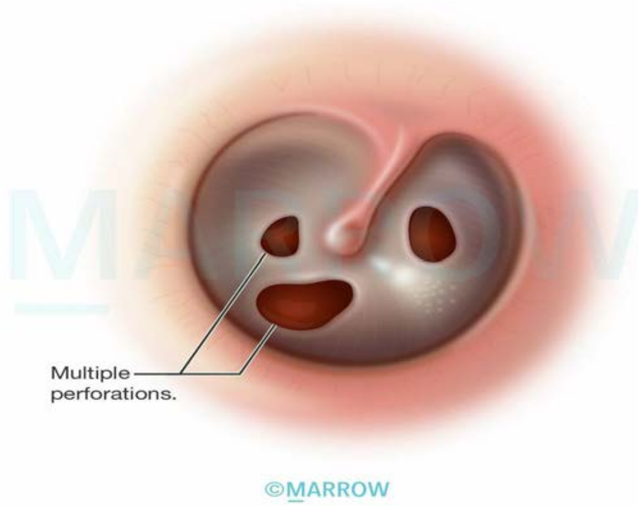
Earache is not a feature of tubercular otitis media. It is a painless condition.

Tuberculosis of the ear is usually secondary to pulmonary tuberculosis. It occasionally spreads hematogenously, from a tubercular focus in the lungs, tonsils, and cervical or mesenteric lymph nodes

It presents as a painless condition with foul-smelling discharge from the ear, that does not respond to standard antibiotic treatment. Patients have severe hearing loss that is out of proportion to symptoms. Although predominantly conductive, sensorineural hearing loss may also occur if the labyrinth is involved.

A thickened tympanic membrane with multiple perforations in pars tensa of the tympanic membrane and pale granulations in the middle ear cleft is seen.

The image below shows the tympanic membrane in tuberculous otitis media:



Histopathology shows submucosal tubercles with Langhans giant cells and caseous necrosis. Smears and culture of ear discharge may be positive for acid-fast bacilli (AFB). Evidence of primary pulmonary tuberculosis may also be seen on chest radiography.

Facial nerve palsy is a common complication. Mastoiditis, postauricular fistula, and osteomyelitis are also seen.

Treatment includes aural toilet, treatment of secondary bacterial infections, and a full course of antitubercular therapy followed by reconstructive surgery.

Solution to Question 2:

All of the above statements are true regarding keratosis obturans.

Keratosis obturans occurs due to failure of migration of desquamated epithelial cells along the posterior meatal wall (Option A).

It is usually seen between the ages of 5-20 years and is commonly unilateral. Clinical features include severe otalgia and difficulty in hearing. It is associated with systemic conditions like bronchiectasis and chronic sinusitis (Option C).

On otoscopic examination, there is a thickening of the tympanic membrane and a keratin plug appearing like a pearly white mass occluding the canal. The keratin that is shed from the entire circumference of the deep ear canal appears to have an onion skin arrangement. There is also hyperemia of canal skin, sometimes with ulcerations and granulations.

Sometimes, the extent of bone erosion is such that it leaves the facial nerve exposed to injury and subsequent paralysis (Option B). Extensive erosion of the bone can also result in automastoidectomy.

Treatment involves the removal of the plug by syringing or instrumentation. Any recurrence can be prevented with the use of keratolytic such as 2% salicylic acid.

Solution to Question 3:

The given image shows Trotter's method in which the patient is made to sit, leaning a little forward over a basin to spit any blood and breathe quietly from the mouth. It is the first step to control epistaxis.

Other options:

Option A: Epley's maneuver is used in the treatment of BPPV (Benign Paroxysmal Positional Vertigo).

Option C: McGovern's technique is used in choanal atresia. A feeding nipple with a large hole provides a good oral airway.

Option D: Valsalva maneuver is performed by forceful attempted exhalation against a closed nose and mouth. It is used as a test of cardiac function and autonomic nervous control of the heart, or to clear the ears and sinuses.

Solution to Question 4:

Pott's puffy tumor is the subperiosteal abscess of frontal bone. It is a complication of frontal sinusitis.

An acute infection of the frontal sinus can cause osteomyelitis of the frontal bone. Suppurative material then breaches the frontal bone leading to subperiosteal abscess formation in the tissues of the forehead. Pus may form externally under the periosteum as a soft boggy swelling with a doughy consistency called Pott’s puffy tumor. It may spread internally causing subdural, epidural, and periorbital abscesses as well as secondary septic thrombosis of the dural sinuses. It is more often seen in adults as frontal sinus is not developed in infants and children.

CT of the brain and paranasal sinuses is the most appropriate investigation. It can show the involvement of sinuses as well as intracranial or soft tissue spread of infection.

Treatment consists of large doses of antibiotics, abscess drainage, and trephining of the frontal sinus through its floor. Sometimes, the removal of sequestra and necrotic bone may also be required.

The images below show the clinical presentation and radiographic findings of the condition: red arrow - Pott's puffy tumor; blue arrow - intracranial spread

Solution to Question 5:

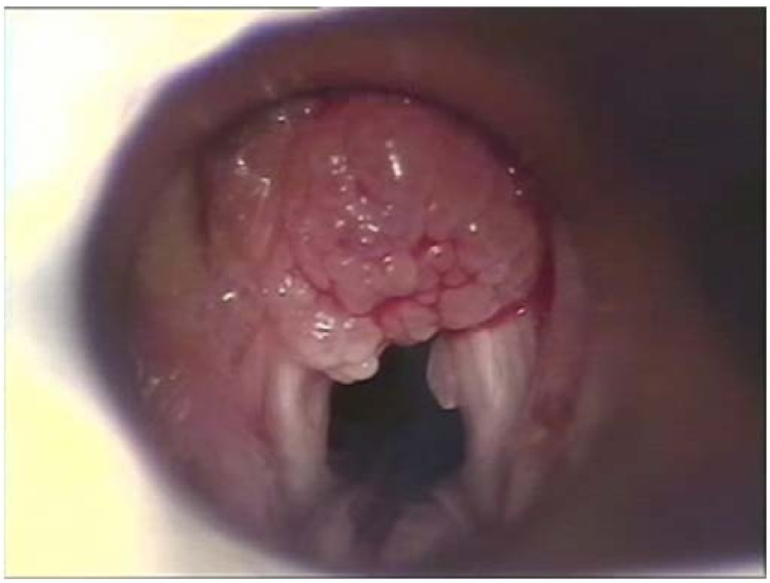
The image showing grape-like projections, involving the anterior part of the vocal cords depicts a laryngeal papilloma.

Laryngeal papilloma is a benign tumor of the larynx caused by the human papillomavirus (HPV type 6 and 11). They can occur in children, most commonly around 3 -5 years of age (juvenile papillomatosis), as well as in adults. They involve the supraglottic and glottic regions of the larynx. Seeding of the lesions can occur during invasive laryngeal procedures, such as intubation. They have a high recurrence rate and very low potential for malignancy.

Adult-onset laryngeal papillomatosis is less common and usually seen in males aged 30-50 years. It is less aggressive and involves the anterior part of the vocal cords. The treatment is the same as juvenile papillomatosis.

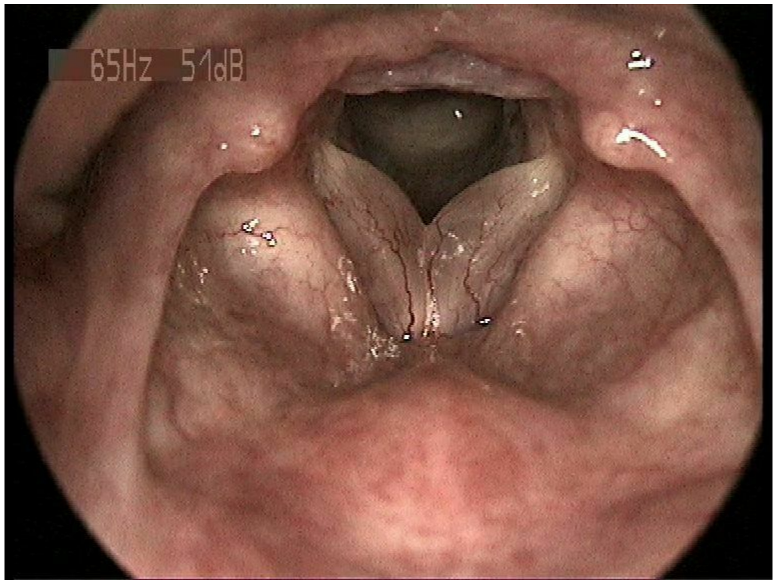
Patients present with hoarseness, aphonia, dyspnea, or stridor. Laryngoscopy shows pinkish to white grape-like projections that can be sessile or pedunculate with a visible central vascular core. The diagnosis is confirmed by direct laryngoscopy and biopsy. Treatment consists of micro-laryngoscopy and CO_2 laser excision.

The image below depicts laryngeal papillomatosis on laryngoscopy:



Reinke's edema (Option A) is a collection of fluid in the subepithelial space of Reinke. It generally occurs in patients with vocal abuse or smoking. The vocal cords will show diffuse, symmetrical swelling.

The laryngoscopic image below shows Reinke's edema:



Solution to Question 6:

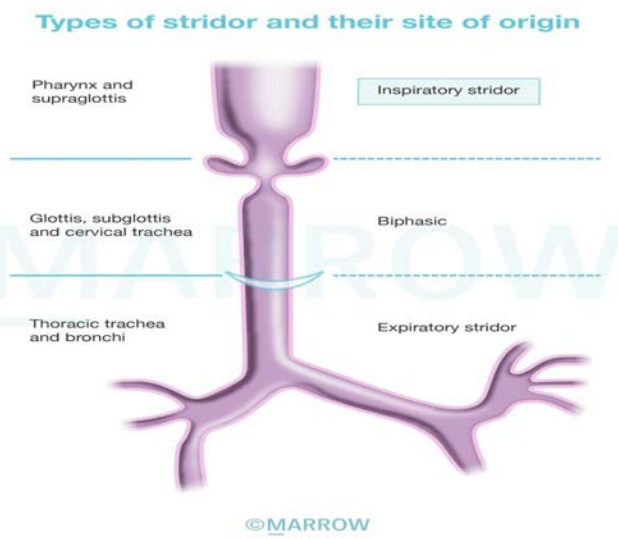
Inspiratory stridor is seen in supraglottic lesions. It is often produced in obstructive lesions of supraglottis or pharynx, e.g. laryngomalacia or retropharyngeal abscess.

Expiratory stridor is produced in lesions of thoracic trachea, primary and secondary bronchi such as bronchial foreign body, and tracheal stenosis.

Biphasic stridor is seen in lesions of glottis, subglottis, and cervical trachea like laryngeal papillomas, vocal cord paralysis, and subglottic stenosis.

Stridor is an audible respiratory noise that gets produced by turbulent airflow through an obstructed upper airway. It is an inspiratory sound that may have an expiratory component. It is distinct from stertor which is a snoring-like noise produced from an obstructed nasopharynx or oropharynx.

The intensity and pitch of sounds originating from the respiratory system vary based on their site of origin within the respiratory tract and may be heard with or without a stethoscope. Usually, the pitch of the sound increases, and the intensity decreases as one goes lower into the respiratory tract.


Types of stridor and their site of origin
Pharynx and supraglottis
Inspiratory stridor
Glottis, subglottis and cervical trachea
Biphasic
Thoracic trachea and bronchi
Expiratory stridor
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ENT NEET 2020

Question 1:

From which of the following structure does the saccule develop?

a) Saculus anterior

b) Saculus posterior

c) Pars superior

d) Pars inferior

Question 2:

What is the surgery done to widen the cartilaginous part of the external auditory canal called?

a) Meatoplasty

b) Tympanoplasty

c) Myringoplasty

d) Otoplasty

Question 3:

Cough on scratching the external acoustic canal is due to?

a) Auriculotemporal nerve

b) Auricular branch of vagus

c) Great auricular nerve

d) Facial nerve

Question 4:

Which of the following is not a feature of tubercular otitis media?

a) Ear ache

b) Multiple perforations

c) Pale granulation

d) Foul smelling ear discharge

Question 5:

A person who met with an accident, and suffered from a skull fracture presents with the following finding. Identify this clinical finding.

a) Battle sign

b) Bezold abscess

c) Mastoiditis

d) Griesinger sign

Question 6:

Surgery where nostrils are partially or completely occluded is done for which condition?

a) Vasomotor rhinitis

b) Atrophic rhinitis

c) Invasive aspergillosis

d) Allergic rhinitis

Question 7:

What is occipitomental view with open mouth also known as?

a) Water's view

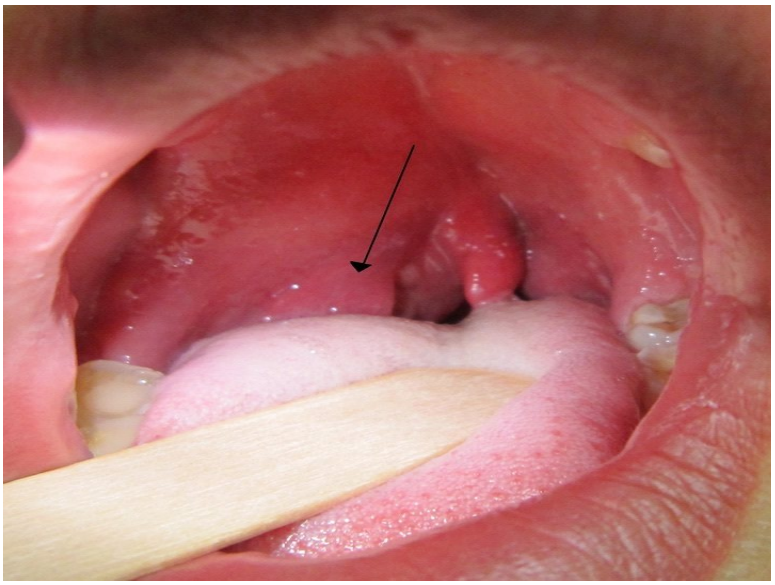
b) Towne's view

c) Law's view

d) Stenver's view

Question 8:

A 9-year-old boy comes with complaints of right ear pain, difficulty in opening the mouth, painful swallowing, and fever. The oral cavity examination reveals the following. The external facial examination is unremarkable. Which of the following is likely?



a) Pharyngitis

b) Parotid abscess

c) Bezold's Abscess

d) Quinsy

Answer Key

Question No.	Correct Option
1	d
2	a
3	b

4	a
5	a
6	b
7	a
8	d

Detailed Explanations

Solution to Question 1:

The saccule develops from pars inferior.

The development of the inner ear begins in the 3rd week of fetal life and is completed by the 16th week. The surface ectoderm on each side of the rhombencephalon thickens to form otic/auditory placodes, which then invaginate to form the otic vesicle (otocysts).

The cells of the otic vesicle differentiate to form ganglion cells, which later develop into the vestibulocochlear ganglia. Later each otic vesicle divides into a dorsal component called pars superior and a ventral component called pars inferior.

Pars superior gives rise to the utricle, semicircular canals, and endolymphatic ducts. Pars inferior gives rise to saccule and cochlear duct. The saccule is connected to the cochlea through a narrow duct called ductus reuniens.

Solution to Question 2:

The surgery done to widen the cartilaginous part of the external auditory canal is meatoplasty.

Meatoplasty is a surgical procedure that involves the removal of a crescent-shaped portion of conchal cartilage to widen the external auditory meatus. It is performed in combination with all canal wall-down procedures like modified radical and radical mastoidectomies, for easy access to the mastoid cavity for periodic inspection and cleaning.

It is also done as an isolated procedure in elderly people with sagging auricles. A sagging auricle causes obstruction to the meatus leading to the retention of wax and difficulty in hearing.

Other options:

Option B: Tympanoplasty is the surgical reconstruction to correct damage in the middle ear and restore the integrity of the tympanic membrane for improved hearing. It is ossicular reconstruction only or ossicular reconstruction with myringoplasty.

Option C: Myringoplasty refers to the reconstruction of the tympanic membrane only.

Option D: Otoplasty is the correction of external ear/deformities of the pinna.

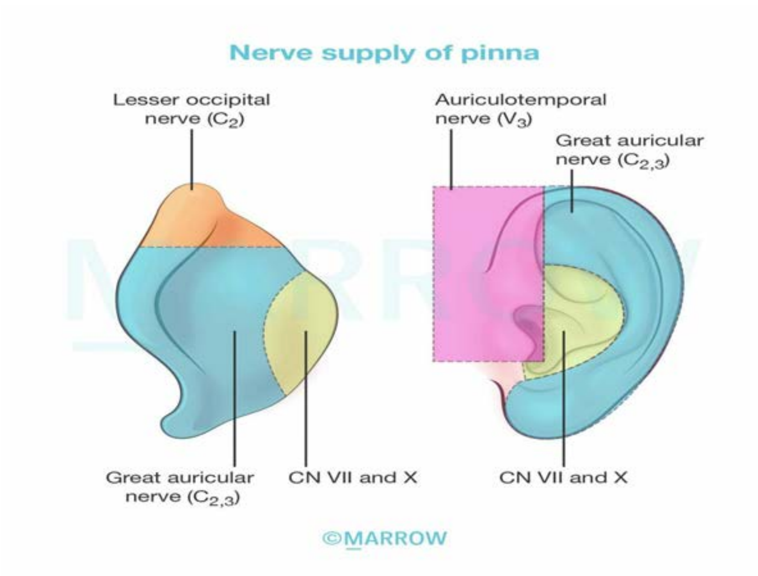
Solution to Question 3:

Cough on scratching the external acoustic canal is due to the auricular branch of the vagus nerve (Arnold's nerve).

Arnold's nerve supplies the concha and corresponding eminence on the medial surface. It also supplies the posterior wall and floor of the external auditory canal. Scratching or mechanical stimulation of the ear leads to activation of Arnold's nerve, which evokes reflex cough (ear-cough reflex/otorespiratory reflex).

The image below shows the nerve supply of the pinna:

Causesofreferred otalgia	Nerveresponsible	Site
Temporomandibular joint and dental conditions, parotid infections/tumor, and anterior 2/3rd of tongue	Auriculotemporal nerve	Tragal area, crus of the helix, and anterior wall and roof of the external auditory canal.
Acute tonsillitis, peritonsillar abscess, carcinoma base of the tongue or tonsils (oropharynx)	The glossopharyngeal nerve (Jacobson's)	Medial surface of the tympanic membrane, and middle ear
Carcinoma of the larynx (cough), hypopharynx, epiglottis, vallecula, esophagus, and thyroid	Auricular branch of the vagus nerve (Arnold's/Alderman's nerve)	Concha, posterior wall, and floor of the external auditory canal
Cervical spondylitis	Great auricular nerve (C2,3)	Lateral and medial surfaces of the pinna
Acoustic neuroma	Facial nerve (nervus intermedius/nerve of Wrisberg)	Concha, retro auricular groove, and posterosuperior part of the external auditory meatus. Numbness in the posterosuperior meatal wall known as Hitzelberger's sign, is seen in acoustic neuroma.



Solution to Question 4:

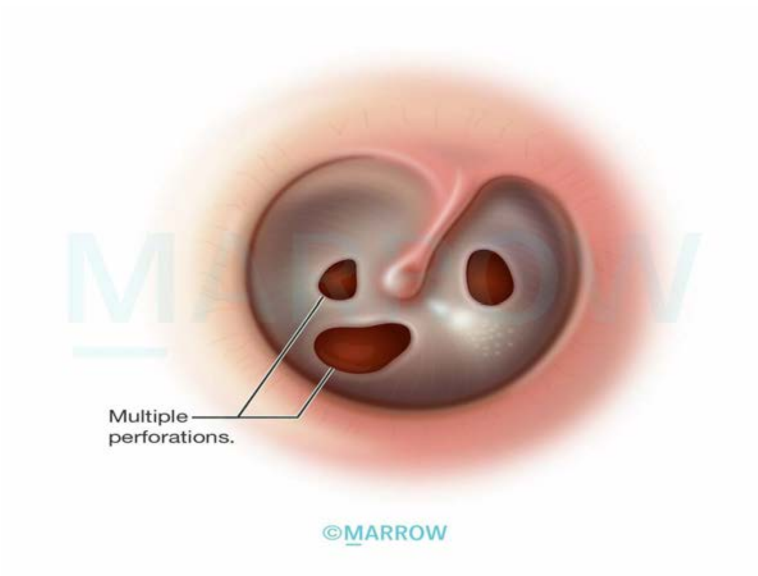
Earache is not a feature of tubercular otitis media. It is a painless condition.

Tuberculosis of the ear is usually secondary to pulmonary tuberculosis. It occasionally spreads hematogenously, from a tubercular focus in the lungs, tonsils, and cervical or mesenteric lymph nodes

It presents as a painless condition with foul-smelling discharge from the ear, that does not respond to standard antibiotic treatment. Patients have severe hearing loss that is out of proportion to symptoms. Although predominantly conductive, sensorineural hearing loss may also occur if the labyrinth is involved.

A thickened tympanic membrane with multiple perforations in pars tensa of the tympanic membrane and pale granulations in the middle ear cleft is seen.

The image below shows the tympanic membrane in tuberculous otitis media:



Histopathology shows submucosal tubercles with Langhans giant cells and caseous necrosis. Smears and culture of ear discharge may be positive for acid-fast bacilli (AFB). Evidence of primary pulmonary tuberculosis may also be seen on chest radiography.

Facial nerve palsy is a common complication. Mastoiditis, postauricular fistula, and osteomyelitis are also seen.

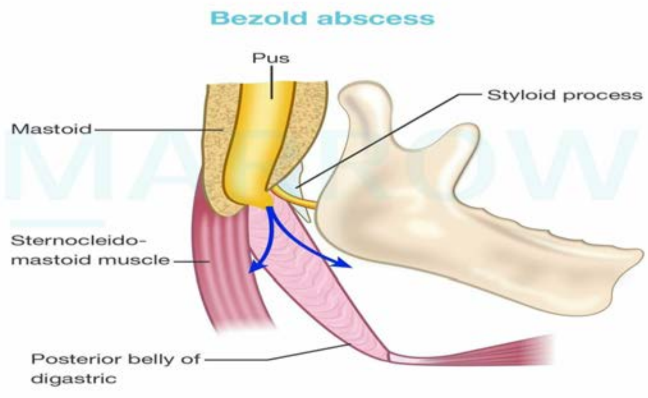
Treatment includes aural toilet, treatment of secondary bacterial infections, and a full course of antitubercular therapy followed by reconstructive surgery.

Solution to Question 5:

The given image shows ecchymosis over the retro auricular and mastoid area suggestive of Battle's sign. It is typically seen in fractures of the middle cranial fossa following head trauma. Battle sign is prominent when there is a fracture of the petrous temporal bone.

Other options:

Option B: Bezold abscess is a complication of acute otomastoiditis where the infection/pus breaks through the thin medial side of the tip of the mastoid and presents as a swelling in the upper part of the neck.



Pus bursting through the medial side of the tip of mastoid and collecting under the sternomastoid or digastric triangle.

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Option C: Acute mastoiditis is used to refer to an acute infection that spreads from the mucosa lining the mastoid air cells to involve the bony walls of the mastoid air cell system. Mastoiditis presents as retro-auricular redness and swelling which is painful.

The image below shows a smooth ironed-out mastoid with erythema seen in acute mastoiditis.

Option D: The Griesinger sign refers to edema of the postauricular soft tissues overlying the mastoid process as a result of thrombosis of the mastoid emissary vein seen in sagittal sinus thrombosis. It is a complication of acute otomastoiditis.

Solution to Question 6:

Complete closure of the nostrils (Young's operation) or partial closure of the nostrils (Modified Young's operation) is done for the treatment of atrophic rhinitis.

In Young's operation, both nostrils are closed completely by raising flaps within the nasal vestibule. They are opened after 6 months. During this time, the mucosa becomes normal, and the crusting is reduced.

In Modified Young's operation, the procedure has been modified to avoid discomfort from complete closure of the nostrils. Here the nostrils are only partially closed. The results are the same when compared to Young's operation.

Atrophic rhinitis (ozaena) is a chronic inflammation of the nose with atrophy of nasal mucosa and turbinate bones. Roomy nasal cavities with foul-smelling crusts are seen. The patient presents with a foul smell from the nose, marked anosmia (merciful anosmia), and paradoxical nasal obstruction (nasal obstruction despite wide nasal passages). Paradoxical nasal obstruction occurs due to degenerative changes in the sensory nerves of the nasal mucosa, leading to an inability to sense airflow.

Medical management includes nasal irrigation and removal of crusts. 25% glucose in glycerine, antibiotics (quinolones or metronidazole), estradiol nasal spray, and mucolytics can be used.

Solution to Question 7:

Occipitomental view is called Water's view. Water's view with an open mouth is known as Pierre's view. Among the given options, Water's view is the best answer.

In Water's view, the patient’s head is positioned such that the nose and the chin touch the film while the X-rays are projected from backward. The mouth is usually kept open, which allows visualizing the sphenoid sinus.

Structures seen in Water’s view are:

- Maxillary sinuses (seen best)
- Frontal sinuses
- Sphenoid sinus (if the film is taken with an open mouth)
- Zygoma
- Zygomatic arch
- Nasal bone
- Frontal process of the maxilla
- Superior orbital fissure
- Infratemporal fossa

Other options:

Option B: Towne's view is an angled anteroposterior radiograph of the skull. Structures seen are the petrous part of the pyramids, arcuate eminence, superior semicircular canal, mastoid antrum, internal auditory canal, tympanic cavity, cochlea, and external auditory canal. This view is taken for acoustic neuroma and apical petrositis.

Option C: Law's view is a lateral oblique X-ray view of the mastoid. Structures seen are the external auditory canal, mastoid air cells, tegmen, lateral sinus plate, and temporomandibular joint.

Option D: Stenver's view is an X-ray view of the skull where the long axis of the petrous bone is parallel to the film. Structures seen are the entire petrous pyramid, arcuate eminence, internal auditory meatus, labyrinth with its vestibule, cochlea, and mastoid antrum.

Solution to Question 8:

The given clinical scenario describes a child with fever, otalgia, trismus, and odynophagia with a normal external facial examination. The oral cavity shows congested and edematous tonsil and tonsillar pillars with edematous uvula pushed to the opposite side. These findings are highly suggestive of quinsy (peritonsillar abscess).

It is the collection of pus in the peritonsillar space between the fibrous capsule of the tonsil and the superior constrictor muscles of the pharynx. Both aerobic and anaerobic organisms are seen.

Patient presents with fever, unilateral throat pain, odynophagia, hot potato voice, torticollis, foul breath, referred otalgia through glossopharyngeal nerve and trismus due to spasm of pterygoid muscles.

Oral cavity examination may show congested and edematous tonsils, pillars, and soft palate on the involved side. Edematous uvula pushed to the opposite side. Pus may be seen on the tonsillar region. Jugulodigastric lymph nodes may be enlarged.

The management of quinsy is by antibiotics and incision and drainage to drain the abscess followed by interval tonsillectomy 4 to 6 weeks later.

Rare complications seen are parapharyngeal abscess, edema of the larynx, septicemia, lung abscess, or jugular vein thrombosis.

Other options:

Option A: Pharyngitis presents with fever, throat pain, enlarged tonsils, and lymphoid follicles on the posterior pharyngeal wall.

Option B: Parotid abscess occurs due to the suppuration of parotid space. It presents with swelling, redness, firmness, and tenderness in the parotid area and at the angle of the mandible. It is usually unilateral, but bilateral abscesses can occur. Due to the thick capsule, fluctuation is often difficult to elicit.

Option C: Bezold abscess is a complication of acute otomastoiditis where the infection/pus breaks through the thin medial side of the tip of the mastoid and presents as a swelling in the upper part of the neck.

ENT INICET 2021

Question 1:

Identify the condition shown in the image.

a) Adenoid facies

b) Goldenhar syndrome

c) Horse facies

d) Frog facies

Question 2:

The given image is of a person with a history of minor trauma with a piece of wood, 20 days back. His face is affected on the left side with facial swelling and orbital edema. CT scan revealed clear sinuses and the presence of subcutaneous nodules. Microscopy of the tissue sections was PAS-positive and stained positive with Grocott's methenamine silver stain. What is the most likely diagnosis?

a) Phycomycosis

b) Midline lethal granuloma

c) Foreign body granuloma

d) IgG4 granuloma

Question 3:

Which of the following are the objective tests for the hearing?

a) 1, 2, and 4

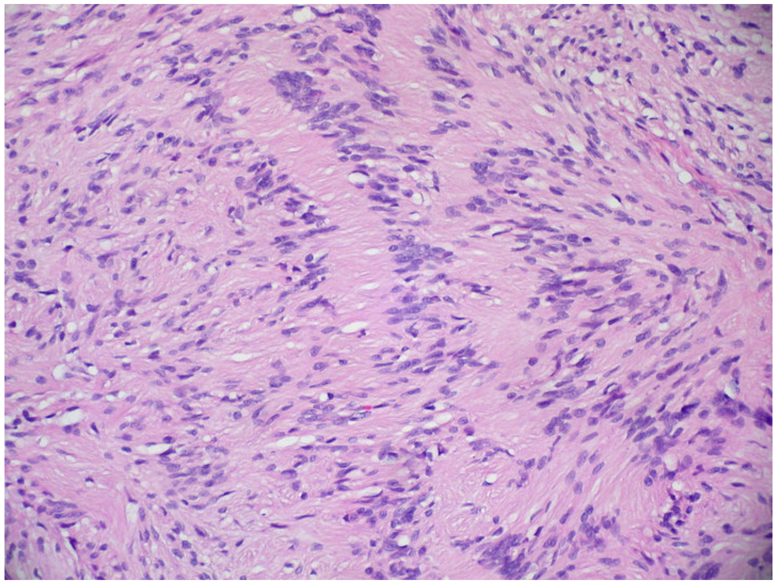
b) 1,2, and 3

c) 2,3, and 4

d) 1,2,3, and 4

Question 4:

A young man presented with hearing loss and tinnitus. Histology image is shown below. What is the diagnosis?



a) Neurofibroma

b) Schwannoma

c) Leiomyoma

d) Rhabdomyoma

Question 5:

All of the following are major diagnostic criteria for allergic fungal sinusitis except?

a) Presence of nasal polyps

b) Eosinophilic mucin without invasion

c) Characteristic CT findings

d) Positive fungal culture

Question 6:

Which among the following is not used in post laryngectomy rehabilitation?

a) Tracheo-esophageal prosthesis

b) Esophageal speech

c) Polite yawning

d) Super supraglottic swallow

Question 7:

Which of the following devices can be used in a patient with bilateral external auditory canal atresia?

a) 1

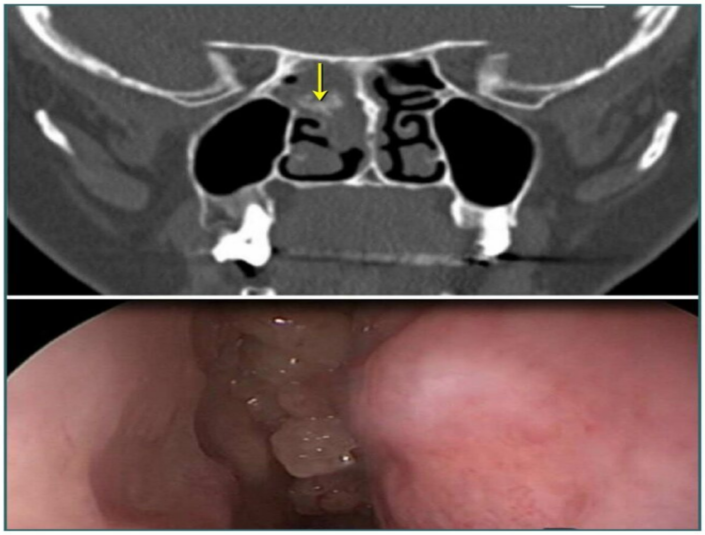
b) 2

c) 3

d) 4

Question 8:

A 65-year-old patient presents with right-sided nasal obstruction and occasional blood-stained nasal discharge. His CT and endoscopy findings are shown in the below image. What could be the underlying cause?



a) Inverted papilloma

b) Carcinoma maxilla

c) Esthesioneuroblastoma

d) Nasopharyngeal angiofibroma

Question 9:

A patient presents with vertigo, tinnitus, and head tilt. He underwent myringoplasty for the safe type of chronic suppurative otitis media(CSOM) 6 months back. What is your diagnosis?

a) Peri-lymphatic fistula

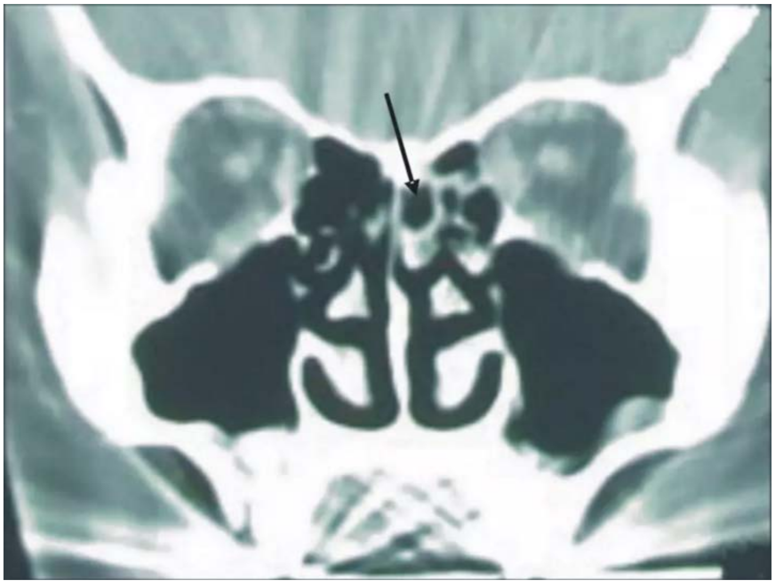
b) Labyrinthitis

c) Paget disease

d) Vestibular schwannoma

Question 10:

What is the probable diagnosis based on the finding given in the CT image?



a) Concha bullosa

b) Pneumatised superior turbinate

c) Onodi cell

d) Haller cell

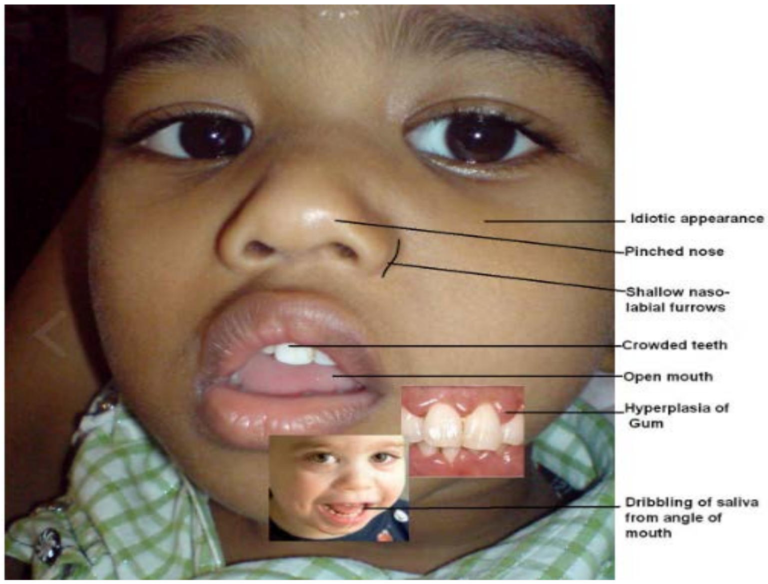
Answer Key

Question No.	Correct Option
1	a
2	a
3	a
4	b
5	d
6	d
7	a
8	a
9	a
10	b

Detailed Explanations

Solution to Question 1:

The given image shows adenoid facies. It is characterized by an elongated face with a dull expression, an open mouth, a hitched-up upper lip, prominent upper teeth causing crowding (malocclusion of teeth), a high arched hard palate (due to loss of tongue's molding action on the palate), and a pinched-in nose due to disuse atrophy of the alae nasi.

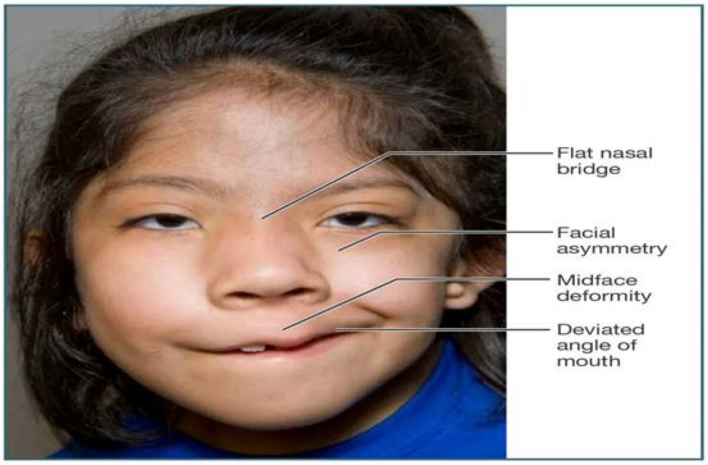


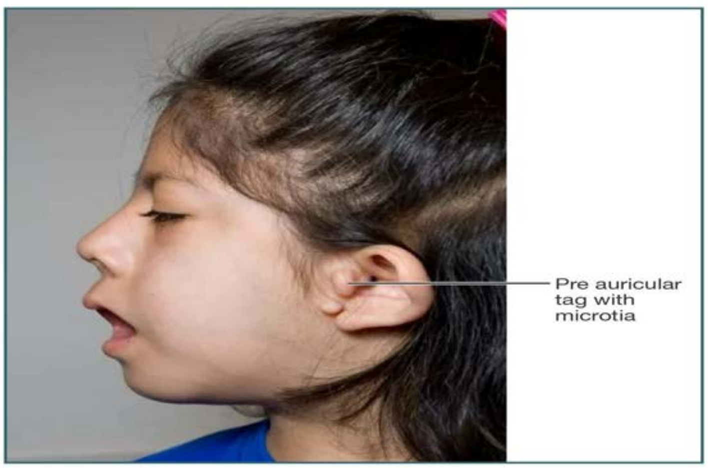
Chronic nasal obstruction resulting in mouth breathing leads to this characteristic facial appearance in children.

Adenoids (nasopharyngeal tonsils) are prone to physiological enlargement during childhood. Affected children present with symptoms of nasal obstruction, nasal discharge, sinusitis, and rhinolalia clausa (loss of nasal tone of voice). They are prone to eustachian tube obstruction, recurrent acute otitis media, chronic otitis media, and otitis media with effusion. They may also experience aprosexia (inability to concentrate) leading to declining scholastic performance. Pulmonary hypertension and cor pulmonale are potential long term complications

Other Options:

Option B: Goldenhar Syndrome, also known as fascio-auriculo-vertebral dysplasia or oculo-auriculo-vertebral (OAV) syndrome, is characterized by facial asymmetry, low set ears, atresia of the ear canal, cardiac abnormalities, pre-auricular tags/pits, hemivertebrae in the cervical region, epibulbar dermoid, coloboma of the upper lid, and mixed or conductive hearing loss.





Option D: Frog facies is a characteristic presentation observed in cases of juvenile nasopharyngeal angiofibroma. It is identified by the swelling of the cheeks, proptosis (protrusion of the eyeballs), and broadening of the nasal bridge.

Solution to Question 2:

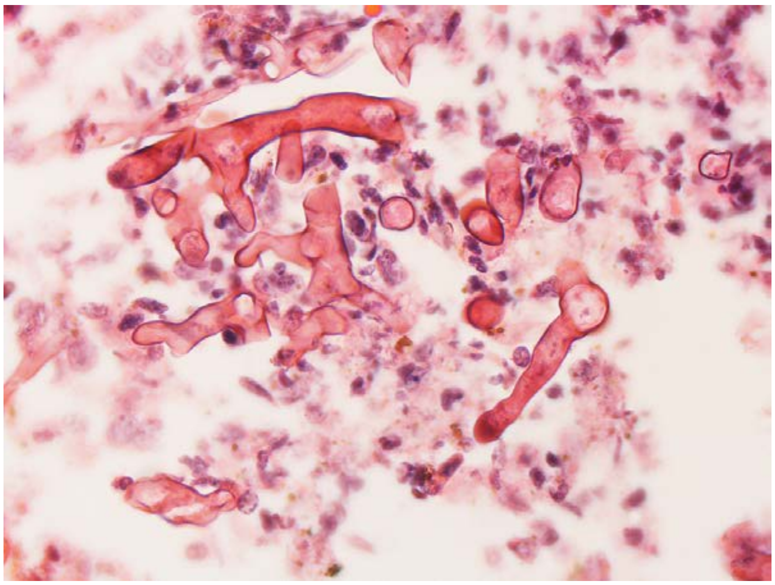
The history of facial swelling and orbital edema following trauma with wood and positive fungal stain (Grocott-Gomori's methenamine silver) points toward phycomycosis (mucormycosis), a potentially invasive fungal infection.

Mucormycosis is an opportunistic mycosis caused by various molds in the Mucorales order which are ubiquitous thermotolerant saprobes. These molds, including Rhizopus, Mucor, and Absidia, are common in the environment. The spores can be found in air and dust.

Mucormycosis can manifest as a localized cutaneous or subcutaneous infection in immunologically normal individuals after traumatic exposure to soil or vegetation. It is a fatal condition due to its angio-invasive nature. Lung involvement is common, and the infection primarily affects immunocompromised patients.

Rhino-orbital-cerebral mucormycosis is frequently seen in individuals with diabetic ketoacidosis. It typically starts with eye and facial pain and may progress to orbital cellulitis, proptosis, and vision loss. A characteristic finding is a black necrotic mass filling the nasal cavity.

Diagnosis involves testing specimens and cultures. In exudate samples, wet preparations with 10% potassium hydroxide (KOH) reveal broad, ribbon-like hyphae. Tissue biopsy confirms the diagnosis by observing non-septate hyphae branching at right angles, characteristic of Mucorales like Rhizopus delemar.



Culture isolation is performed on Sabouraud agar without cycloheximide, resulting in grey-white colonies with a thick cottony, and fluffy surface.

Treatment is primarily with liposomal amphotericin B. Surgical debridement of the affected tissues and control of underlying predisposing cause is equally important.

Other options:

Option D: IgG4-related disease (IgG4-RD) is characterized by tissue infiltrates dominated by IgG4 antibodies, which is the hallmark of the disease. IgG4-RD leads to fibrosis and obliterative phlebitis. Patients may present with Riedel's thyroiditis, autoimmune pancreatitis, or idiopathic orbital inflammation, but not with ptosis. Diagnosis is confirmed through positive fibrosis and the presence of plasma cells on biopsy. Treatment with Rituximab is done.

Option C: Foreign body granulomas are due to inflammation induced by inert foreign bodies. These granulomas develop without T-cell-mediated immune responses. Sutures are a common site for their formation. The foreign material is at the center of the granuloma, surrounded by epithelioid cells and specific giant cells opposing the surface.

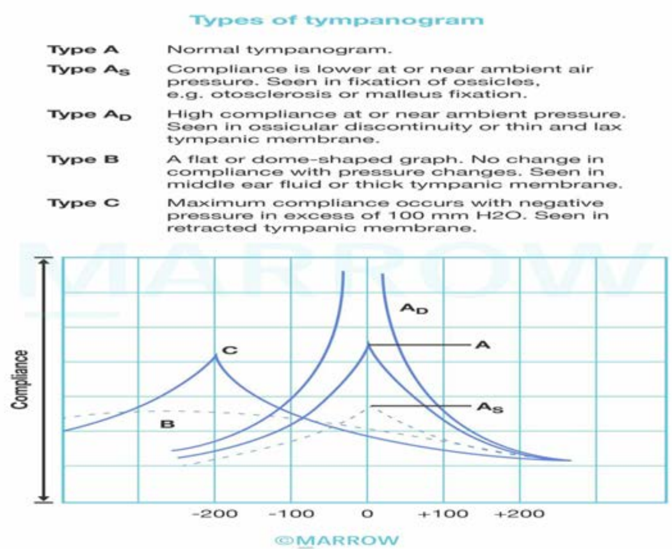
Option B: Midline lethal granuloma is characterized by necrotizing destruction and damage to the nose and other respiratory structures. It is rare and usually a diagnosis of exclusion. CT scans can detect the lesions associated with this condition. Negative c-ANCA (cytoplasmic antineutrophil cytoplasmic antibodies) and fungal stain results contribute to the diagnosis.

Solution to Question 3:

BERA, OAE, and tympanometry are the objective tests for hearing. PTA (pure tone audiometry) is a subjective test for hearing as it relies on patient responses to pure tone stimuli. Other examples of subjective tests include tuning fork tests and speech audiometry.

Tympanometry is an objective test of the middle ear and eustachian tube functions (conductive compartment of hearing). Here, the compliance of the TM, i.e., how much it moves to the pressure differentials applied from the external ear is measured.

The image below shows the types of tympanogram:



Solution to Question 4:

The history of hearing loss and tinnitus and the histology showing Verocay bodies are suggestive of Schwannoma.

Vestibular Schwannoma (acoustic neuroma), is a benign tumor that predominantly originates from the Schwann cells in the vestibular division of the eighth cranial nerve, but it rarely arises from the cochlear division. It commonly affects individuals aged 40-60, with no gender preference. Bilateral schwannomas occur in neurofibromatosis type 2 (NF2).

The most common presenting manifestations are progressive, unilateral sensorineural hearing loss, often accompanied by tinnitus.

It can involve all cranial nerves except the first and second. Apart from the 8th cranial nerve, the earliest cranial nerve to be involved is the 5th cranial nerve leading to reduced corneal sensitivity and paraesthesia of the face. This indicates that the tumor is roughly 2.5 cm in diameter and occupies the cerebellopontine angle.

Facial nerve involvement causes hypoaesthesia of the posterior meatal wall (Hitzelberger sign), taste loss, and delayed blink reflex. The earliest sign is a delayed blink reflex.

Glossopharyngeal (IX) and vagus (X) nerve involvement result in
dysphagia and hoarseness. Brainstem involvement manifests as ataxia, limb weakness, numbness, and increased tendon reflexes. Cerebellar involvement leads to an unsteady gait and dysdiadochokinesia.

Histopathologically, the tumor comprises dense Antoni A areas with spindle cells in a palisading pattern and Antoni B areas with loosely arranged spindle cells in a myxoid matrix. Verocay bodies are formed in the Antoni A region, characterized by nuclear-free zones surrounded by palisaded nuclei.

Pure tone audiometry reveals sensorineural hearing loss, particularly in high frequencies. Speech audiometry shows the roll-over phenomenon. The absence of the recruitment phenomenon is noted. Short Increment Sensitivity Index (SISI) ranges from 0-20%.

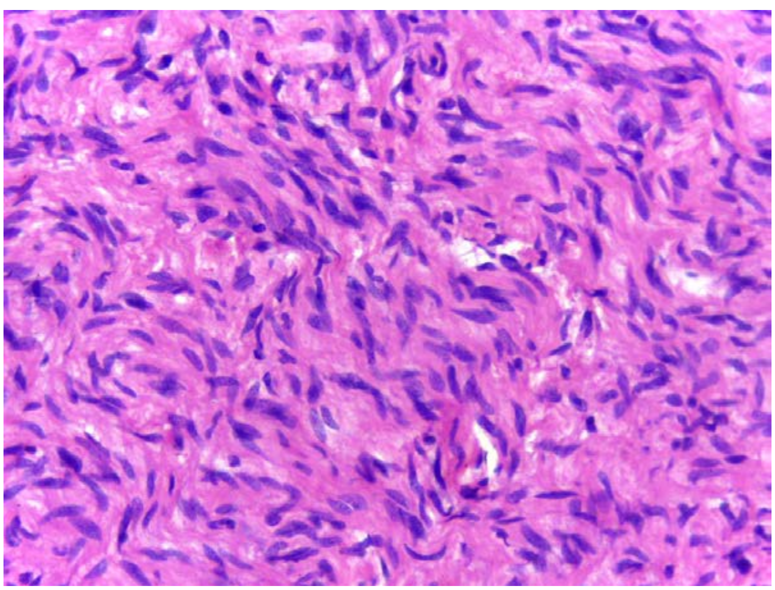
Brainstem Evoked Response Audiometry (BERA) demonstrates a significant delay of >0.2 ms in wave V between the two ears. Magnetic Resonance Imaging (MRI) with gadolinium contrast is considered the gold standard for diagnosis.

The primary management is the surgical removal of the tumor via the middle cranial fossa approach, the trans-labyrinthine approach, the suboccipital (retro sigmoid) approach, or a combined trans-labyrinthine-suboccipital approach.

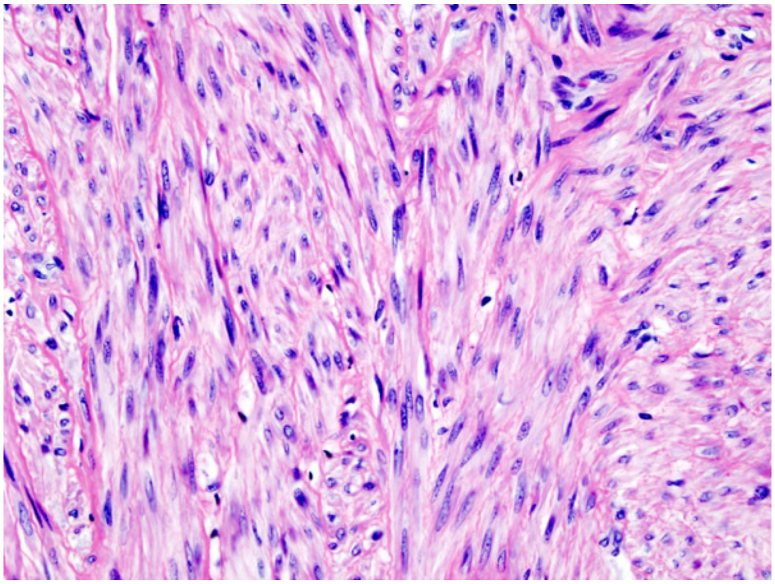
Conventional radiotherapy using external beam radiation is not effective. In cases where surgery is contraindicated, refused by the patient, or when a residual tumor remains, alternative treatments such as the x-knife or gamma knife can be considered.

Other options:

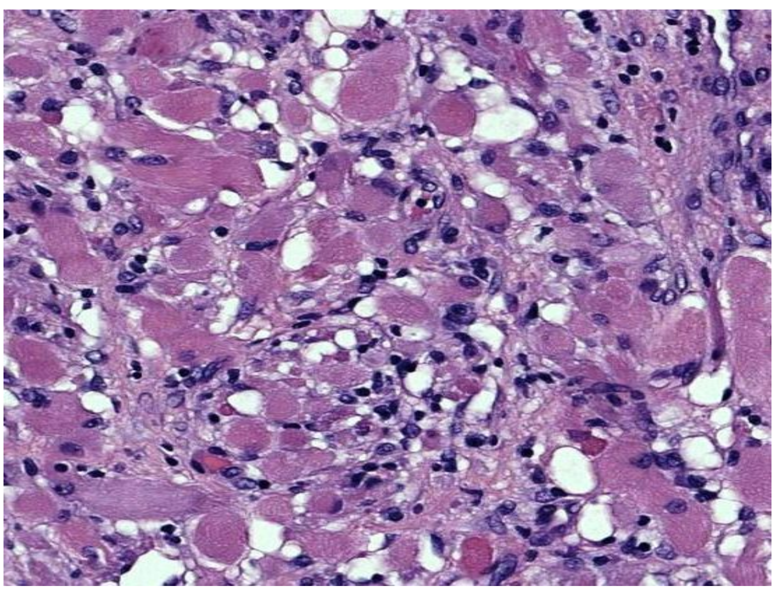
Option A: Neurofibromas are also benign nerve sheath tumours and the histological picture shows the presence of bland Schwann cells admixed with stromal cells such as mast cells, perineural cells, and fibroblasts. The stroma of the tumour contains loose collagen arranged in wavy bundles resembling carrot shavings.



Option C: Leiomyomas are benign smooth muscle neoplasms that may occur single or multiple. Microscopically, leiomyomas are comprised of well-differentiated, regular, spindle-shaped smooth muscle cells associated with hyalinization. Very less mitotic figures are seen.



Option D: Rhabdomyomas are the most frequent primary tumours of the heart in children. Microscopically, they are composed of bizarre, markedly enlarged myocytes with abundant glycogen. But due to routine histological processing, the glycogen content is reduced and looks like thin strands that stretch from the nuclei to the surface membrane, which appears like spider cells.



Solution to Question 5:

A positive fungal culture is a minor and not a major diagnostic criterion for allergic fungal rhinosinusitis (AFRS).

The Bent and Kuhn diagnostic criteria for allergic fungal rhinosinusitis include:

Major criteria:

• Type I hypersensitivity

• Nasal polyposis

• Characteristic CT findings

• Eosinophilic mucin without invasion

• Positive fungal stain

Minor criteria:

• Asthma

• Unilateral disease

• Bone erosion

• Fungal cultures

• Charcot-Leyden crystals

• Serum eosinophilia

All the major criteria must be met by the patient while the minor criteria serve to support the diagnosis and describe individual patients.

Characteristic CT findings in AFRS include heterogeneous signal intensities within the paranasal sinuses filled with allergic mucin (double density sign), expansion of the paranasal sinuses/nasal cavity, unilateral or asymmetric disease load, and bony erosion.

The treatment of allergic fungal sinusitis includes avoidance of allergen exposure, antihistamines, corticosteroid nasal sprays, aggressive surgical intervention for mucin removal which improves sinus drainage, post-surgery oral corticosteroids, and immunotherapy against fungal and non-fungal allergens.

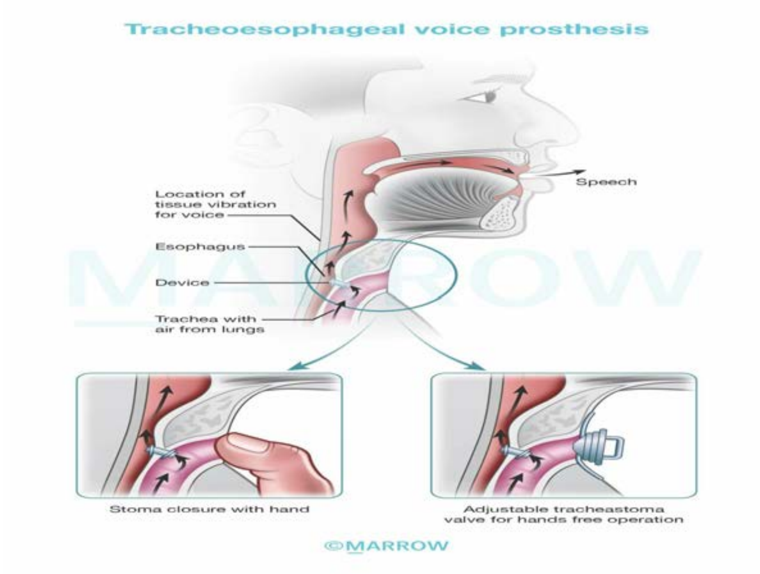
Solution to Question 6:

Super supraglottic swallow is not used in post-laryngectomy rehabilitation.

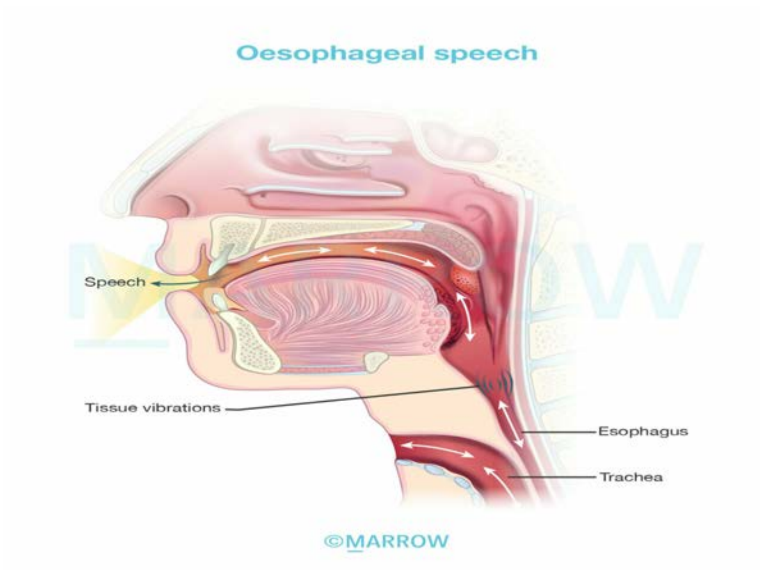
Super-supraglottic swallow is a maneuver that prevents the aspiration of food or liquid by closing the airway before swallowing. It is used in the management of dysphagia. This method involves holding one's breath to close the airway prior to swallowing, followed by coughing voluntarily immediately after swallowing to clear any residual food or liquid from the airway entrance. It is an effortful breath-holding technique, which not only results in Valsalva and subsequent leakage of bolus but can also result in arrhythmias.

Other options:

Option A: A tracheoesophageal prosthesis(TEP) involves a one-way valve that is inserted through a previously-created tracheoesophageal puncture. When the patient occludes the stoma, the exhaled air is shunted through the TEP into the oesophagus, where it induces vibration of the upper oesophagal sphincter, which is responsible for phonation. It generates the most intelligible, fluent, natural-sounding voice in contrast to other laryngeal speech methods.

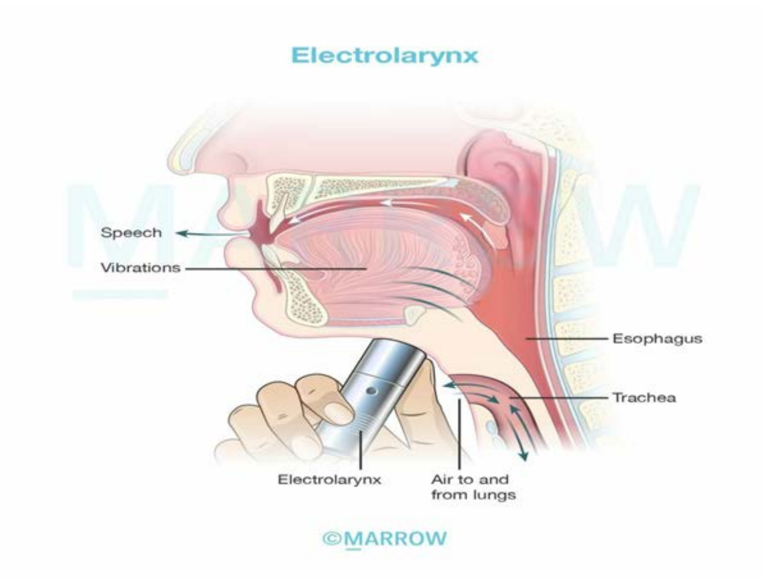


Option B: Oesophagal speech also involves vibration of the upper oesophagal sphincter and subsequent phonation.



Option C: Polite yawning also called the nasal airflow-inducing maneuver is an important olfaction rehabilitation tool after a laryngectomy.

Electrolarynx produces vibrations that are transmitted through the skin toward the throat. The sound generated is converted into low-pitch speech by the vocal tract—palate, tongue, and teeth. The advantages are that is easy and quick to learn, while the disadvantages are that the voice is monotonous and the constant occupation of one hand.



Solution to Question 7:

Device 1 shown in the above image represents bone anchored hearing aid (BAHA), which can be used in a patient with bilateral external auditory canal atresia. Cochlear implants (Device 2) are primarily used to restore sensorineural hearing loss, whereas BAHA is used in cases with defects in the outer or middle ear.

External auditory canal atresia refers to the absence of a patent ear canal and this can either be congenital or acquired.

Congenital atresia: The external ear develops from the first and second pharyngeal arch and the first pharyngeal cleft. Atresia of the external canal can occur in association with microtia (under-developed pinna) or with abnormalities of the middle ear, internal ear, and other structures. It is usually bilateral.

Acquired atresia: A rare condition, following external ear trauma, such as road traffic accidents, gunshot wounds, or recent otologic surgery. In rare instances, canal stenosis and atresia have been observed in association with neoplastic changes and idiopathic inflammatory processes.

Bilateral aural atresia will present with failed audiological testing and require early bone conduction hearing aids, one of which is BAHA, that combines a sound processor with a small titanium fixture implanted behind the ear and allows the bone to transfer sound to a functioning cochlea rather than via the middle ear – a process known as direct bone conduction.

A picture of BAHA is shown below. It consists of three parts, an innermost titanium fixture in the bone, a middle abutment, and an outer hearing aid fixed to the abutment.

Bone-anchored Hearing Aids (BAHA) are used:

• When air-conduction (AC) hearing aid cannot be used:

• People who have chronic inflammation or infection of the ear canal associated with discharge and cannot wear standard “in the ear” air-conduction hearing aids.

• Children with malformed or absent outer ear and ear canals as in microtia or canal atresia.

• In conductive or mixed hearing loss, e.g. otosclerosis and tympanosclerosis where surgery is contraindicated

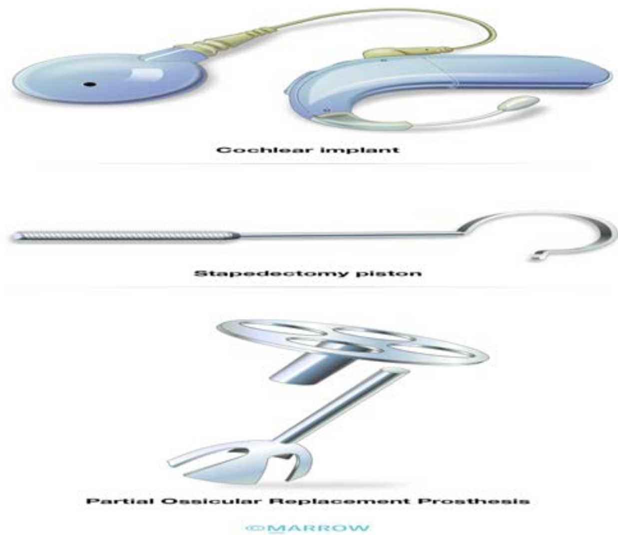
• In unilateral deafness

Other options:

Option B: Device 2 is a picture of a cochlear implant. It is a medical device that uses electricity to stimulate the spiral ganglion cells of the auditory nerve to restore sensorineural hearing loss. It converts sound to an electrical signal (transduction) and delivers this to the hearing nerve, which bypasses the damaged hearing apparatus. It cannot be used in patients with cochlear aplasia or without the VIIIth cranial nerve.

Option C: Device 3 is a picture of a stapedectomy piston, used in patients with otosclerosis.

Option D: Device 4 is a picture of a partial ossicular replacement prosthesis that is designed for situations when the incus and/or malleus are eroded but the stapes is structurally intact.



Solution to Question 8:

In an elderly patient, unilateral nasal obstruction, epistaxis, and pale polypoidal mass arising from the lateral wall of the nose are suggestive of inverted papilloma (Ringertz tumor).

Inverted papilloma (transitional cell papilloma) is a tumor originating from the nonolfactory mucosa of the nose (Schneiderian membrane) and paranasal sinuses. The most common site of origin is the lateral wall of the nose in the middle meatus.The unilateral growth of hyperplastic papillomatous tissue into the stroma characterizes this condition. Human papillomavirus (HPV) is believed to be the cause.

It primarily affects men aged 40-70 and presents unilaterally with symptoms such as nasal obstruction, nasal discharge, and epistaxis. Invasive growth into the sinuses or orbit can lead to proptosis, diplopia, and lacrimation.

On nasal endoscopy, it appears as a pale polypoidal mass resembling a nasal polyp. Computed tomography (CT) and magnetic resonance imaging (MRI) are used to determine the location and extent of the lesion. MRI can also help differentiate between associated sinus secretions and the tumor mass. Biopsy is essential for diagnosis, as it is important to differentiate it from simple nasal polyps, which may be associated with or mistaken for the tumor.

The treatment of choice is medial maxillectomy, which can be performed through lateral rhinotomy, sublabial degloving approach, or preferably, an endoscopic approach. In 10%-15% of cases, the tumor is associated with malignancy. Recurrence is possible, and radiotherapy is generally not recommended due to the potential risk of inducing malignancy.

Other options:

Option B: Carcinoma maxilla arises from the lining of the maxillary sinus and its early features include additional symptoms like facial paraesthesias, pain, or epiphora and depending on the direction of spread of the growth, would cause additional symptoms.

Option C: Esthesioneuroblastoma (olfactory neuroblastoma) arises from the olfactory epithelium in the upper third of the nose. It has bimodal peaks of incidence at 10–20 years and 50–60 years. Unilateral nasal obstruction and epistaxis are the most common symptoms. The tumour also invades the orbit and the surrounding structures, which results in other symptoms like proptosis, headache, epiphora, diplopia, and blurred vision.

Option D: Juvenile nasopharyngeal angiofibroma is seen almost exclusively in males in the age group of 10–20 years and is very rarely seen in older people and females. Profuse, recurrent, and spontaneous epistaxis is the most common presentation.

Solution to Question 9:

The presentation of vertigo, tinnitus, and head tilt following myringoplasty suggests a diagnosis of perilymphatic fistula.

A perilymphatic fistula is an abnormal connection between the inner ear and middle ear, resulting in leakage of perilymph from the cochlea or vestibule. This most commonly occurs through the round or oval window.

Causes of perilymphatic fistula include post-stapedectomy, ear surgery with dislocation of the stapes, head trauma, inner ear injury due to forceful Valsalva or barotrauma, raised intracranial pressure, and occasionally spontaneous or idiopathic.

Clinically, intermittent vertigo, fluctuating sensorineural hearing loss, tinnitus, and a sense of fullness in the affected ear are present. The fistula test is positive, and high-resolution computed tomography (HRCT) of the temporal bone is highly sensitive for detecting the fistula. A combination of CT and magnetic resonance imaging (MRI) can diagnose almost all cases. Audiometry and electrocochleography may be performed additionally.

Management of perilymphatic fistula can be conservative or surgical. Conservative measures involve avoiding activities that increase inner ear or intracranial pressure and using intra-tympanic steroids for acute decompensation.

Other options:

Option B: Labyrinthitis presents with unilateral vestibular symptoms like sudden onset of persistent vertigo with associated symptoms such as nausea, vomiting, sweating, and pallor. Symptoms may progress resulting in nystagmus and hearing loss. Tinnitus is not a common finding in labyrinthitis.

Option C: Paget's disease of bone usually presents with hearing loss as a prominent feature, when the skull is involved. It is not associated with a history of surgery, trauma, or vestibulocochlear symptoms.

Option D: Vestibular Schwannoma (acoustic neuroma), originates from Schwann cells in the vestibular division of the vestibulocochlear (VIIIth cranial) nerve within the internal auditory canal. The main symptoms are progressive unilateral hearing loss and tinnitus. Other cranial nerves (V, IX, X, XI) may also be involved.

Solution to Question 10:

The given CT image reveals a pneumatised superior turbinate.

The superior turbinate, located posteriorly and superiorly to the middle turbinate can become pneumatized. It forms an important landmark to identify the ostium of the sphenoid sinus which lies medial to it. This pneumatization is common in the lateral nasal wall, although it is less frequently seen in the superior turbinate compared to the middle turbinate.

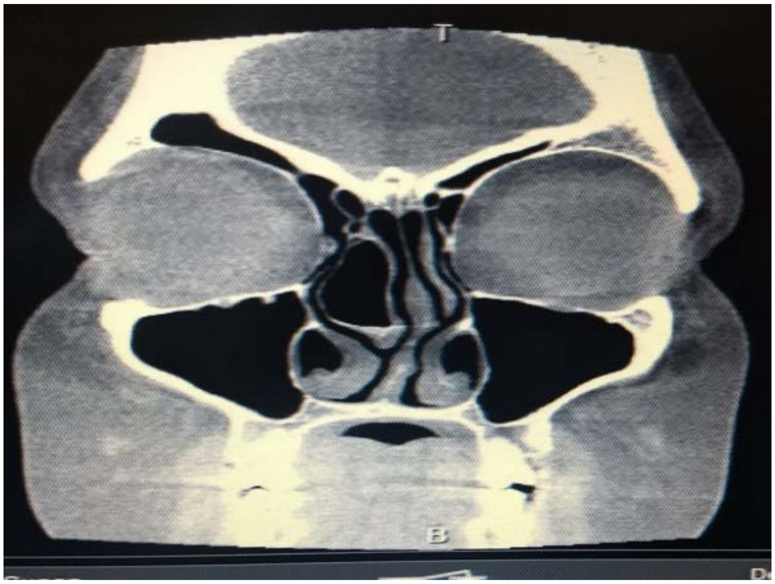
In most cases, pneumatized superior turbinate is an incidental finding during radiological examinations. It can lead to nasal obstruction, headache, epistaxis, anosmia, and postnasal drainage.

Asymptomatic cases of superior turbinate pneumatization generally do not require treatment. However, when symptoms are present, various surgical techniques can be considered, including outfracture, crushing, radio-frequency ablation, turbinoplasty, and microdebrider surgery.

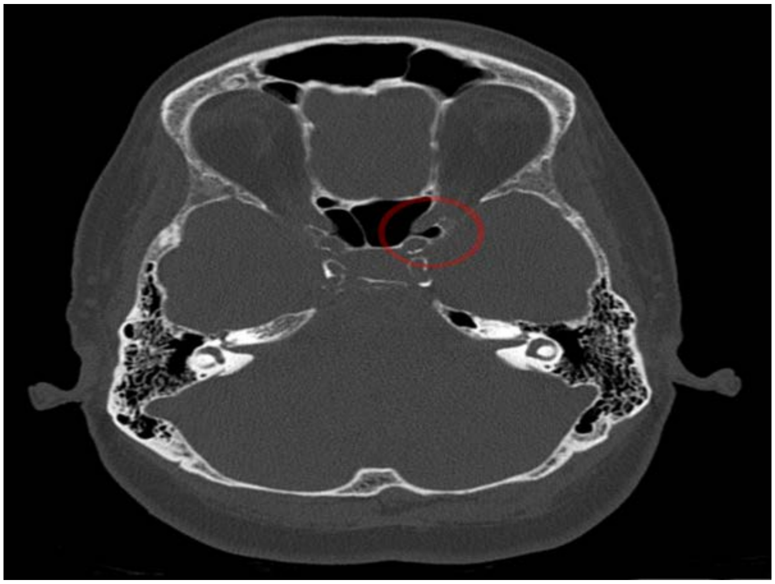
Turbinoplasty is contraindicated if inferior turbinate pneumatization is associated with the maxillary sinus because it may result in the formation of a second ostium. In addition, total turbinectomy is not performed because it may cause atrophic rhinitis.

Other options:

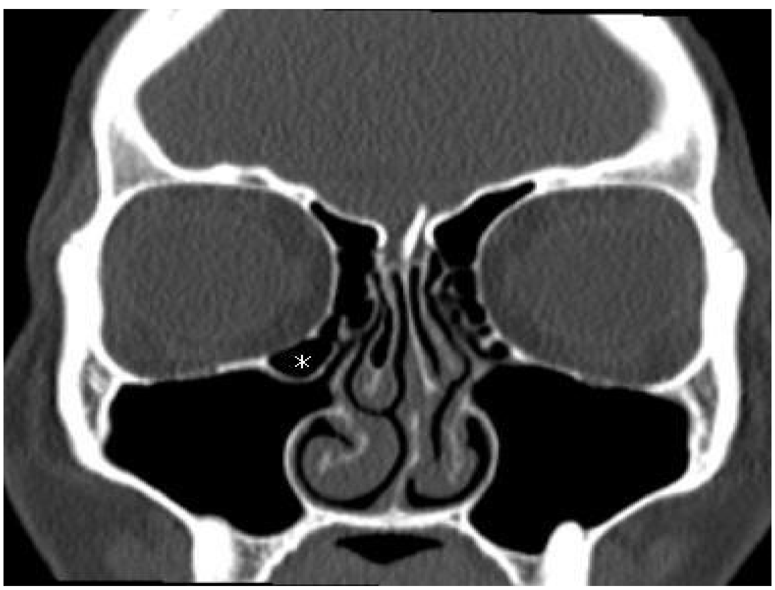
Option A: Pneumatization of the middle turbinate leads to an enlarged middle turbinate that is ballooned out, this is called concha bullosa. The CT image shown below depicts concha bullosa.



Option C: Onodi cell is a posterior ethmoidal cell that may grow posteriorly, either by the side of the sphenoid sinus or superior to it. Onodi cell is surgically important as the optic nerve may be related to its lateral wall. A CT image of an onodi cell is shown below.



Option D: Haller cells are air cells situated in the roof of the maxillary sinus. They are pneumatized from anterior or posterior ethmoid cells. Enlargement of Haller cells may lead to maxillary sinusitis as it may encroach the ethmoid infundibulum, impeding draining of the maxillary sinus. A CT image of Haller cell is shown below.



ENT NEET 2021

Question 1:

The image of a drug is given below. It is used in which of the following conditions?

a) Subglottic stenosis

b) Rhinocerebral mucormycosis

c) Adenoidectomy

d) Tympanoplasty

Question 2:

Following total thyroidectomy, a patient started having difficulty in breathing, and repeated attempts to extubate were unsuccessful. The most probable cause is ___

a) Superior laryngeal nerve injury

b) Unilateral recurrent laryngeal nerve injury

c) Bilateral recurrent laryngeal nerve injury

d) Hematoma

Question 3:

A female patient presents with nasal obstruction, nasal discharge, and loss of smell. On examination, foul-smelling discharge and yellowish-green crusts are present in the nasal cavity. She is found to have merciful anosmia. Which of the following finding can also be seen during the examination of her nose?

a) Roomy nasal cavity

b) Nasal polyps

c) Inferior turbinate hypertrophy

d) Foreign body

Question 4:

A patient presents to the emergency with epistaxis. There was no relief on pinching the nostrils. Nasal packing was done but the patient still continues to bleed. What would be the next appropriate step in the management of this patient?

a) Ligation of external carotid artery

b) Ligation of internal carotid artery

c) Ligation of sphenopalatine artery

d) Ligation of maxillary artery

Question 5:

A patient post-tonsillectomy in the recovery room starts bleeding from the operative site. On examination, blood clots are seen. What will be your immediate management?

a) Shift to OT, remove the clots, and cauterize/ligate the vessel

b) Shift to OT, start IV antibiotics, and pack the tonsillar fossa

c) Give anticoagulants, repeated gargling, and wait for 24 hours

d) Do blood transfusion and wait and watch

Question 6:

A patient with a history of chronic ear infection presents with fever, headache, vomiting, irritability, and confusion. His CT brain is shown in the image below. What is the possible diagnosis?

a) Temporal lobe abscess

b) Cerebellar abscess

c) Subdural abscess

d) Meningitis

Answer Key

Question No.	Correct Option
1	a
2	c
3	a
4	c
5	a
6	a

Detailed Explanations

Solution to Question 1:

The given drug is Mitomycin-C. It is used topically in the management of subglottic stenosis.

Mitomycin-C is derived from the Streptomyces caespitosus bacteria. It can modify wound healing at a molecular level by inhibiting DNA synthesis by cross-linking DNA, acting like an alkylating agent (cell cycle non-specific). It has been used to inhibit post-surgical scar formation and as a chemotherapeutic agent for solid tumors.

Mitomycin-C is employed in the treatment of superficial bladder tumors, postoperative glottic and subglottic stenosis, and tracheal scarring after tracheal reconstruction surgery.

Subglottic stenosis is the narrowing of the airway in the subglottis, clinically presenting with biphasic stridor. Diagnosis is made if the subglottic diameter is <4mm in full-term infants and if the subglottic diameter is <3mm, in the case of premature infants.

Myer-Cotton grading of subglottic stenosis is as follows:

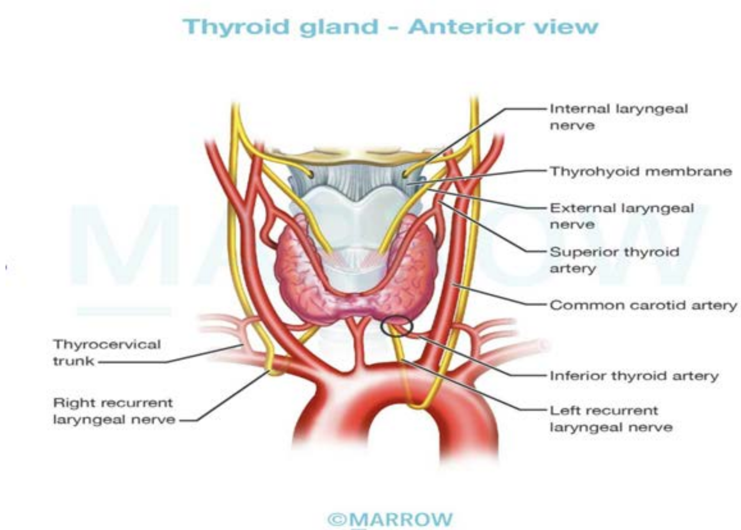
Grade	Obstruction
Grade I	≤50%
Grade II	51-70%
Grade III	71-99%
Grade IV	No detectable lumen

Solution to Question 2:

The clinical picture of breathing difficulty and unsuccessful attempts at extubation after total thyroidectomy is suggestive of bilateral recurrent laryngeal nerve injury.

Thyroid surgery may cause injury to both external (ELN) and recurrent laryngeal nerves (RLN). The upper part of the RLN is closely related to the inferior thyroid artery. Unilateral RLN injury may be asymptomatic but in a few may cause hoarseness of voice.

The relation of RLN with the inferior thyroid artery is shown in the image below:



Bilateral recurrent laryngeal nerve palsy can also occur postoperatively and the presentation is acute. Other causes of it include carcinoma of the cervical esophagus, carcinoma thyroid, and cervical lymphadenopathy. All the intrinsic muscles are paralyzed in this condition.

The vocal cords lie in the median or paramedian position due to the unopposed action of the cricothyroid muscles. This leads to dyspnoea and stridor, necessitating emergency tracheostomy.

For long-standing cases of bilateral recurrent laryngeal nerve palsy, there are two main treatment options. One approach involves performing a tracheostomy with a speaking valve. Alternatively, cord lateralizing procedures can be considered, such as transverse cordotomy, partial arytenoidectomy, injection of botulinum toxin, thyroplasty type II, or re-innervation procedures.

Transverse cordotomy (Kashima operation) involves excising laterally the soft tissue at the junction of the membranous cord and the vocal process of the arytenoid with laser.

Partial arytenoidectomy involves excising the medial part of the arytenoid with a laser.

Thyroplasty type II creates a lateral expansion of the larynx which is similar to vocal cord lateralization.

Reinnervation procedures innervate paralyzed posterior cricoarytenoid muscle by implanting a nerve-muscle pedicle of sternohyoid or omohyoid muscle with its nerve supply from ansa hypoglossi.

Solution to Question 3:

The clinical vignette of anosmia, foul-smelling discharge, and yellowish-green crusts in the nasal cavity points to a diagnosis of atrophic rhinitis. The other examination finding that is observed in atrophic rhinitis is a roomy nasal cavity.

Atrophic rhinitis is a chronic inflammation of the nose characterized by atrophy of nasal mucosa and turbinate bones. The exact etiology of this condition is not known.

The patients present with nasal obstruction, foul odor, merciful anosmia, greenish-grey crusting of the nasal passage, pale nasal mucosa, atrophied nasal turbinates, and epistaxis following the removal of the crusts. Roomy nasal cavity is seen due to the resorption of bones of the turbinates.

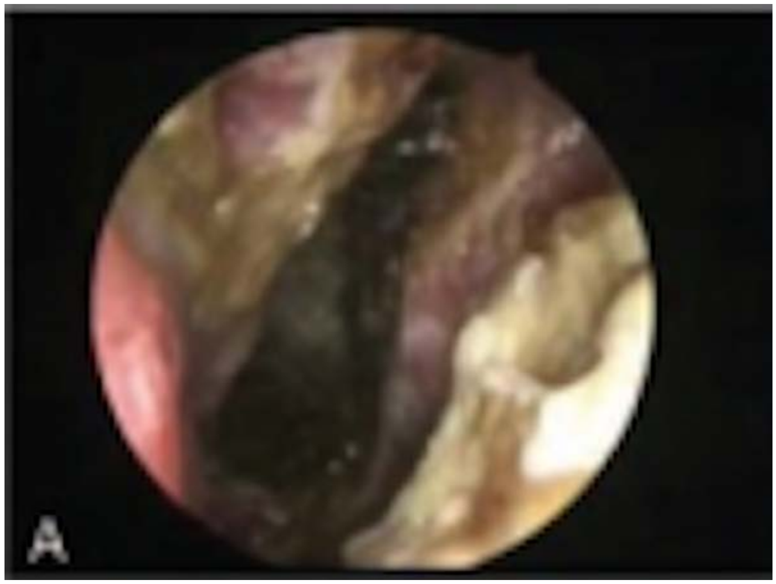
Treatment options for atrophic rhinitis include nasal irrigation using warm saline or an alkaline solution composed of sodium bicarbonate, sodium biborate, and sodium chloride (1:1:2) mixed with water.

Local antibiotics such as Kemicetine antiozaena solution, which contains chloromycetin, oestradiol, and vitamin D2, can be used. Topical applications of 25% glucose in glycerine, estradiol spray, and systemic administration of streptomycin, potassium iodide, or placental extract may also be employed.

Surgical management options include Young's operation, where flaps are raised within the nasal vestibule and both nostrils are completely closed, then opened after 6 months. During this time, the nasal mucosa may revert to a more normal state, and crusting is reduced. Another option is the modified Young's operation, where the nostrils are partially closed to avoid discomfort. The results of this modified approach are comparable to the original Young's operation.

Additionally, the nasal cavity can be narrowed through submucosal injection of teflon paste or by inserting fat, cartilage, or teflon strips beneath the mucoperiosteum.

The image below shows the endoscopic appearance of atrophic rhinitis:



Other options:

Option B: Nasal polyps are non-neoplastic masses of the edematous nasal or sinus mucosa. Nasal polyps can be antrochoanal polyps or ethmoidal polyps.

Option C: Inferior turbinate hypertrophy is associated with various mucosal diseases which include allergic rhinitis, infectious rhinitis, and vasomotor rhinitis. It can cause chronic nasal obstruction.

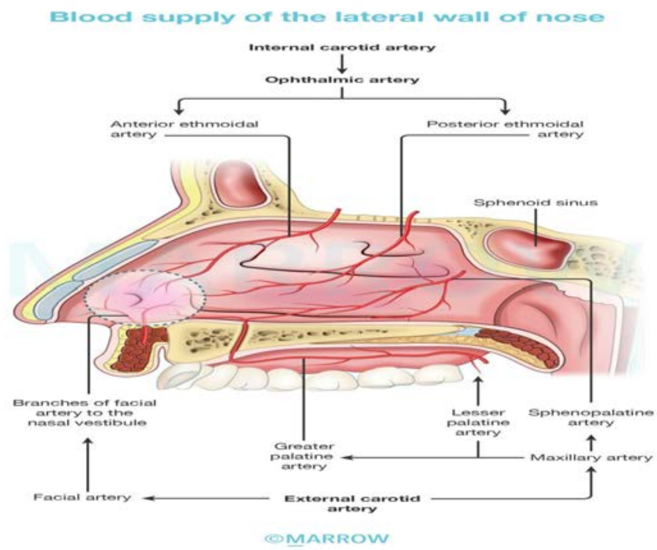
Option D: Foreign body nose is commonly seen in children and the child presents with unilateral, foul-smelling nasal discharge.

Solution to Question 4:

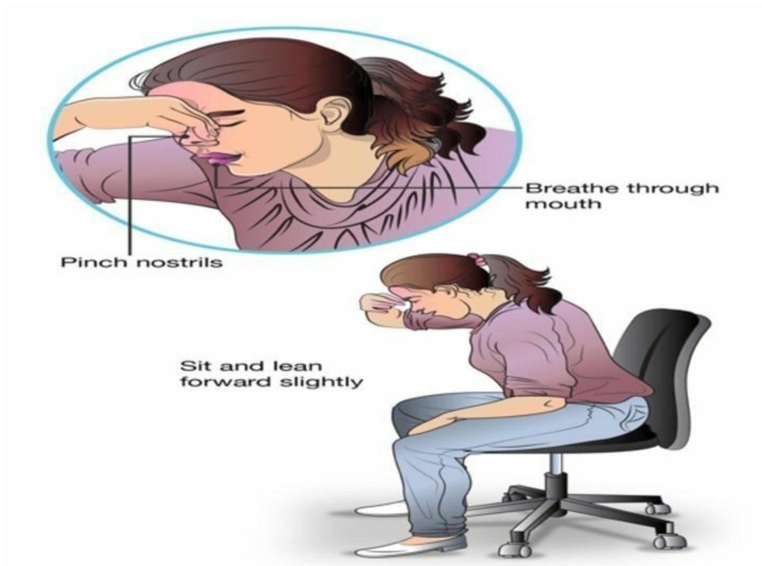
The next appropriate step for the management of epistaxis which was not controlled by pinching of nostrils and nasal packing would be ligation of the sphenopalatine artery, as it is the main artery contributing to Kiesselbach’s plexus.

Little's area is the most common site of epistaxis. It is present in the anteroinferior part of the septum. Four arteries anastomose here to form Kiesselbach’s plexus.

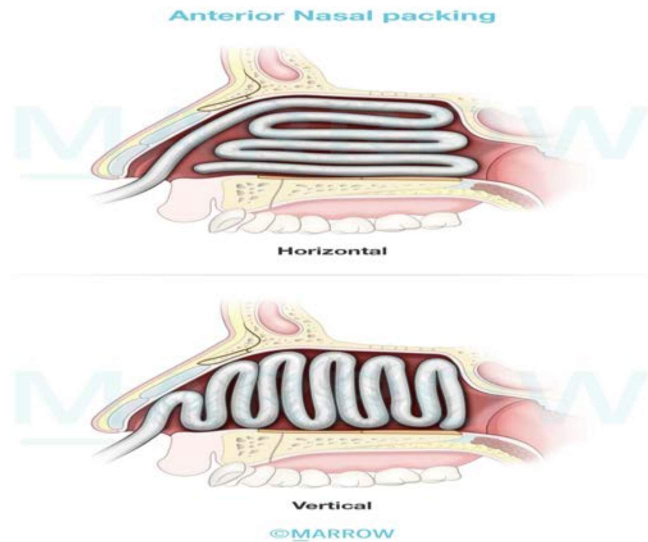
The image below shows the blood supply of the nasal septum and areas of epistaxis.

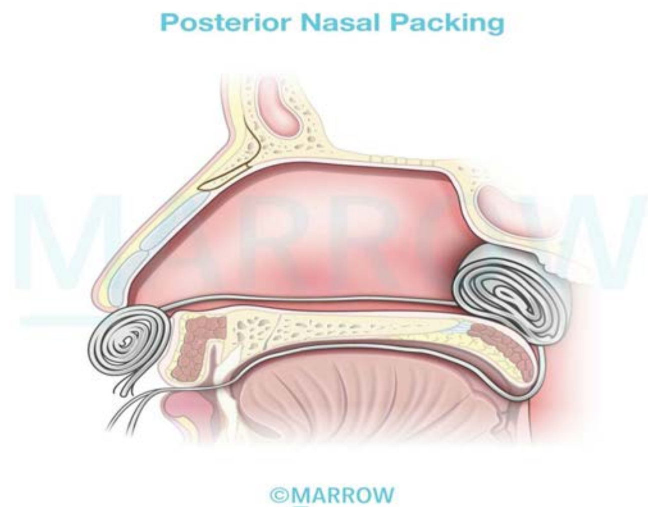


The first step in treating epistaxis is to pinch the nose with the thumb and index finger for about 5 minutes. The patient should sit leaning slightly forward over a basin to spit out blood and breathe through the mouth (Hippocratic or Trotter's method).



If the bleeding point is localized, cauterization can be done with electrocautery. In cases of uncontrolled bleeding or when the bleeding is posterior into the throat, anterior or posterior nasal packing may be required, respectively. If bleeding continues despite these measures, endoscopic procedures or vessel ligation close to the bleeding point can be considered.





The recommended hierarchy of ligation of vessels in a patient of epistaxis is the sphenopalatine artery first, the internal maxillary artery followed by the external carotid artery, and finally the anterior and posterior ethmoidal arteries.

The sphenopalatine artery is ligated first through a trans-nasal endoscope (transnasal endoscopic sphenopalatine artery ligation) at the sphenopalatine foramen which is situated at the posterior end of the middle turbinate.

However, if the bleeding persists, internal maxillary artery ligation is done at the pterygopalatine fossa via the Caldwell-Luc approach. If bleeding is still not controlled, external carotid artery ligation is done just above the origin of the superior thyroid artery.

If the bleeding site is located superior to the level of the middle turbinate, the anterior and posterior ethmoidal arteries should be ligated via Lynch incision in the medial wall of the orbit.

Solution to Question 5:

The given clinical picture suggests reactionary hemorrhage post-tonsillectomy. The immediate management is to shift the patient to OT, remove the clots, and cauterize/ligate the bleeding vessel.

Hemorrhage is a common complication of tonsillectomy. It can be primary, reactionary, or secondary.

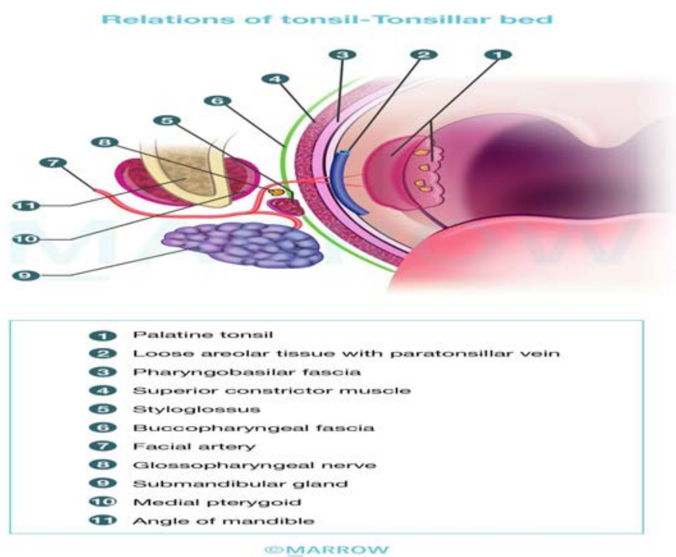
Primary hemorrhage occurs at the time of operation. It can be controlled by pressure, ligation, or electrocoagulation of the bleeding vessels.

Reactionary hemorrhage occurs within 24 hours. It may be due to improper clipping action of the superior constrictor muscle.

The tonsillar artery perforates the superior constrictor and ramifies in the tonsil. Therefore, the presence of a clot can hinder the clipping action of the superior constrictor muscle, leading to hemorrhage. Slippage of ligature, or blood pressure returning to normal are other causes.

It is managed by removing the clot and application of pressure or vasoconstrictor. Ligation or electrocoagulation of the bleeding vessels is done under anesthesia if the above measures fail.

The image below shows the relation between the superior constrictor muscle and the tonsillar artery at the tonsillar bed.



Secondary hemorrhage is usually seen between the 5th to 10th postoperative day, due to sepsis and premature separation of the membrane. Clots are removed, IV antibiotics are given and the bleeding vessel is electrocoagulated or ligated if bleeding continues.

Solution to Question 6:

The clinical vignette depicting features of CNS infection (fever, headache, vomiting, and confusion) in a patient with a chronic ear infection and the CT image showing a ring-enhancing lesion point to a diagnosis of a temporal lobe abscess.

Temporal lobe abscess is an intracranial complication that can arise from chronic suppurative otitis media (CSOM).

Brain abscess typically progresses through four stages. The first stage is the stage of invasion (initial encephalitis), which often goes unnoticed. Symptoms such as headache, malaise, low-grade fever, and drowsiness may be present.

The second stage is the stage of localization (latent abscess), during which a capsule forms around the abscess, effectively localizing the pus.

The third stage is the stage of enlargement (manifest abscess). The abscess enlarges, leading to increased intracranial pressure (ICP) and the development of edema surrounding the abscess. As a result, symptoms worsen.

The final stage is the stage of termination (rupture of abscess). At this point, the abscess expands and eventually ruptures into the ventricle or subarachnoid space, which can result in potentially fatal meningitis.

Signs of raised ICP may include headache, nausea, vomiting, lethargy, drowsiness, confusion, papilledema, and coma.

Characteristic findings associated with temporal lobe abscess include nominal aphasia, homonymous hemianopia, seizures, contralateral motor paralysis, pupillary changes, and oculomotor palsy.

The preferred diagnostic imaging modality for temporal lobe abscess is a CT scan of the brain. This imaging study reveals a ring-enhancing lesion indicative of a brain abscess.

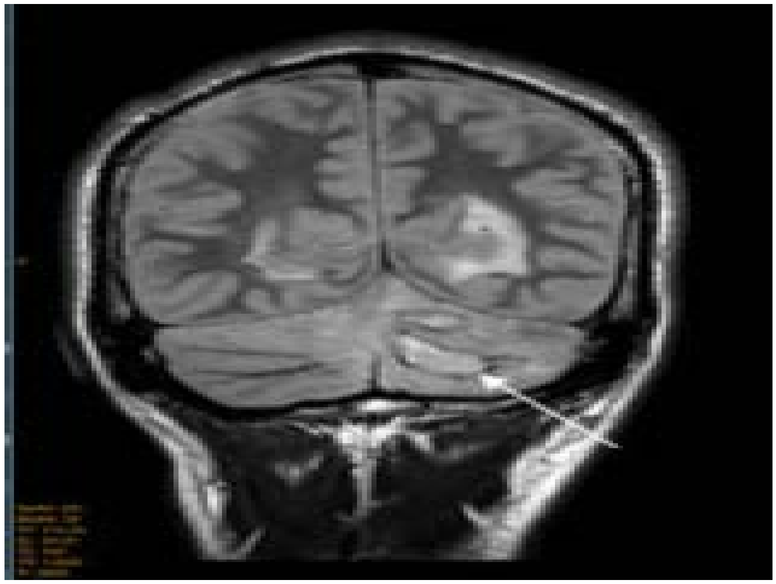
Medical treatment involves initiating intravenous antibiotics, typically metronidazole. Raised ICP can be addressed using dexamethasone or mannitol.

Surgical intervention is the definitive management and involves drainage of the abscess followed by radical mastoidectomy to address the underlying CSOM.

Other options:

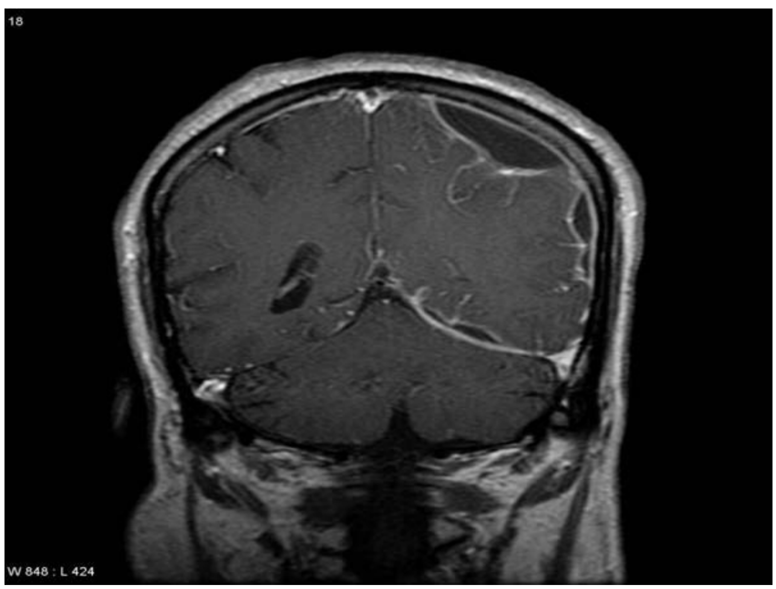
Option B: Cerebellar abscess is a form of brain abscess and is usually associated with an otogenic source of infection. It can occur as a direct extension through Trautmann's triangle or via retrograde thrombophlebitis.

An MRI image showing cerebellar abscess is given below:



Option C: Subdural abscess refers to the focal collection of purulent material between the dura mater and the arachnoid mater.

An MRI image showing subdural abscess is given below:



Option D: Meningitis is the inflammation of the leptomeninges which includes the arachnoid mater and pia mater. It can be due to an infectious or noninfectious etiology. It presents with fever, vomiting, neck stiffness, and signs of meningeal irritation. Ring enhancing lesion on neuroimaging is not a feature of meningitis.

ENT INICET 2022

Question 1:

CT scan of a patient with laryngeal carcinoma is shown below. What is the staging?

a) T1N0M0

b) T4N1M0

c) T2N1M0

d) T3N0M0

Question 2:

Arrange the following in the sequence of auditory pathway:

a) 1-2-3-4-5

b) 5-4-3-2-1

c) 2-1-3-4-5

d) 3-4-5-1-2

Question 3:

What is the grade of tympanic membrane retraction shown in the image?

a) Grade 1

b) Grade 2

c) Grade 3

d) Grade 4

Question 4:

A patient presents with vertigo associated with horizontal nystagmus. The slow component is towards the left. What is the most likely diagnosis?

a) Posterior canal BPPV

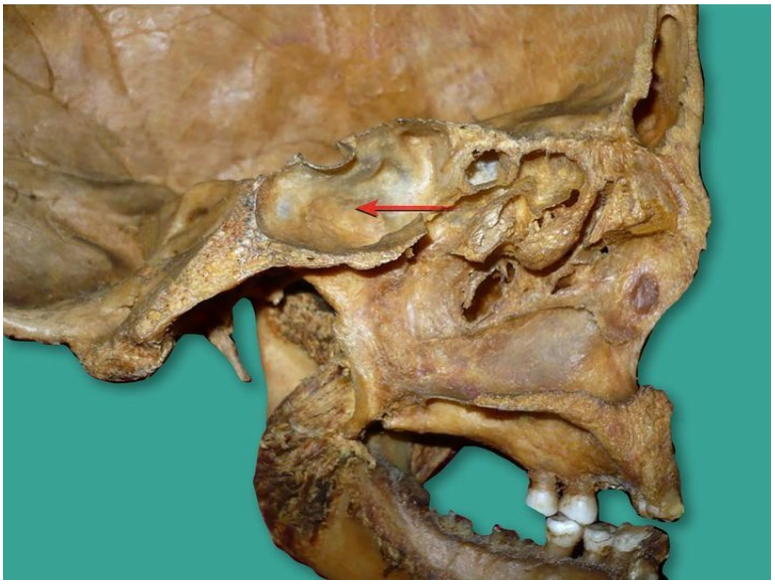
b) Superior canal BPPV

c) Right hypoactive labyrinth

d) Left hypoactive labyrinth

Question 5:

What is the site of drainage of the marked structure?



a) Superior meatus

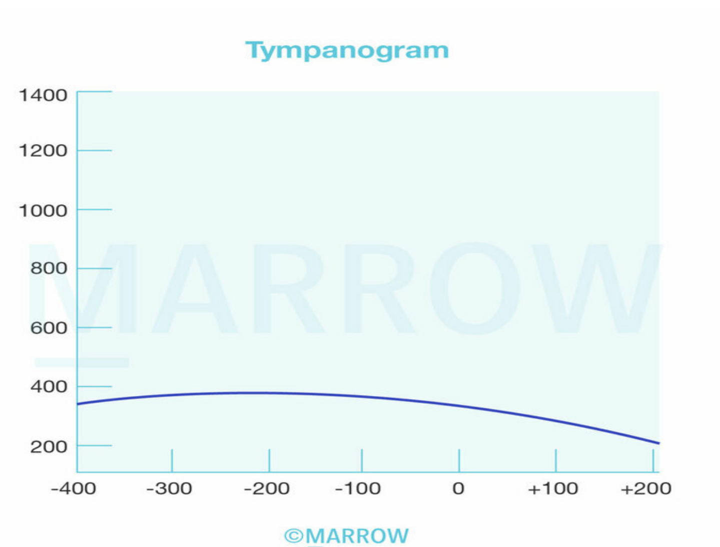
b) Middle meatus

c) Inferior meatus

d) Sphenoethmoidal recess

Question 6:

A 6-year-old boy presented with recurrent URTI, poor growth, high-arched palate, and impaired hearing. Tympanogram is given as follows. What would be the most appropriate management?



a) Grommet insertion

b) Adenoidectomy with grommet insertion

c) Myringotomy with grommet insertion

d) Myringotomy

Question 7:

A gentleman presents to the OPD with a saddle nose deformity. He gives a past history of cough, on/off fever, and hemoptysis. On examination, septal perforation is present with pale granuloma. His chest x-ray showed multiple cavitary lesions, and the biopsy showed granuloma, multinucleate giant cells, and caseous necrosis. What would be the probable cause?

a) Wegener's granulomatosis

b) Tuberculosis

c) Syphilis

d) Sarcoidosis

Question 8:

Foreign body entry into the laryngeal inlet is prevented by the cough reflex. This reflex is sluggish in alcohol ingestion and some other conditions. Which nerve is likely injured?

a) Internal laryngeal nerve

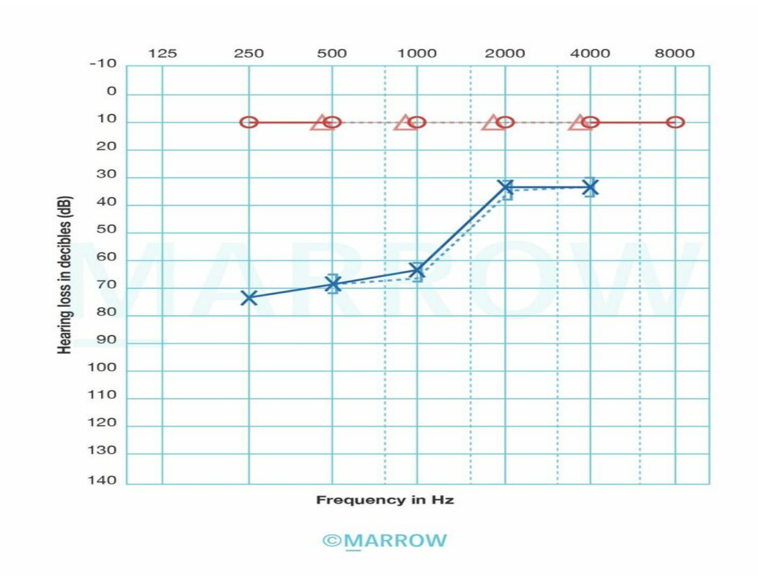
b) Glossopharyngeal nerve

c) External laryngeal nerve

d) Recurrent laryngeal nerve

Question 9:

A woman presents with impaired hearing. Her tympanogram findings are given below. Pick the right combination of tuning fork test results that would be seen in this patient.



a) Left Rinne's test negative, Weber's test lateralized to right ear

b) Left Rinne's test negative, Weber's test lateralized to left ear

c) Left Rinne's test positive, Weber's test lateralized to right ear

d) Left Rinne's test positive, Weber's test lateralized to left ear

Question 10:

A 35-year-old female patient presents with complaints of nasal obstruction and post-nasal drip. There is a past history of FESS for failed conservative management 5 years ago. Uncinectomy and maxillary ostium dilation was done during the previous FESS. A DNE done now shows patent ostia and mucosal edema of the maxillary sinus lining. What is the next best step in management?

a) Repeat surgery

b) Macrolides for 4 months

c) Steroid irrigation and antihistamines

d) Biological therapy

Question 11:

Which of the following tests cannot be used to assess nasal mucociliary clearance?

a) Saccharine test

b) Electron microscopy

c) Charcoal powder method

d) Scintigraphy

Question 12:

Match the following.

a) a-1, b-2, c-3, d-4

b) a-2, b-3, c-4, d-1

c) a-2, b-4, c-3, d-1

d) a-1, b-3, c-4, d-2

Question 13:

What test is not used for testing olfaction?

a) Smell diskettes

b) Arnold stick test

c) UPSIT

d) CC-SIT

Question 14:

Which of the following tests can be used if otoacoustic emissions is absent for screening?

a) 2 and 4

b) 1, 3, 4

c) 2, 3, 4

d) 2 and 3

Answer Key

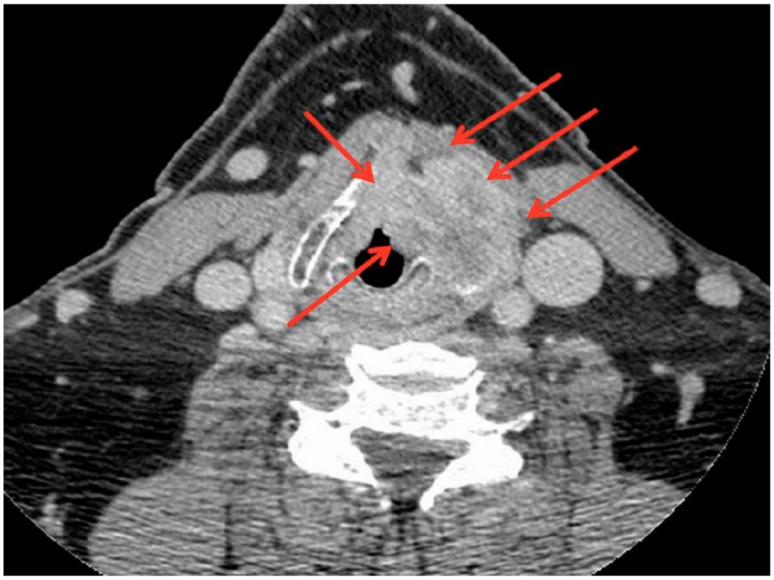
Question No.	Correct Option
1	b
2	c
3	b

4	d
5	d
6	b
7	b
8	a
9	c
10	c
11	b
12	c
13	b
14	a

Detailed Explanations

Solution to Question 1:

The given CT image depicts a mass invading the thyroid cartilage as pointed by the red arrows. This indicates a tumor (T) stage of T4. Among the given options, T4N1M0 is the only option with a T4 staging and hence the best answer.



Laryngeal cancer is more commonly seen in men. Alcohol, tobacco, and previous radiation to the neck region are some of the risk factors for the development of laryngeal cancer. The most common type is squamous cell carcinoma.

Larynx is divided into three regions - supraglottis, glottis, and subglottis. In the TNM system: T indicates the extent of the tumor, N indicates lymph node involvement, and M indicates distant metastasis.

TNM classification of laryngeal carcinoma:

Tumor (T)

• T1: Only one site/subsite is involved (one named structure) with both vocal cords mobile.

• T1a: Only one vocal cord involved

• T1b: Involvement of both vocal cords

• T2: Spread to adjacent sites/subsites

• T2a: Both vocal cords are mobile

• T2b: Slight impairment in the mobility of vocal cord

• T3

• Fixed vocal cord

• Involvement of pre-epiglottic space, para-glottic space, post-cricoid space, or inner cortex of thyroid

• T4a: Local invasion.

• Anteriorly: Thyroid cartilage, thyroid gland, strap muscles

• Superiorly: Tongue muscles

• Inferiorly: Trachea

• Posteriorly: Esophagus

• T4b: Unresectable

• Involvement of prevertebral space, carotid sheath, or mediastinum

Regional lymph nodes (N)

• Nx: Cannot be assessed

• N0: No regional lymph node metastasis

• N1: Metastasis in a single ipsilateral lymph node, ≤3cm

• N2: Metastasis in a single ipsilateral lymph node, >3cm but <6cm or multiple ipsilateral lymph nodes or bilateral lymph nodes none more than 6cm

• N3: Metastasis in a lymph node more than 6cm

Distant metastasis (M)

• Mx: Metastasis cannot be assessed

• M0: No distant metastasis

• M1: Distant metastasis

Solution to Question 2:

The correct sequence of the auditory pathway is spiral ganglion (2) - cochlear nucleus (1) - superior olivary nucleus (3) - inferior colliculus (4) - medial geniculate body (5).

The spiral ganglion contains bipolar cells whose dendrites form the afferent supply for inner and outer hair cells of the organ of Corti and whose axons form the cochlear division of CN VIII that end in the cochlear nuclei.

The auditory pathway can be remembered with the mnemonic E.C.O.L.I.M.A

Eighth nerve

Cochlear nuclei

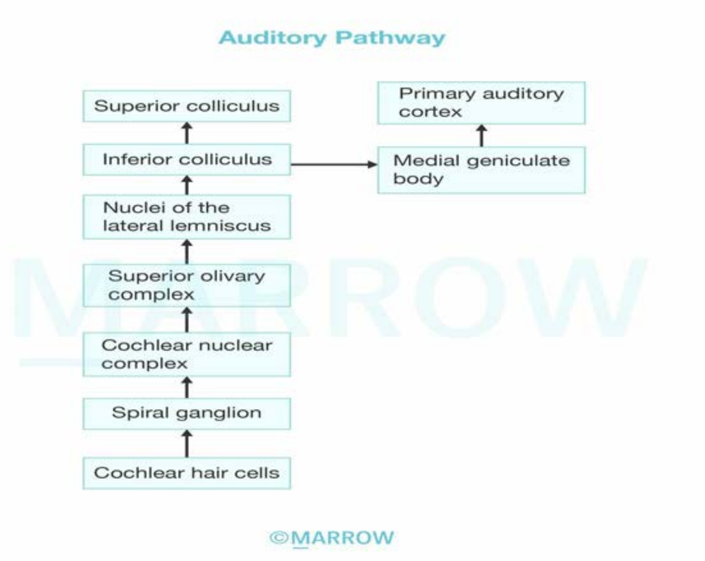
Olivary complex

Lateral lemniscus

Inferior colliculus

Medial geniculate body

Auditory cortex



Solution to Question 3:

The given image shows retraction in both pars tensa and pars flaccida. It is grade 2 retraction as the pars flaccida retraction is in contact with the neck of malleus while that in parts tensa is in contact with the incus.

Retraction pockets on tympanic membrane

Tympanic membrane can be divided into two parts. Pars tensa makes up most of the tympanic membrane. It is peripherally thickened to form annulus tympanicus (a fibrocartilaginous ring). It contains the cone of light. Pars flaccida (Shrapnell's membrane), is slightly pinkish and is located above the lateral process of malleus.

Tympanic membrane retraction can be either in the pars tensa or pars flaccida.

Tos classification is used to grade the retraction in pars flaccida or attic retraction

• Stage 1: The par flaccida is dimpled but not in touch with the malleus.

• Stage 2: The tympanic membrane is in contact with the neck of malleus and the entire extent of retraction can be visualized.

• Stage 3: The retraction is not visible fully, and there may be partial erosion of bony attic wall.

• Stage 4: The retraction is definite erosion of the bony attic wall and the full extent of retraction is not visible.

Due to difficulty in distinguishing the latter two, they are grouped as stage 3/4.

Sade classification is used to grade the retraction in pars tensa

• Stage 1: The tympanic membrane is retracted but not in touch with the incus

• Stage 2: The tympanic membrane is in contact with the long process of the incus

• Stage 3: Middle ear atelectasis - the tympanic membrane is in contact with the promontory, but not adherent. It can be raised from the promontory with suction

• Stage 4: Adhesive otitis media - the tympanic membrane is adherent to the promontory

Solution to Question 4:

The most likely diagnosis for a patient presenting with horizontal nystagmus and fast component towards the right (slow component towards the left) is left hypoactive labyrinth.

Nystagmus is an involuntary oscillatory movement of the eyes, which can occur in horizontal, vertical, or torsional directions. The direction of nystagmus is determined by the fast component.

In horizontal nystagmus, the direction of nystagmus (fast phase) is always away from the hypoactive lesions and towards the active or hyperactive lesions of the labyrinth. Benign paroxysmal positional vertigo (BPPV) is a hyperactive disease while vestibular neuronitis and vestibular Schwanomma are hypoactive diseases.

The involvement of the vertical semicircular canals (superior and posterior) leads to vertical or mixed vertical-torsional nystagmus, ruling out posterior canal BPPV (Option A) and superior canal BPPV (Option B).

Alternatively, if the right hypoactive labyrinth was one of the options, it could be considered as a differential for horizontal nystagmus with the slow component towards the right.

BPPV is a condition characterized by recurrent episodes of transient vertigo (<1 min) triggered by changes in head position relative to gravity. It is caused by dislodged calcium carbonate crystals (otoconia) from the utricular macula into one of the semicircular canals, usually the posterior semicircular canal. A diagnosis of BPPV can be made by Dix-Hallpike test and the condition is treated by repositioning maneuvers for the otoconia such as the Epley maneuver.

Solution to Question 5:

The marked structure in the given image is the sphenoid sinus and it drains into the sphenoethmoidal recess.

Paranasal sinuses are air-containing cavities that are present in the skull bones. These paranasal sinuses have been divided into two groups:

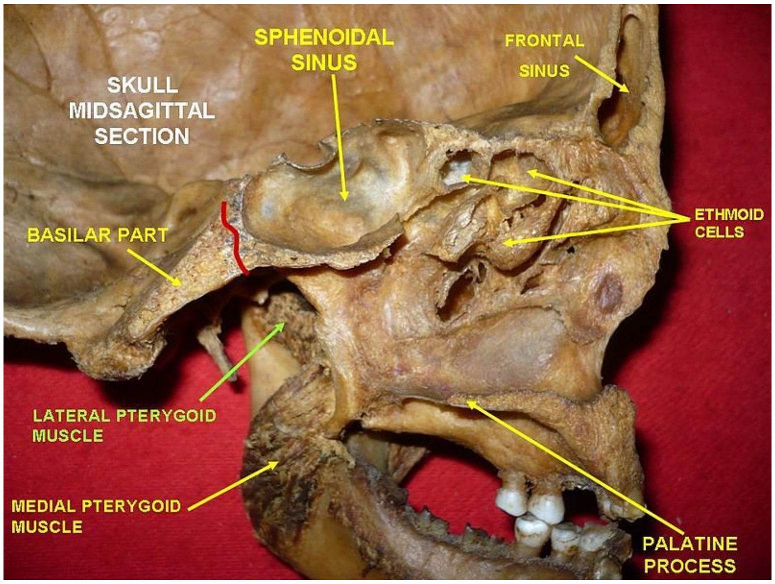
Anterior group: Their ostia drain into the middle meatus (Option B) through a common drainage pathway known as osteomeatal unit. This group includes:

- Maxillary sinus (antrum of Highmore)
- Frontal sinus
- Anterior ethmoidal sinus

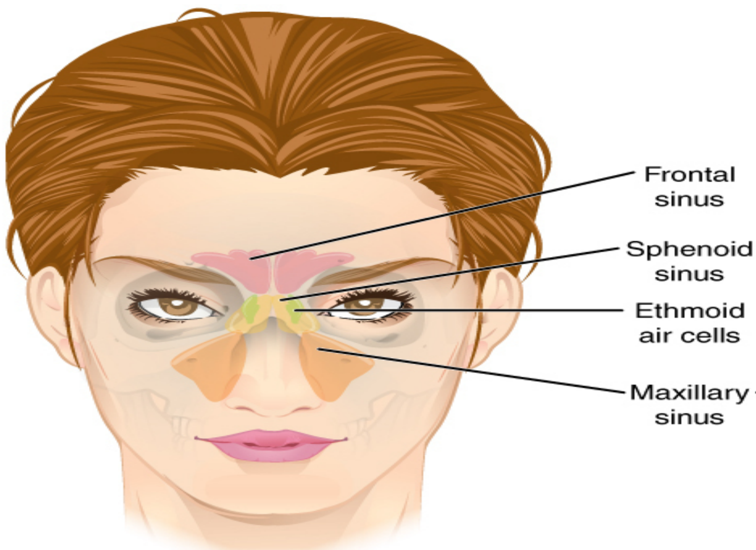
Posterior group:

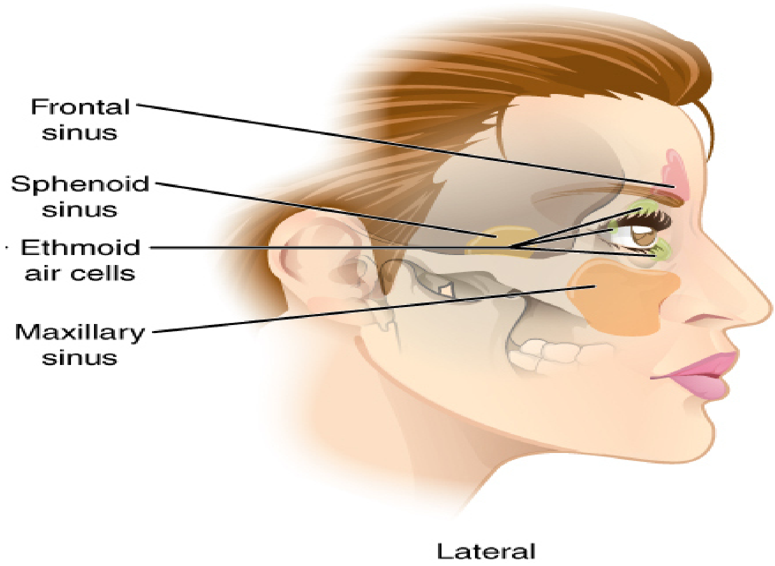
- Posterior ethmoidal sinuses: Opens in the superior meatus (Option A)
- Sphenoid sinus: Opens in sphenoethmoidal recess

The inferior meatus (Option C) has the Nasolacrimal duct draining into it with its terminal end being guarded by the Hasner's valve.



The images below show the location of paranasal sinuses:





Solution to Question 6:

The given clinical scenario along with type B tympanometry curve is suggestive of serous otitis media secondary to adenoid hypertrophy. Adenoidectomy with grommet insertion would be the most appropriate management.

Enlarged adenoids in children can present as nasal obstruction, rhinolalia clausa, recurrent acute otitis media, and serous otitis media. Adenoid facies, characterized by an elongated face, open mouth, teeth crowding, high arched hard palate, and pinched-in nose, may also be present.

Serous otitis media also known as secretory otitis media, mucoid otitis media, or glue ear is characterized by the accumulation of nonpurulent, serous effusion in the middle ear cleft.

Glue ear can occur either due to eustachian tube dysfunction (adenoid hyperplasia, chronic sinusitis or tonsillitis or rhinitis, cleft palate, tumors of nasopharynx) or due to increased secretory activity of the middle ear mucosa (allergy, unresolved otitis media, viral infections).

Glue ear is commonly seen in children aged 5-8 years and is associated with insidious onset of hearing loss (not exceeding 40 dB), mild ear aches, and a history of upper respiratory tract infections. Delayed or defective speech may also result from hearing loss.

Otoscopy shows:

- Dull and opaque tympanic membrane with loss of light reflex
- Thin leash of vessels along the handle of the malleus
- Fluid levels and air bubbles can be seen when the tympanic membrane is transparent.

The otoscopy image of serous otitis media is shown below:

The tuning fork test reveals a conductive type of hearing loss and audiometry reveals hearing loss in the range of 20–40 dB. The fluid accumulation in glue ear causes decreased mobility (compliance) of the tympanic membrane leading to a flat or dome-shaped curve or type B curve on a tympanogram.

Treatment is mainly aimed at the removal of fluid and preventing recurrence.

• Medical management includes decongestants, antihistamines, and antibiotics.

• Surgical management is done if the condition cannot be controlled with medications.
It includes myringotomy and aspiration of fluid with grommet insertion. Adenoidectomy and tonsillectomy may be required and are usually done at the time of myringotomy.

Solution to Question 7:

In a patient with saddle nose deformity, septal perforation with pale granuloma and caseous necrosis on biopsy is consistent with tuberculosis (TB). The history of fever, cough, and hemoptysis along with cavitary lesions on chest x-ray support the diagnosis of TB.

While other conditions like Wegener's granulomatosis (Option A), syphilis (Option C), and sarcoidosis (Option D) can also cause saddle nose deformity, caseous necrosis on biopsy makes TB a more likely diagnosis.

Saddle-nose deformity refers to a depressed nasal dorsum that can involve either the bony, cartilaginous, or both components.

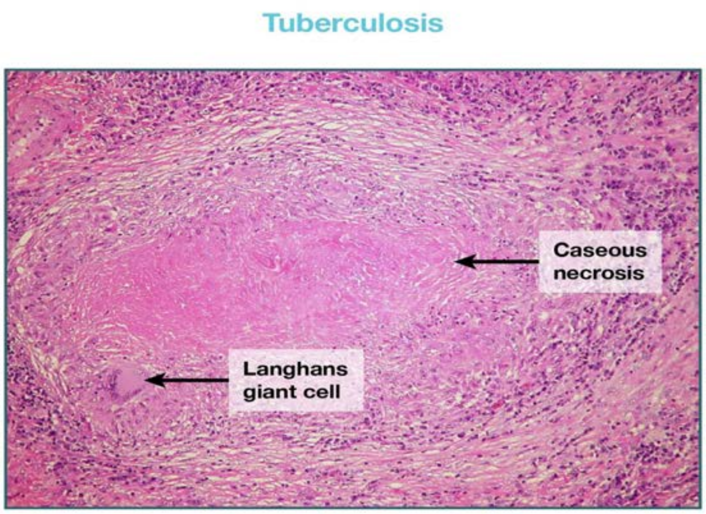
The most common cause is nasal trauma resulting in depressed fractures. Other causes include excessive submucosal resection, destruction of septal cartilage due to hematoma or abscess, and infections such as leprosy, tuberculosis, or syphilis, as well as inflammatory conditions like granulomatosis with polyangiitis (Wegener granulomatosis) and sarcoidosis.

Chronic granulomatous conditions such as lupus, tuberculosis, and leprosy can affect the cartilaginous part, syphilis primarily affects the bony part, and Wegener's granulomatosis and T-cell lymphoma can lead to total (bony and cartilaginous) septal destruction involving both the bony and cartilaginous portions.

TB is an infectious disease caused by Mycobacterium tuberculosis and primarily affects the lungs but can also involve the nose and sinuses.

Common symptoms of TB include fever, productive cough, weight loss, night sweats, and hemoptysis. Chest x-ray often shows multiple cavitation lesions affecting the upper lobe and hilar lymphadenopathy. Biopsy findings in tuberculosis typically reveal granulomas, caseous necrosis, and multinucleate giant cells (Langhan's giant cells).

The following images show the histopathological image of tuberculosis:



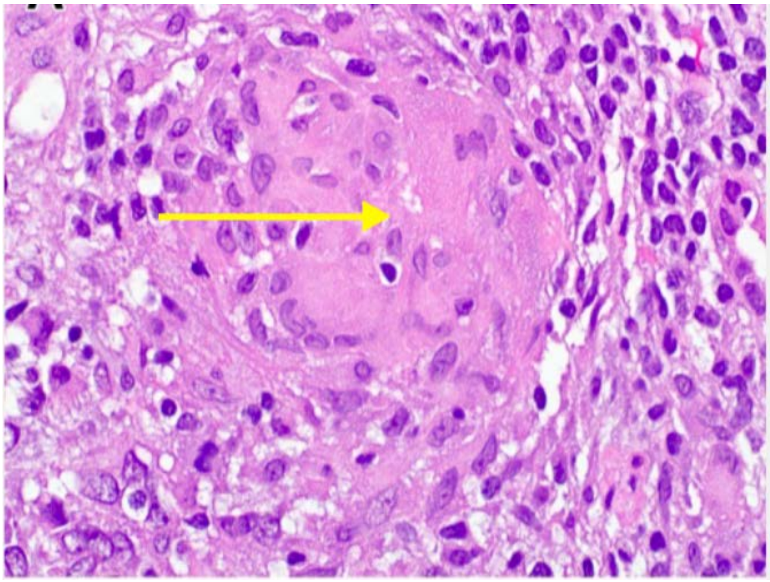
Other options:

Option A: Wegener's granulomatosis is characterized by vasculitis affecting small to medium-sized vessels. It affects the upper respiratory tract (ear, nose, sinuses, and throat), lower respiratory tract (lung), and kidney and is characterized by necrotizing granuloma which is non-caseating.

Option C: Although nose deformity is seen in syphilis, gummatous lesions (caseating granulomas) are present in tertiary syphilis which is not seen in this patient. Fever, cough, and hemoptysis are not features of syphilis.

Option D: Sarcoidosis most commonly affects the lungs forming non-caseating granuloma with no central necrosis. Laminated calcium and protein-containing bodies called Schaumann bodies and star-shaped inclusions called asteroid bodies are also found in the giant cells of most of these granulomas. Chest x-ray would show interstitial fibrosis and hilar lymphadenopathy.

The image below shows the characteristic non-caseating sarcoid granuloma:



Solution to Question 8:

The internal laryngeal nerve is the nerve implicated in the inhibition of the cough reflex in an alcoholic. Alcohol can cause chemical irritation of the nerve and it is also a potent inhibitor of the central nervous system.

The cough reflex is a protective mechanism of the larynx to expel irritants and keep the airway clear. Chemical or mechanical irritation of the respiratory mucosa stimulates the afferent internal laryngeal nerve, which is the branch of the superior laryngeal nerve of the vagus nerve.

Internal laryngeal nerve supplies the mucosa of the larynx above the level of the vocal cords (supraglottis). Hence, any injury to the internal laryngeal nerve results in aspiration and depressed cough reflex with no change in voice.

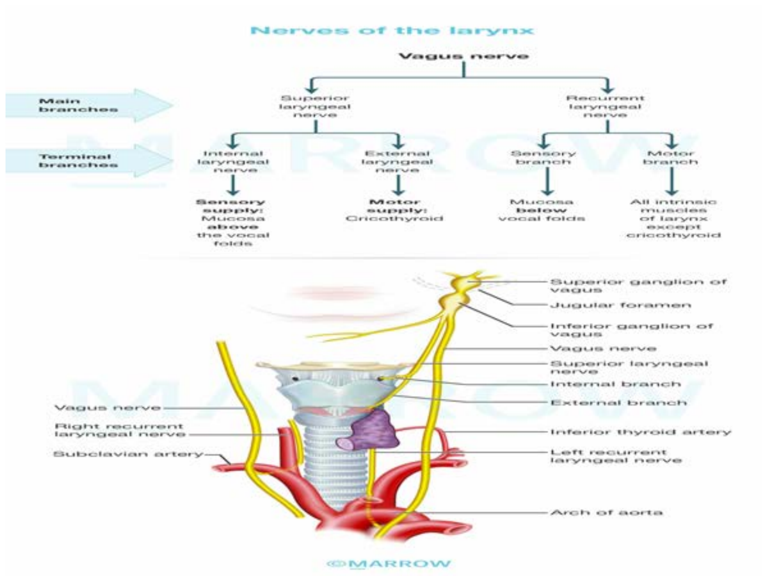
Recurrent laryngeal nerve (Option D) gives sensory supply to the mucosa of the larynx at the level of vocal cords (glottis) and below the level of vocal cords (subglottis) and also motor supply to all the intrinsic muscles of the larynx except cricothyroid.

Unilateral injury to the recurrent laryngeal nerve (RLN) may cause minimal or no change in voice, but it does not lead to aspiration or airway obstruction.

Bilateral injury to RLN can result in dyspnea and stridor due to the median or paramedian position of the vocal cords, which narrows and obstructs the airway. However, the voice remains unaffected.

The glossopharyngeal nerve (Option B) provides sensory supply to the oropharynx, posterior third of the tongue, and middle ear cavity. Damage to the glossopharyngeal nerve leads to the absence of the gag reflex.

The external laryngeal nerve (Option C) innervates the cricothyroid muscle, which is responsible for adjusting the pitch of sound during speech. Paralysis of the external laryngeal nerve results in the inability to increase the pitch of sound.



Solution to Question 9:

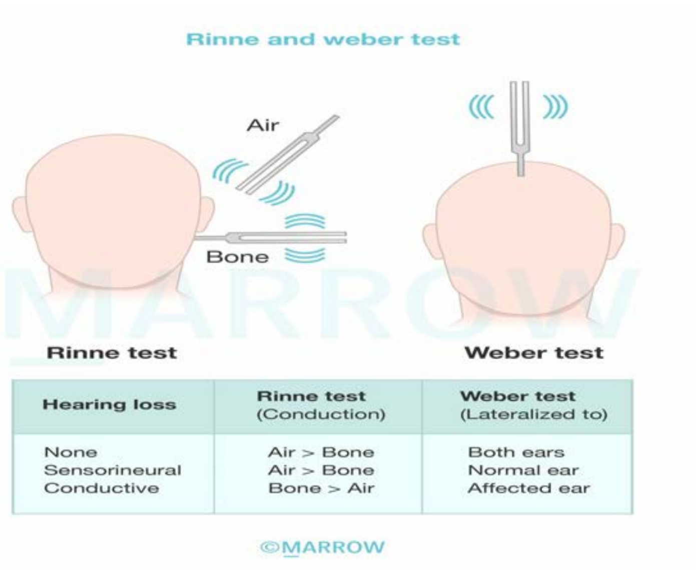
The given tympanogram findings indicate sensorineural hearing loss in the left ear. Tuning fork tests in this patient would show left Rinnie's positive and Weber lateralized to the right ear.

In left sensorineural hearing loss, Rinne's test will be positive on both left and right ears, and Weber's test will be lateralized to the better ear, which is the right.

Tuning fork tests are performed commonly with tuning forks of frequencies such as 128, 256, 512, 1024, etc. The tuning fork of 512 Hz is considered ideal as forks of lower frequencies produce a sense of bone vibration, while those of higher frequencies have a shorter decay time.

Rinne’s test is done by placing a vibrating tuning fork on the patient’s mastoid and when he/she stops hearing, it is brought beside the external auditory meatus (EAC). If he/she still hears the sound, air conduction (AC) is better than bone conduction (BC).

Weber test is done by placing a vibrating tuning in the middle of the forehead or the vertex and the patient is asked in which ear the sound is better heard.



Solution to Question 10:

The history of nasal obstruction and post-nasal drip and endoscopy showing mucosal edema is suggestive of acute rhinosinusitis. The next best step in management is symptomatic treatment with nasal steroid irrigation, decongestants, and antihistamines, regardless of the patient's past history of FESS operation.

Acute viral rhinosinusitis causes mucosal edema which causes obstruction of the sinus ostia. There is also reduced mucociliary transport in the sinuses resulting in stasis of secretions within the sinuses causing headache, facial pain, congestion, nasal obstruction or discharge, post-nasal drip, and hyposmia or anosmia.

If the sinuses remain obstructed or the mucociliary transport system does not return to normal, a bacterial infection can ensue, causing fever, purulent discharge, and persistent or worsening symptoms beyond 10 days.

In the case of bacterial sinusitis, antibiotics like amoxicillin and macrolides (erythromycin) are used for a short course (Option B).

Procedures like antral lavage and surgery (Option A) are done in cases not responding to medical therapy.

Biological therapy (Option D) is not used in the treatment of sinusitis.

Solution to Question 11:

Among the given options, electron microscopy will be the least useful in assessing nasal mucociliary clearance. Other tests mentioned can be used in humans to detect mucociliary transit time and total nasal clearance in humans.

Electron microscopy can be used to best assess the normalcy of cilia structure and its function, but it might not be able to detect any other abnormality that might interfere with nasal mucociliary clearance like any obstruction in the nasal cavity.

Test for mucociliary clearance include:

• Saccharine with or without color test in vivo (Option A)

• Radioisotope transport testing in vivo

• Measuring ciliary activity in vitro

• Nasal nitric oxide

• Scintigraphy (Option D) is the nuclear medicine equivalent of the dye disappearance test to assess functional and anatomic obstruction

The charcoal method (Option C) is a less commonly used method to detect nasal mucociliary clearance. In this method, a very small amount of charcoal is placed behind the head of the inferior turbinate. The pharynx is then observed to detect the presence of charcoal.

Mucociliary clearance tests are used to diagnose inborn disorders of mucociliary transport such as primary ciliary dysfunction and cystic fibrosis.

Solution to Question 12:

The correct match for Wullstein's classification is a-2, b-4, c-3, d-1 (Option C).

Solution to Question 13:

Arnold stick test is not a test used for olfaction.

Tests used for olfaction are as follows:

• Smell diskettes are used as a screening test for olfaction. Reusable diskettes are used with the application of 8 different odorants. A questionnaire with illustrations is used to score the patient.

• The University of Pennsylvania smell identification test (UPSIT) is a commercially available, self-administered test used to assess the function of an individual's olfactory system via smell identification. It uses microencapsulated odorants, which are released on scratching standardized odor-impregnated test booklets.

• The 12-item version of UPSIT, also known as the cross-cultural smell identification test (CC-SIT), is useful for quantifying smell loss when less time is available. This test serves as a means to reliably assess olfactory function in less than 5 minutes.

• The Sniffin' sticks test is a new test of nasal chemosensory performance based on pen-like odor-dispensing devices.

Solution to Question 14:

Among the given options, tympanometry (Test 2) and speech audiometry (Test 4) are used in cases where otoacoustic emission (OAE) is absent for screening.

Otoacoustic emissions are sounds produced by the outer hair cells of cochlea. These sounds then travel in the reverse direction from the outer hair — basilar membrane and perilymph of cochlea— oval window — ossicles and tympanic membrane (middle ear) — ear canal (external ear).

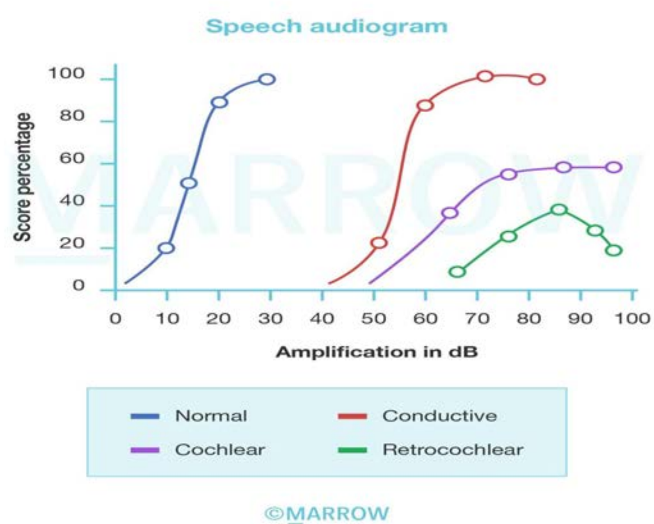
Hence in the absence of OAE, pathologies of the middle and external ear should be ruled out before making a definitive diagnosis of cochlear pathology.

Tympanometry is an objective test that can be performed in patients with absent OAE to rule out any middle ear pathologies. Speech audiometry is a subjective test that can be performed in these patients to confirm if the hearing loss is due to cochlear or retro cochlear pathology (neural hearing loss).

Impedance Audiometry also known as Tympanometry is an objective test that is based on the principle that when the sound strikes the tympanic membrane, certain amount of sound is absorbed, while the rest is reflected So a stiffer tympano-ossicular system would reflect greater sound than a compliant one. By marking the compliance of tympano-ossicular system against various pressure changes, different types of tympanograms are obtained which are diagnostic of various middle ear pathologies.

In speech audiometry, the patient's ability to hear and understand speech is measured. The speech reception threshold (SRT) is the lowest intensity of sound at which 50% of the words are repeated correctly by the patient. Speech discrimination scores test the capacity of the patient to understand speech. Hence, speech audiometry is a subjective test.

The rollover phenomenon is diagnostic of retrochochlear pathologies. Here, the phonetically balanced word score falls with an increase in speech intensity above a particular level due to nerve fatigue. The graph in cochlear disorders will rather show a plateau after a certain level unlike the fall seen in the rollover phenomenon.



A better objective test that could be used to diagnose retro-cochlear pathologies is brainstem auditory evoked response audiometry (BERA). It is a noninvasive test that can find any pathologies of central auditory pathways through the VIIIth nerve to pons and midbrain.

Other options:

1. Pure tone audiometry (Test 1) is a test that measures the threshold of hearing by air and bone conduction and thus the degree and type of hearing loss. Pure tone audiometry is a subjective test for hearing and is not employed in cases where OAE is absent when better objective tests such as tympanometry are available.

3. Free field audiometry (Test 3), a part of behavior observational audiometry is used in the age group where children start localizing sounds when an auditory signal is presented. This test is not useful in this case as the possibility of hearing loss is already established with absent OAE.

ENT NEET 2022

Question 1:

A post-COVID patient, who is a known diabetic, develops unilateral facial pain and loosening of teeth. Which investigation would you do to confirm the diagnosis on this patient?

a) MRI

b) Biopsy with histopathologic examination

c) Serum ferritin

d) HbA1c

Question 2:

A patient presents with the complaint of inability to close the eye, drooling of saliva, and deviation of the angle of the mouth. Which of the following nerves is most likely to be affected?

a) Facial nerve

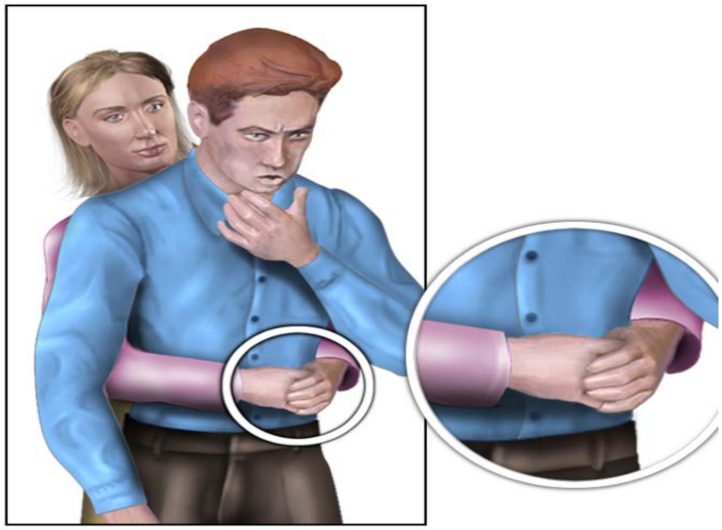
b) Trigeminal nerve

c) Occulomotor nerve

d) Glossopharyngeal nerve

Question 3:

An adult man in a restaurant suddenly begins choking on his food. He is conscious. The following procedure was performed. Identify the procedure.



a) Heimlich's maneuver

b) Back slap

c) Chest thrust

d) Blind insertion of finger

Question 4:

A patient presents with a firm, tender, slow-growing mass below the ear as shown in the image below. What could be the diagnosis?

a) Bezold abscess

b) Parotid abscess

c) Upper cervical lymphadenopathy

d) Osteoma of the mandible

Question 5:

A 5-year-old child presents with reduced hearing for the past 2-3 months. The otoscopy finding is given below. What is the most likely diagnosis?

a) Myringitis bullosa

b) Serous otitis media

c) Acute otitis media

d) Pneumotympanum

Question 6:

A patient comes with a history of asthma and sinusitis. On looking into his medical records, you notice this has been attributed to Samter's triad. Which drug should be avoided in this patient?

a) Cotrimoxazole

b) Co-amoxiclav

c) Aspirin

d) Chloramphenicol

Question 7:

A 70-year-old male patient presents with decreased hearing in higher frequencies. It was noted that the basilar membrane was affected. Which of the following structures lie near the affected structure?

- **a)** Modiolus
- **b)** Stria vascularis
- **c)** Oval window
- **d)** Helicotrema

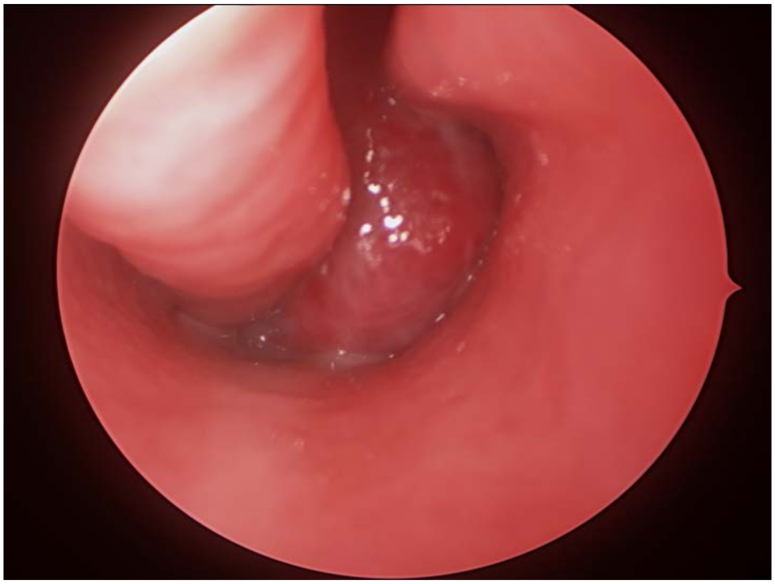
Question 8:

Identify the structure given in the image.

- **a)** Malleus
- **b)** Incus
- **c)** Stapes
- **d)** Vomer

Question 9:

A 20-year-old male patient presents with unilateral nasal obstruction and recurrent bleeding for the past 1 year. Transnasal endoscopic results are shown below. A contrast-enhanced CT revealed a mass extending from the posterior choana to the nasopharynx. What is the most likely diagnosis?



a) Nasopharyngeal angiofibroma

b) Antrochoanal polyp

c) Rhinoscleroma

d) Concha bullosa

Answer Key

Question No.	Correct Option
1	b
2	a
3	a
4	b
5	b
6	c
7	c
8	b
9	a

Detailed Explanations

Solution to Question 1:

The given scenario of a post-COVID diabetic patient with facial pain and loosening of teeth is suggestive of rhino-orbital-cerebral mucormycosis. Among the given options, Biopsy with histopathologic examination would be the investigation for definitive diagnosis on this patient.

Rhino-orbital-cerebral mucormycosis is the most common form of mucormycosis. It commonly occurs in patients with diabetes mellitus.

The presenting features are facial pain, numbness, loosening of teeth, conjunctival swelling, congestion, and blurry vision. When the disease spreads to the orbit, proptosis can be present. It can later spread to the frontal lobe of the brain and cavernous sinus.

A definitive diagnosis is done with a biopsy and histopathologic examination. It is the most sensitive and specific method. It reveals aseptate fungi with thick-walled, ribbon-like, hyphae branching at right angles.

Among the imaging techniques used for diagnosis, MRI is more sensitive to detecting orbital and nervous involvement than CT. Contrast-enhanced MRI is the imaging technique of choice. 1,3 beta D- glucan assay, which is helpful for the identification of most fungi, cannot be used for the diagnosis of mucormycosis.

Management is by amphotericin B and surgical debridement of the affected tissues. The underlying cause and risk factors should also be addressed.

Solution to Question 2:

The given clinical features of inability to close the eye, drooling of saliva, and deviation of the angle of the mouth are suggestive of paralysis of the facial nerve.

Facial nerve paralysis can be central or peripheral. Peripheral is more common and is usually idiopathic.

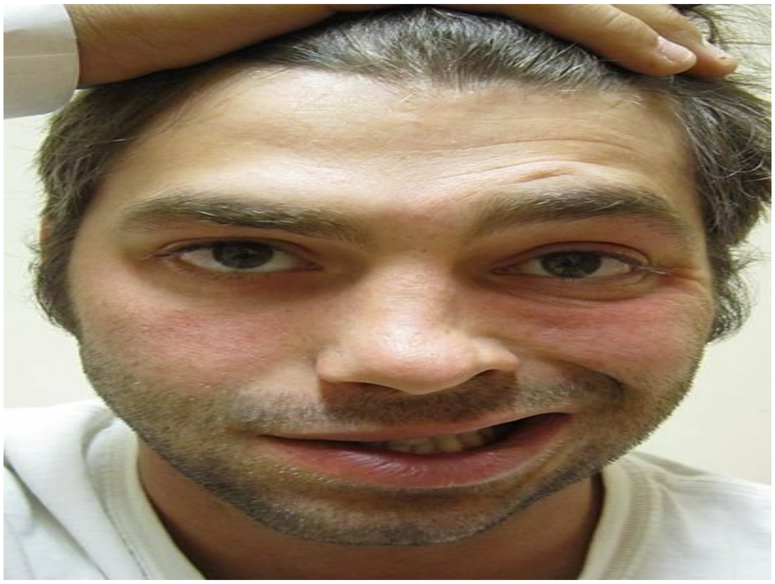
Bell's palsy is an acute onset idiopathic, peripheral facial paralysis. It is a lower motor neuron lesion of the facial nerve and involves the ipsilateral side of the upper and lower halves of the face.

It presents with an inability to close the eye, and if the patient attempts to close the eye, it turns up and out. This is known as the Bell's phenomenon. Other features include drooling of saliva and facial asymmetry. There may be noise intolerance and a loss of taste sensation.

Bell's palsy is a diagnosis of exclusion. All other possible causes of facial paralysis should be ruled out by investigations.

Treatment of Bell's palsy involves prednisone 60-80 mg daily during the first 5 days and then tapered over the next 5 days. Tape to depress the upper eyelid during sleep and artificial tears can be used to prevent corneal drying. Most patients recover within a few weeks or months.

The following image shows a patient with Bell's palsy.



Other options:

Option B: Lesions of the trigeminal nerve produce numbness and paresthesias, and loss of sensation in the area supplied by the trigeminal nerve.

Option C: In oculomotor nerve palsy, there is ptosis, a dilated pupil, and the eye appears to be down and out due to the unopposed lateral rectus and superior oblique.

Option D: Glossopharyngeal nerve lesions produce difficulty swallowing, and impairment of taste over the posterior one-third of the tongue and palate. There is also impaired sensation over the posterior one-third of the tongue, palate, and pharynx, and an absent gag reflex.

Solution to Question 3:

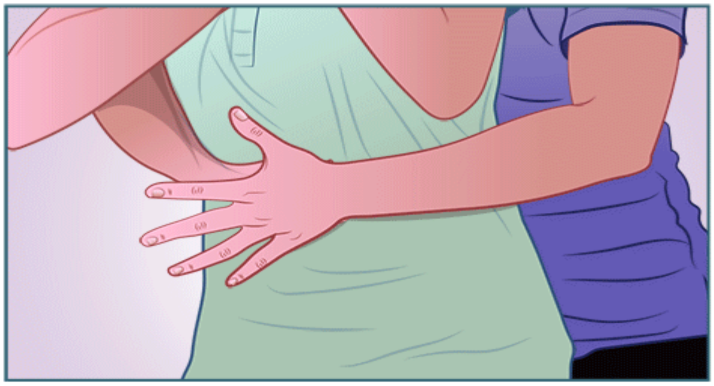
The image shows a person standing behind with arms placed around the choking person's lower chest, giving abdominal thrusts, which is suggestive of Heimlich's maneuver.

The Heimlich maneuver (also known as subdiaphragmatic abdominal thrusts or abdominal thrusts) is done for an adult or a child (1-8 years) with foreign body airway obstruction (FBAO). It is not recommended for FBAO relief in infants.

The rescuer should stand or kneel behind the patient and wrap his arms around the patient’s waist. A fist is made with one hand and using the other hand, the fist is driven into the patient’s abdomen with a rapid, forceful inward and upward thrust to dislodge the foreign body. This is continued until the foreign body is expelled or if the patient becomes unresponsive.

If the patient becomes unresponsive, immediately start cardiopulmonary resuscitation (CPR).

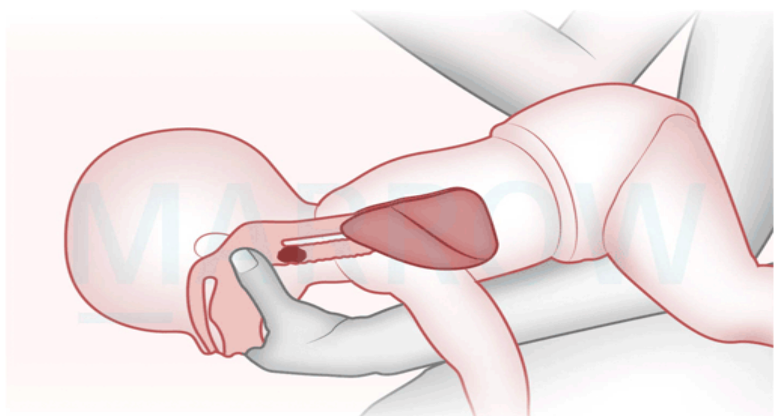
Heimlich maneuver

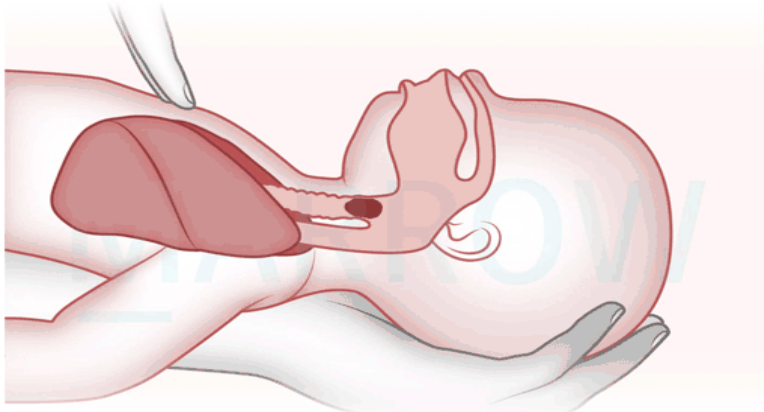


Other options:

Option B: Back slaps or back blows can also be used to relieve FBAO. They are given between the shoulder blades using the heel of the rescuer’s hand, and up to 5 are given. If there is a failure of the back slaps to relieve the obstruction, up to 5 abdominal thrusts can be given.

Infants (<1 year) should be treated with alternating 5 back blows and then 5 chest thrusts. The infant's torso is placed on the rescuer's nondominant arm. It is shown below.





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Option C: Chest thrusts can be used as an alternative to the Heimlich maneuver for pregnant women or markedly obese patients. The patient is supine, and the rescuer should kneel close to the victim's side. Hands are placed in the same manner as is done to perform CPR chest compressions. Each thrust is given in such a way so as to relieve the obstruction.

Option D: The blind finger sweep should be avoided in infants to avoid pushing the object further inside the airway. However, it can be done for unresponsive adults with complete FBAO. With the patient's face up, the mouth is opened by grasping both the tongue and lower jaw and lifting the mandible. The index finger is inserted along the inside of the cheek and using a hooking action, the foreign body is dislodged and removed.

Solution to Question 4:

The given clinical scenario and the image are suggestive of a parotid abscess.

A parotid abscess refers to an infection and pus formation in the parotid space. It is caused most commonly by Staphylococcus aureus. Post-surgical cases and debilitated patients with stasis of salivary flow are at risk of developing parotid abscesses.

Patients typically present with unilateral swelling, redness, induration, and tenderness over the parotid gland and angle of the mandible. Dehydration and fever are also present. Diagnosis is by ultrasonography or a CT scan. Multiple small loculi of pus are seen in the parenchyma. Aspiration of abscess can be done to determine the culture and sensitivity of the infective agent.

Correction of dehydration, improving oral hygiene, and promoting salivary flow. Intravenous antibiotics are also used in treatment. Surgical drainage of the abscess is done via a preauricular incision.

The parotid gland is enclosed by deep cervical fascia which splits into superficial and deep layers. Parotid space lies deep in the superficial layer. Contents of parotid space include :

• parotid gland

• lymph nodes associated with parotid gland

• facial nerve

• external carotid artery

• retromandibular vein

Other options:

Option A: Bezold abscess is a complication of acute coalescent mastoiditis and is formed when pus breaks inferiorly, through the tip of the mastoid and collects in the upper part of the neck. The onset is sudden with pain, fever, tender neck swelling, and torticollis. There will be a history of purulent otorrhea.

Option C: Osteoma of the mandible is usually slow growing and asymptomatic. It usually occurs in the mandibular condyle area. They are usually detected incidentally during a radiographic examination.

Option D: Upper cervical lymphadenopathy occurs in the neck region following infection in the area drained by these structures.

Solution to Question 5:

The given clinical scenario of a child presenting with hearing loss for the past 2-3 months and the otoscopic finding is suggestive of serous otitis media. The given otoscopic image shows a dull and opaque tympanic membrane with clear effusion.

Serous otitis media is also known as secretory otitis media or mucoid otitis media or glue ear. It is characterized by the accumulation of nonpurulent effusion in the middle ear cleft.

Glue ear is thought to happen due to either malfunctioning (adenoid hyperplasia, chronic sinusitis or tonsillitis or rhinitis, palatal defects, tumors of nasopharynx) or increased secretory activity of the eustachian tube (allergy, unresolved otitis media, viral infections).

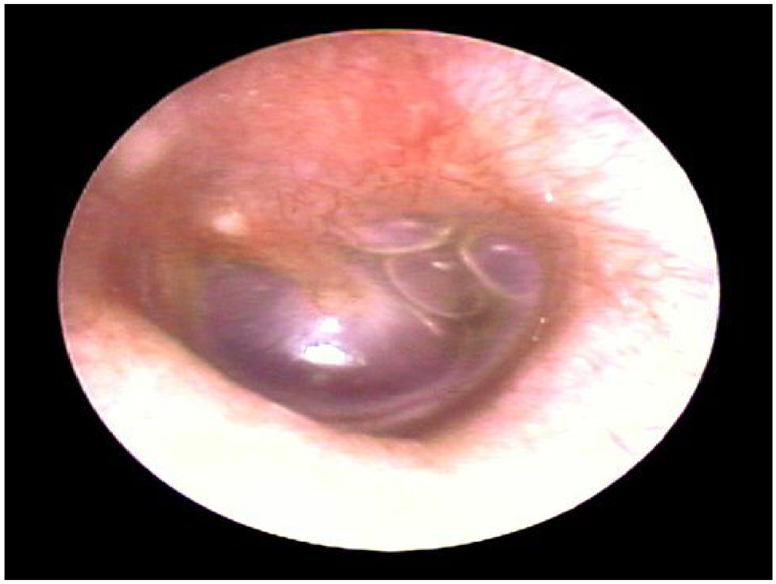
It is often seen in children of 5-8 years of age. They present with hearing loss not exceeding 40 dB, which is insidious in onset. Speech may be delayed and defective due to hearing loss. Mild earaches with a history of upper respiratory tract infection are usually seen.

On otoscopy,

• Dull and opaque tympanic membrane with loss of light reflex

• Thin leash of vessels along the handle of the malleus

• If the tympanic membrane is transparent fluid levels and air bubbles can be seen



The tuning fork test reveals a conductive type of hearing loss and audiometry reveals hearing loss in the range of 20-40 dB. On impedance audiometry, there is reduced compliance and a flat curve shifted to the negative side due to the presence of fluid. X-ray mastoid reveals clouding of air cells.

Treatment is mainly aimed at the removal of fluid and preventing recurrence.

• Medical management includes decongestants, antihistamines, antibiotics, and middle ear aeration techniques.

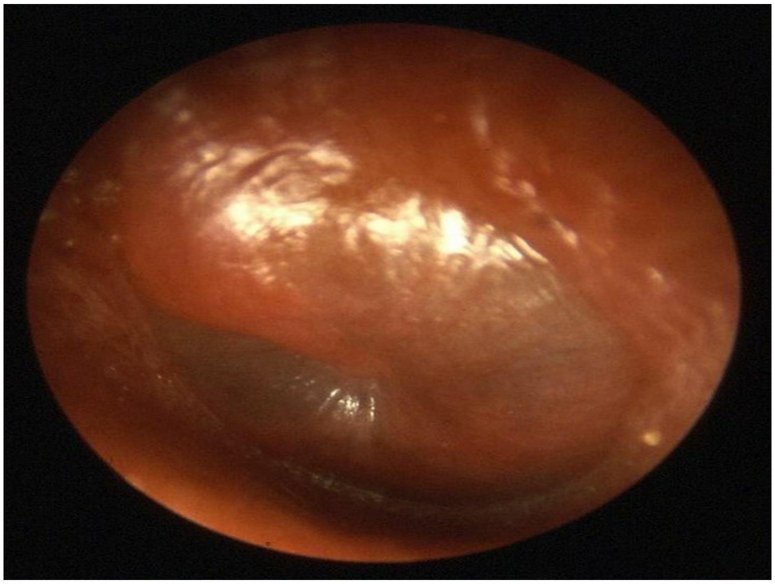
• Surgical management includes myringotomy and aspiration of fluid, grommet insertion, tympanotomy, or cortical mastoidectomy.

Chronic glue ear sequelae include atrophic tympanic membrane and middle ear atelectasis, ossicular necrosis, tympanosclerosis, cholesterol granuloma, and retraction pockets.

Other options:

Option A: Myringitis bullosa is characterized by painful hemorrhagic blebs on the tympanic membrane and deep meatus. It occurs due to infection by a virus or mycoplasma.

Option C: Acute otitis media is an acute infection of the middle ear by pyogenic organisms. It presents with marked earache and hearing loss is seen in later stages. Otoscopy shows a red and bulging tympanic membrane.



Solution to Question 6:

In a patient with Samter's triad, aspirin should be avoided as he will have aspirin intolerance.

Samter's triad is also known as aspirin-induced asthma or aspirin-sensitive asthma. It consists of:

- Nasal polyposis
- Bronchial asthma
- Aspirin intolerance

Patients with Samter's triad will have persistent nasal congestion, rhinorrhea, anosmia, nasal polyps, and recurrent chronic rhinosinusitis. Asthma and aspirin intolerance develop later.

Acute asthma attacks usually occur 3 hours after intake of aspirin or NSAIDs. Severe rhinorrhea, conjunctival injection, facial flushing, and periorbital edema are also seen. Aspirin-induced hypersensitivity reactions also may occur.

Patients should avoid aspirin and NSAIDs that inhibit COX-1. Treatment may involve aspirin desensitization and endoscopic sinus surgery.

Solution to Question 7:

The affected structure in high-frequency hearing loss is the basal turn of the cochlea. Among the given options, the structure that lies near it is the oval window.

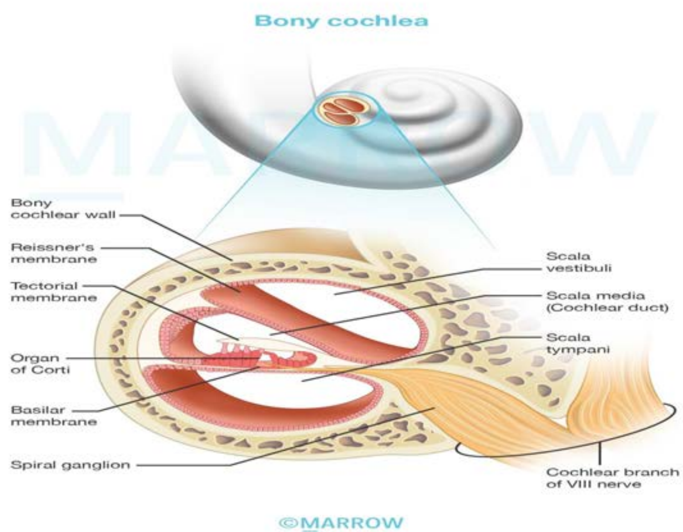
The traveling wave theory of von Bekesy states that a sound wave, depending on its frequency, reaches maximum amplitude at a segment on the basilar membrane and stimulates that segment. The high frequencies are represented in the basal turn of the cochlea and the progressively lower ones towards the apex.

The cochlea has a bony core called the modiolus, and a spiral canal that runs around it. At the base of the canal, the two main openings are the oval window and the round window. The footplate of stapes rests against the oval window. The base of the modiolus transmits nerves and vessels to the cochlea. (option A)

The inner ear consists of a bony labyrinth and a membranous labyrinth. The bony labyrinth consists of the vestibule, semicircular canals, and the cochlea. The membranous labyrinth consists of the cochlear duct, the utricle and saccule, semicircular ducts, and the endolymphatic duct and sac.

The cochlear duct has three walls- the basilar membrane, Reissner's membrane, and the stria vascularis. The basilar membrane supports the organ of Corti, which is the sensory hearing epithelium. The length of the basilar membrane increases from the base to the apex. As a result of this, higher frequency sound is heard at the basal coil while lower frequencies are heard at the apex.

The stria vascularis is involved in the secretion of endolymph and contains vascular epithelium. (Option B)



The helicotrema (option D) is an opening at the apex of the cochlea, at which the scala vestibuli and scala tympani communicate. The scala vestibuli, tympani, and media are three compartments of the bony cochlea.

Solution to Question 8:

The structure given in the image is the incus.

It is one of the three ossicles that are in the middle ear cavity i.e, malleus, incus, and stapes. Ossicles conduct sound energy from the tympanic membrane to the oval window and then to the inner ear fluid.

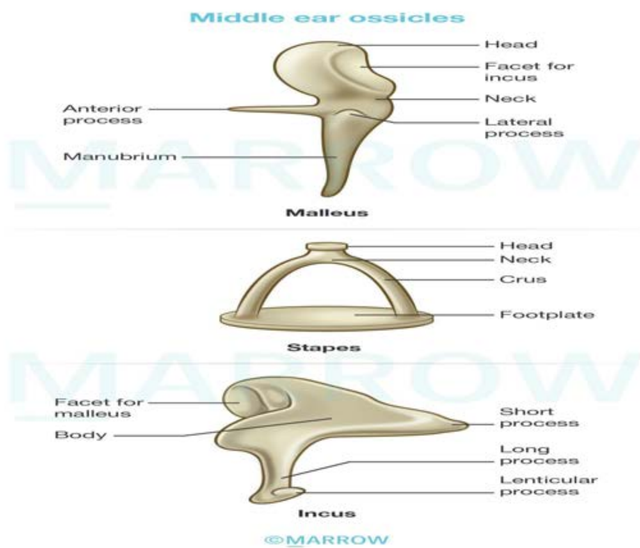
Incus is shaped like an anvil. It has a body, a short process, and a long process. It has a saddle-shaped articular facet on its anterior surface that articulates with the malleus. The long process descends vertically and ends as a rounded lenticular process.

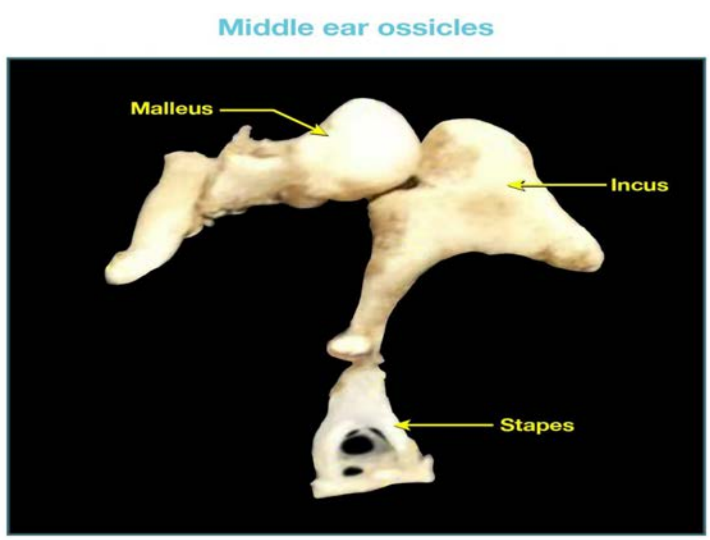
Incus has a single nutrient vessel within the long process and no collateral circulation. Thus, it is more susceptible to aseptic necrosis following a middle ear infection.

The malleus (option A) is the largest ossicle and is shaped like a mallet. The parts of the malleus are the head, neck, handle (manubrium), and anterior and lateral processes. The head and neck lie in the attic. The tympanic membrane is connected to the handle of the malleus.

The stapes (option C) has a head, neck, footplate, and anterior and posterior crura. The annular ligament holds the footplate of stapes in the oval window.

The joints between the ear ossicles are synovial joints. A saddle-shaped joint is present between the incus and malleus. Incus and stapes articulate via a ball and socket joint.





Option D- The vomer is a small bone in the posteroinferior part of the nasal septum. Along with perpendicular plate of the ethmoid, and the septal cartilage, it forms the nasal septum.

Solution to Question 9:

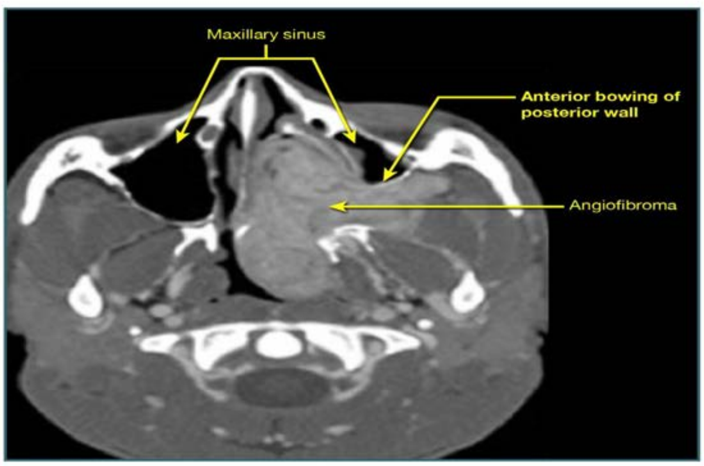
The above clinical scenario with recurrent spontaneous bleeding with unilateral nasal obstruction and contrast-enhanced CT showing a growth protruding into the nasopharynx, along with the transnasal endoscopic image, is suggestive of nasopharyngeal angiofibroma.

Nasopharyngeal angiofibroma is a benign but locally invasive tumor. It is predominantly seen in adolescent males and is thought to be testosterone-dependent. It arises from the posterior part of the nasal cavity near the sphenopalatine foramen.

The clinical features include epistaxis, progressive nasal obstruction, and denasal speech. Other symptoms include conductive hearing loss, broadening of the nasal bridge, proptosis leading to frog-face deformity, cheek swelling, and involvement of II, III, IV, and VI cranial nerves.

Contrast-enhanced CT scan of the head is the investigation of choice that reveals anterior bowing of the posterior wall of the maxillary sinus (Antral sign/Holman Miller sign), which is pathognomonic of angiofibroma as shown below.

Antral sign / Holman-Miller sign



Carotid angiography shows the extent of the tumor with its vascularity. The feeding vessels mostly come from the external carotid artery. In large tumors with intracranial extension, vessels come from the internal carotid system also.

Biopsy of the mass is avoided due to the risk of bleeding.

Surgical approach is the treatment of choice. Embolization of the vessels can be done prior to surgery to reduce the risk of bleeding during the surgery.

Radiotherapy, chemotherapy, and hormonal therapy can also be considered.

Other options:

Option B: Antrochoanal polyp (Killian's polyp) arises from the mucosa of the maxillary antrum. They are usually unilateral and are seen in children and young adults. They present with unilateral nasal obstruction with minimal to no bleeding. X-rays of the paranasal sinuses show opacity of the maxillary sinus.

Option C: Rhinoscleroma is caused by a gram-negative coccobacillus Klebsiella rhinoscleromatis. It causes this progressive granulomatous infection starting in the nose and extending into the upper respiratory tract. It presents initially with foul-smelling nasal discharge, and then with granulomatous nodules in the nose giving it a woody feeling. The last stage is the cicatricial stage, with stenosis of nares, and adhesions in the nose.

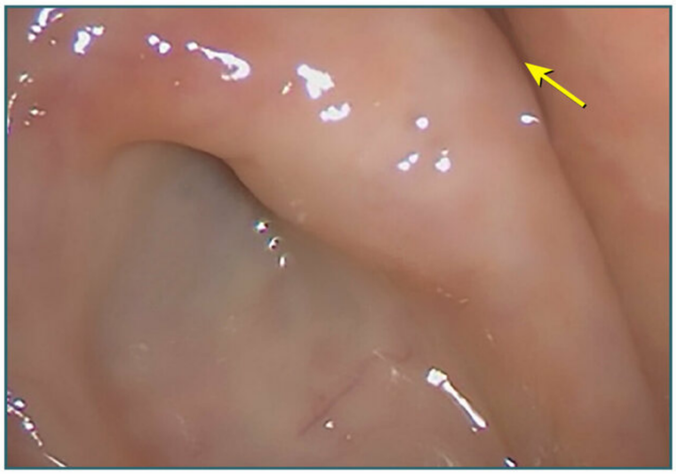
Option D: Concha bullosa is pneumatization of the middle turbinate that leads to an enlarged middle turbinate, which is ballooned out.

The CT image shown below depicts concha bullosa

ENT NEET 2023

Question 1:

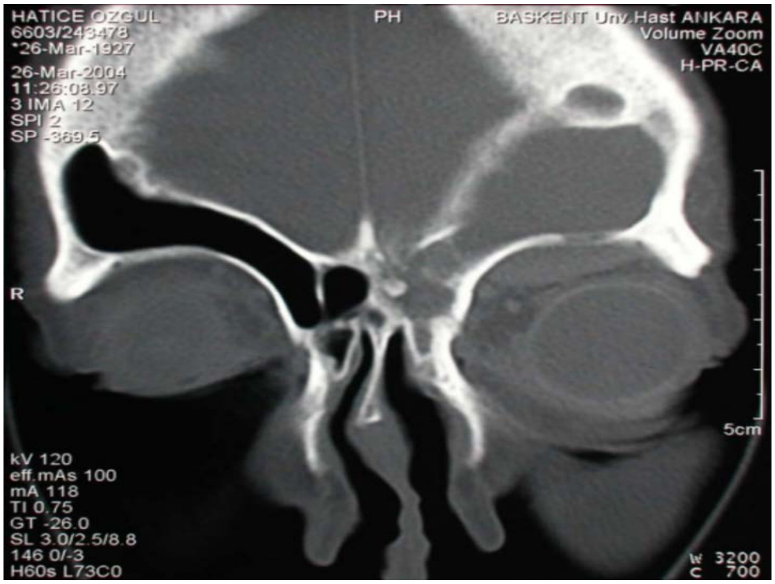
Identify the structure marked in the image.



- **a)** Fossa of Rosenmuller
- **b)** Tubal tonsil
- **c)** Opening of eustachian tube
- **d)** Adenoid

Question 2:

A 30-year-old male presents with nonaxial proptosis of the left eye. The patient gives a history of a road traffic accident 15 years back. The CT image is given below. What is the most likely diagnosis?



a) Frontal mucocele

b) Frontal meningioma

c) Juvenile nasopharyngeal angiofibroma

d) Pseudotumor of orbit

Question 3:

The instrument shown in the image is not used for which of the following?

a) Airway toileting

b) Upper airway examination

c) Acute nasopharyngeal obstruction

d) Prolonged mechanical ventilation

Question 4:

A 10-year-old child presents with throat pain, fever, and ear pain. He is diagnosed with recurrent tonsillitis. Which nerve is responsible for the ear pain in this patient?

a) Tympanic branch of glossopharyngeal nerve

b) Greater auricular nerve

c) Auriculotemporal nerve

d) Auricular branch of vagus nerve

Question 5:

A 55-year-old patient comes with hoarseness of voice and difficulty swallowing. The patient was diagnosed with laryngeal carcinoma and surgical management was done. The post-operative image of the patient is given below. Which of the following surgery was done on this patient?

a) Partial laryngectomy

b) Percutaneous tracheostomy

c) Standard tracheostomy

d) Total laryngectomy

Question 6:

A female patient with hearing loss is examined and is found to be Rinne negative at 256 Hz and 512 Hz, while Rinne positive at 1024 Hz. What is the expected air conduction and bone conduction gap?

a) 30-45 dB

b) 15-30 dB

c) 45-60 dB

d) >60 dB

Answer Key

Question No.	Correct Option
1	a
2	a
3	b
4	a
5	d
6	a

Detailed Explanations

Solution to Question 1:

The image shown is the endoscopic view of lateral wall of the nasopharynx, and the marked structure is the fossa of Rosenmuller.

The nasopharynx is present above the soft palate and is made up of the roof, floor, posterior wall, and two lateral walls.

Important structures seen in the nasopharynx are

• Pharyngeal opening of the eustachian tube

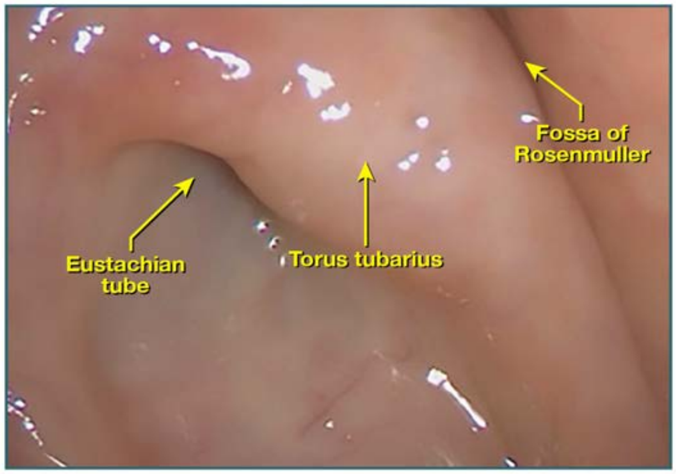
• Torus tubarius- tubal elevation which is present behind the opening of the eustachian tube.

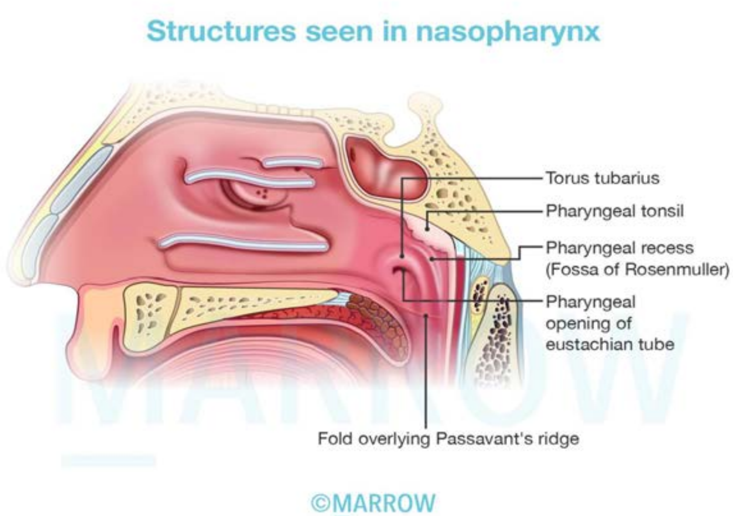
• Fossa of Rosenmuller - A lateral pharyngeal recess that is present above and behind the torus tubarius. It is the most common site for nasopharyngeal carcinoma

• Adenoids or nasopharyngeal tonsil, which is present in the roof and posterior wall

• Passavant's ridge

The tubal tonsil is a collection of subepithelial lymphoid tissue situated at the tubal elevation. It is a part of the Waldeyer’s ring and infection can cause eustachian tube occlusion.





Solution to Question 2:

The given clinical scenario and the image findings are suggestive of a frontal mucocele.

Mucoceles are slow-growing, mucous-filled benign masses that can develop years after head injury. Post-traumatic mucocele formation occurs due to chronic obstruction of the sinus ostium. The frontal sinus is most commonly involved in mucocele formation.

The sinuses commonly affected by mucocele in the order of frequency are the frontal > ethmoidal > maxillary > sphenoid.

Frontal mucocele presents with headache, diplopia, and proptosis. A cystic non-tender swelling is noticed with eggshell crackling.

The image given below shows a coronal MRI demonstrating a frontal mucocele and an orbital cystic mass.

Other options:

Option B: Frontal meningiomas are benign brain tumors presenting with headaches, seizures, vomiting, sensory and motor weakness, and visual impairment.

Option C: Juvenile nasopharyngeal angiofibroma is a benign neoplasm of the nasopharynx presenting with spontaneous recurrent epistaxis, nasal obstruction, and nasal discharge.

Option D: Pseudotumor of the orbit is an idiopathic inflammation of the orbit presenting with painful proptosis, diplopia, and orbital congestion. Imaging shows swollen eye muscles (orbital myositis) and enlarged tendons.

Solution to Question 3:

The image shows a tracheostomy tube. It is not used for upper airway examination.

Tracheostomy involves making an opening and placing a stoma in the anterior wall of the trachea.

Functions:

• Provides an alternative pathway for breathing by bypassing the obstruction.

• Improves alveolar ventilation by reducing dead space by 30-50% and reducing airway resistance

• Uses a cuff to protect the airways from secretions and blood

• Allows suction of secretions in the setting of the impaired cough reflex.

• Allows positive pressure respiration

• Allows administration of anesthesia

Indications:

• Respiratory obstruction due to

• Infections (laryngo-tracheo-bronchitis, epiglottitis, diphtheria)

• Trauma due to external injury, mandibular or maxillofacial fractures

• Neoplasms

• Foreign bodies or edema

• Bilateral abductor paralysis

• Congenital anomalies

• Retained secretions

• Inability to cough due to coma, spasm, or paralysis of respiratory muscles

• Aspiration of pharyngeal secretions

• Painful cough

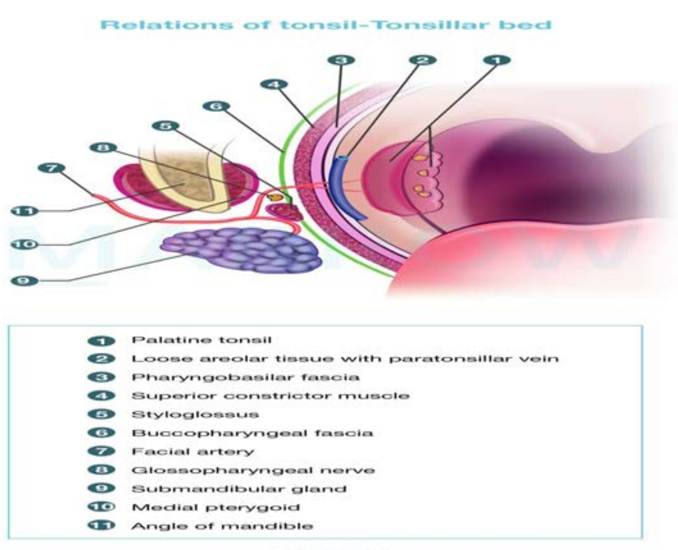
• Respiratory insufficiency due to emphysema, chronic bronchitis

Solution to Question 4:

The tympanic branch of the glossopharyngeal nerve (cranial nerve IX) is responsible for referred otalgia from tonsillitis.

The tonsils and tonsillar fossa are supplied by the glossopharyngeal nerve. The glossopharyngeal nerve and the styloid process lie in relation to the lower part of the tonsillar fossa. Therefore, any irritation or pain can be referred to the ear along the tympanic branch of the glossopharyngeal (Jacobson's) nerve.

The image below shows the anatomic relations of the tonsil and tonsillar bed.



The ear receives extensive sensory innervation via four cranial nerves (V, VII, IX, and X) and two spinal segments (C2 and C3).

Referred otalgia transmitted by the glossopharyngeal nerve may be secondary to lesions and/or inflammatory processes of the nasopharynx, palatine tonsil, soft palate, or posterior one-third of the tongue.

Causes of referred otalgia via glossopharyngeal nerve:

- Acute pharyngitis
- Tonsillitis
- Peritonsillitis
- Peritonsillar abscess
- Post-tonsillectomy
- Foreign bodies embedded in the base of the tongue or tonsillar area
- Elongated styloid process

Other options:

Option B: The great auricular nerve provides sensory sensation for the lower two-thirds of the pinna.

Option C: Causes of referred pain via the auriculotemporal nerve which is the branch of the trigeminal nerve include caries tooth, ulcerative lesions of the oral cavity, temporomandibular joint disorders, and sphenopalatine neuralgia.

Option D: Causes of referred pain via the auricular branch of the vagus nerve include malignancy or ulcerative lesion of the vallecula, epiglottis, larynx, and esophagus.

Solution to Question 5:

The given image shows a post-operative tracheal stump created for breathing in patients undergoing total laryngectomy for laryngeal carcinoma.

In total laryngectomy, the entire larynx, including the hyoid bone, pre-epiglottic space, strap muscles, and one or more rings of trachea, are removed. The pharyngeal wall is repaired, and the tracheal stump is sutured to the skin to allow respiration.

The indications of total laryngectomy are as follows:

- T3 lesions
- All T4 lesions
- Invasion of thyroid or cricoid cartilage
- Bilateral arytenoid cartilage involvement
- Lesions of the posterior commissure
- Failure after radiotherapy or conservation surgery

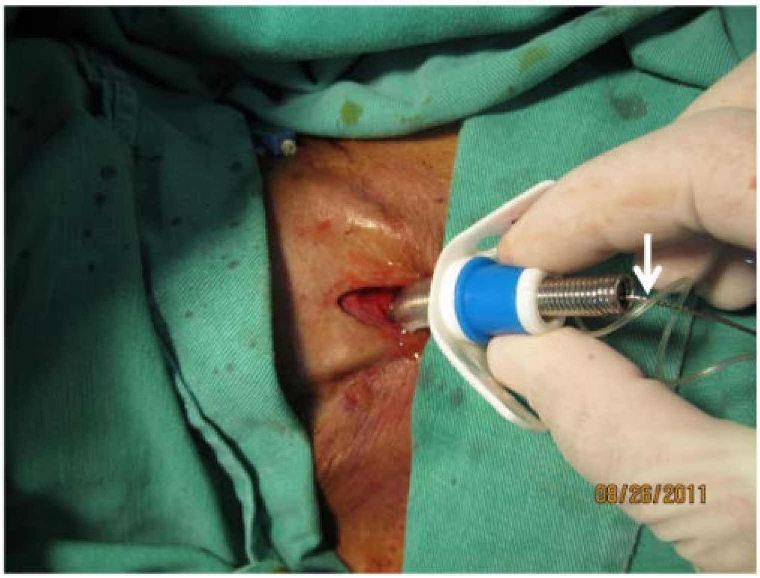
• Transglottic cancers with cord fixation

It is contraindicated in patients with distant metastasis.

Other options:

Option A: Partial laryngectomy is a conservation procedure that involves the excision of the vocal cord and anterior commissure region in partial front lateral laryngectomy and the excision of supraglottis in partial horizontal laryngectomy. This surgery is performed to conserve the voice and avoid a permanent tracheal opening.

Option B: Percutaneous tracheostomy is done in ICU patients under sedation when the patient is already intubated and being monitored. The image below shows the procedure of percutaneous tracheostomy.



Option C: Standard tracheostomy can be therapeutic (used for respiratory obstruction, remove tracheobronchial secretions or assisted ventilation) and prophylactic (for anticipated respiratory obstruction or aspiration of blood or pharyngeal secretions).

Solution to Question 6:

The given clinical scenario of hearing loss with a negative Rinne at 256 Hz and 512 Hz and positive Rinne at 1024 Hz indicates air conduction and bone conduction gap of 30-45 dB.

Tuning fork tests are performed commonly with tuning forks of frequencies such as 128, 256, 512, and 1024. The tuning fork of 512 Hz is considered ideal as forks of lower frequencies produce a sense of bone vibration, while those of higher frequencies have a shorter decay time.

Rinne's test is done by placing a vibrating tuning fork on the patient's mastoid and when he stops hearing, it is brought beside the external auditory meatus (EAC). If he still hears the sound, air conduction (AC) is better than bone conduction (BC).

The inferences are as follows:

• Rinne's test is considered positive (normal) when the air conduction AC>BC is seen in normal people and people with sensorineural deafness.

• It is considered negative when the BC>AC, and is seen in patients with conductive deafness. It suggests a minimum hearing loss of 15-20 dB.

A prediction of the air-bone gap can be made using tuning forks of 256, 512, and 1024 Hz:

• A Rinne test negative for 256 Hz but positive for 512 Hz indicates an air-bone gap of 20-30 dB.

• A Rinne test negative for 256 and 512 Hz but positive for 1024 Hz indicates an air-bone gap of 30-45 dB.

• A Rinne negative for 256, 512, and 1024 Hz indicates an air-bone gap of 45-60 dB.

ENT INICET 2023

Question 1:

A 22 year old male presented with progressive left sided facial swelling for 2 weeks. There is history of traumatic injury with wood two months back. CECT nose and paranasal sinuses shows mass in subcutaneous tissue of left cheek. Biopsy of specimen shows foreign body granuloma with positive PAS and GMS stain. What is the diagnosis?

a) Phycomycosis

b) Midline lethal granuloma

c) NK/T cell lymphoma

d) IgG4 granuloma

Question 2:

Arrange the following structures in the hearing pathway from the receptors to the cortex -

a) 3-2-4-6-7-5-1

b) 2-3-4-6-7-5-1

c) 2-3-6-4-5-7-1

d) 3-2-6-4-7-5-1

Question 3:

Identify the nerve block shown in the given image

- **a)** Nasociliary nerve
- **b)** Anterior ethmoidal nerve
- **c)** Sphenopalatine nerve
- **d)** Greater palatine nerve

Question 4:

Arrange the following in the sequence of auditory pathway:

- **a)** 1-2-3-4-5
- **b)** 5-4-3-2-1
- **c)** 2-1-3-4-5
- **d)** 3-4-5-1-2

Question 5:

A 14-year-old is brought to the opd with complaints that she is able to hear the sound of speech but has difficulty understanding words. Her PTA and BERA were inconsistent with each other, with PTA being more or less normal. Her middle latency and cortical responses were absent. What is the likely diagnosis?

a) Michel aplasia

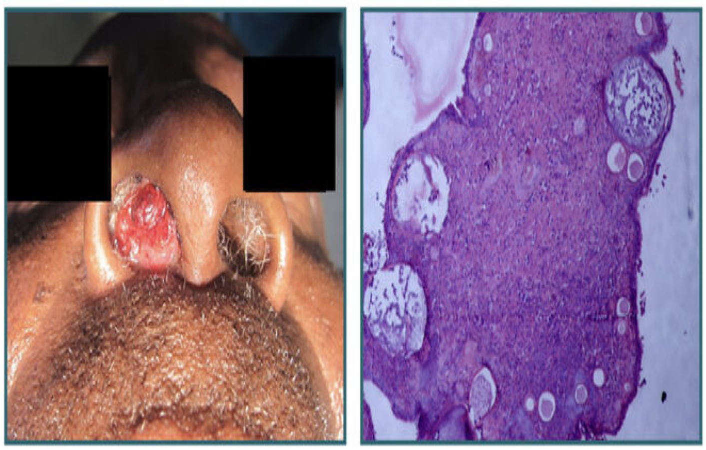
b) Auditory neuropathy

c) Malingering

d) Cochlear otosclerosis

Question 6:

A South Indian male cattle breeder comes with complaints of frank intermittent epistaxis. The examination of the nose is given below. A biopsy was taken and HPE appears as shown below. What is the likely diagnosis?



a) Nasal polyp

b) Rhinosporidiosis

c) Hemangioma

d) Rhinopyma

Question 7:

Which of the following statements are correct about hair cells?

a) i, ii, iv

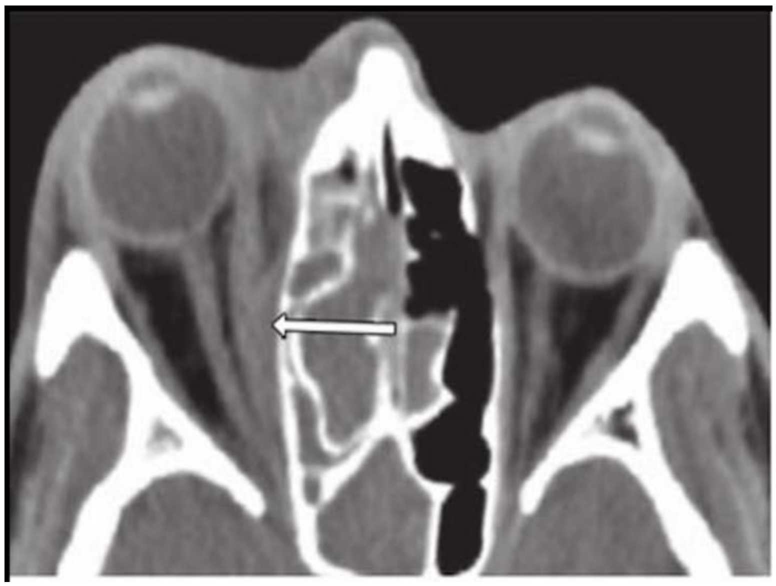
b) i, ii, iii

c) i, iii, iv

d) ii, iii, iv

Question 8:

A 7-year-old male presents with progressive bulging of the right eye with a diminision of vision since 1 day. He has a history of URTI and on examination, it is seen that he also has nasal congestion with active discharge. The CT scan of the PNS is seen. What is the diagnosis?



a) Sinusitis with preseptal cellulitis

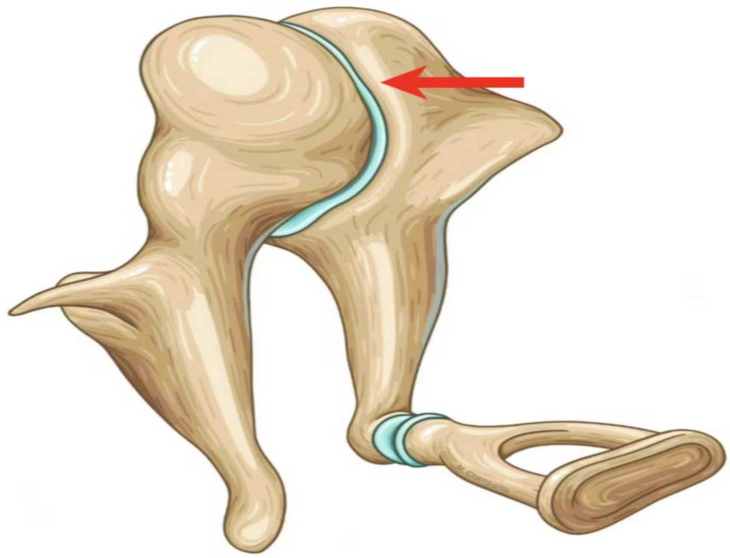
b) Sinusitis with orbital cellulitis

c) Sinusitis with subperiosteal abcess

d) Sinusitis with orbital abscess

Question 9:

Which type of synovial joint is seen in the picture?



a) Ball and socket

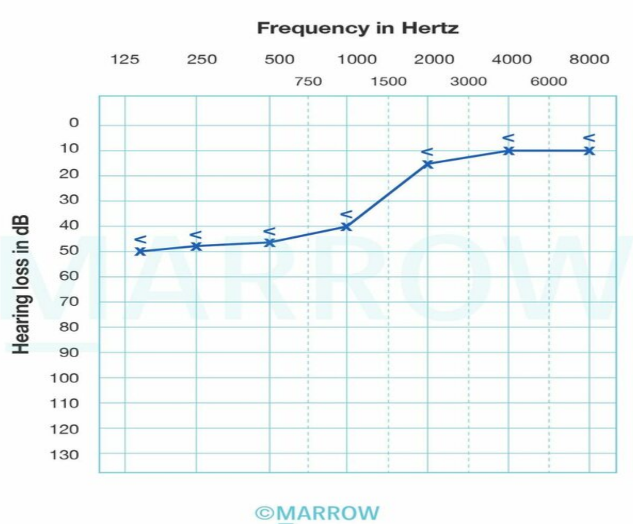
b) Saddle

c) Pivot

d) Plane

Question 10:

The audiogram given in the image below is seen in ______.



a) Ototoxicity

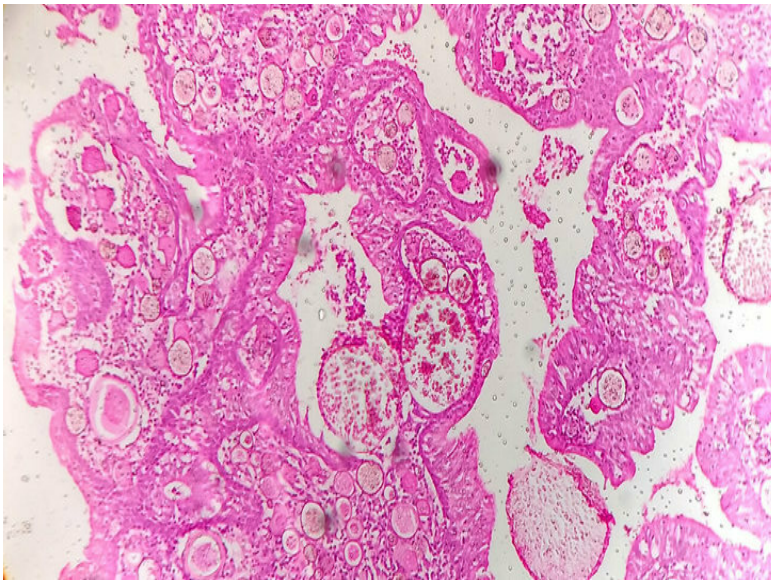
b) Presbycusis

c) Noise induced hearing loss

d) Endolymphatic hydrops

Question 11:

A farmer presents with unilateral obstruction of the nasal cavity and nasal bleeding. A mass is found and removed. A biopsy is shown below. What is the likely diagnosis?



a) Rhinosporidiosis

b) Rhinoscleroma

c) Inverted papilloma

d) Angiofibroma

Question 12:

Out of the four given statements, which is true about coughing?

a) i, ii, iii

b) i, iv

c) ii, iv

d) ii, iii

Question 13:

A 32-year-old female presents with progressive bilateral hearing loss, in the right ear more than left. After the initial evaluation, a stapedotomy of the right ear was planned. What might've been the tuning fork test results of the patient?

a) Bilateral Rinne's negative, Weber's lateralized to the right

b) Bilateral Rinne's negative, Weber's lateralized to the left

c) Bilateral Rinne's positive, Weber's lateralized to the right

d) Bilateral Rinne's positive, Weber's lateralized to the left

Question 14:

A 60-year-old man can only hear when words are shouted. What is the degree of his hearing impairment?

a) Moderate hearing loss < 50%

b) Moderately severe hearing loss < 70%

c) Profound hearing loss > 80%

d) Severe hearing loss > 70%

Answer Key

Question No.	Correct Option
1	a
2	c
3	a
4	c
5	b
6	b
7	b
8	c
9	b
10	d
11	a
12	b
13	a
14	d

Detailed Explanations

Solution to Question 1:

The given clinical scenario is strongly suggestive of Phycomycosis/mucormycosis. A positive fungal stain(Grocott-Gomori's methenamine silver stain) narrows down the diagnosis. The cutaneous disease can be highly invasive, penetrating into muscle, fascia, and even bone.

Mucormycosis is an opportunistic mycosis caused by a number of molds classified in the order ‘Mucorales’. These fungi are ubiquitous thermotolerant saprobes. The leading pathogens among this group are species of the genera Rhizopus, Mucor, and Absidia. Their spores are present in air and dust.

Mucormycosis can occur as isolated cutaneous or subcutaneous infection in immunologically normal individuals after traumatic implantation of soil or vegetation. Angio-invasiveness is one of the characteristics of zygomycosis which makes it fatal. Lungs are usually the primary site of infection. Almost all patients are immunocompromised. Rhino-orbital-cerebral mucormycosis occurs commonly in patients with diabetic ketoacidosis. It is the most common form; starts as eye and facial pain may progress to cause orbital cellulitis, proptosis, and vision loss. The typical finding is the presence of a black necrotic mass filling the nasal cavity.

Diagnosis:

• Exudate: wet preparation is made in 10% potassium hydroxide. Broad, ribbon-like hyphae are seen.

• Tissue: Hematoxylin & eosin stain shows non-septate, broad hyphae, often invading into blood vessels. Special stains help to identify the fungus in tissue sections.

• Biopsy confirms this diagnosis by revealing non-septate hyphae branching at right-angles, which are characteristic of Mucorales such as Rhizopus delemar. The hyphae are broad (10–15 µm) with uneven thickness, irregular branching, and sparse septations.

• Isolation is by culture on Sabouraud agar with an antibacterial agent but without cycloheximide. The colonies are grey-white with a thick cotton, fluffy surface.

Treatment is primarily with liposomal amphotericin B. Surgical debridement of the affected tissues and control of underlying predisposing cause is equally important.

Other options:

Option D: IgG4-related disease (IgG4-RD) is characterized by tissue infiltrates dominated by IgG4 antibodies (the hallmark of the disease). The pathogenesis of this condition is not understood. This condition leads to fibrosis and obliterative phlebitis. The patients may present as Riedel's thyroiditis, autoimmune pancreatitis, or idiopathic orbital inflammation. But not with ptosis as shown in the image. Diagnosis is established by positive fibrosis and the presence of plasma cells on biopsy. Rituximab is known to be beneficial. Sometimes, as a rare presentation, this can mimic Wegener's granulomatosis.

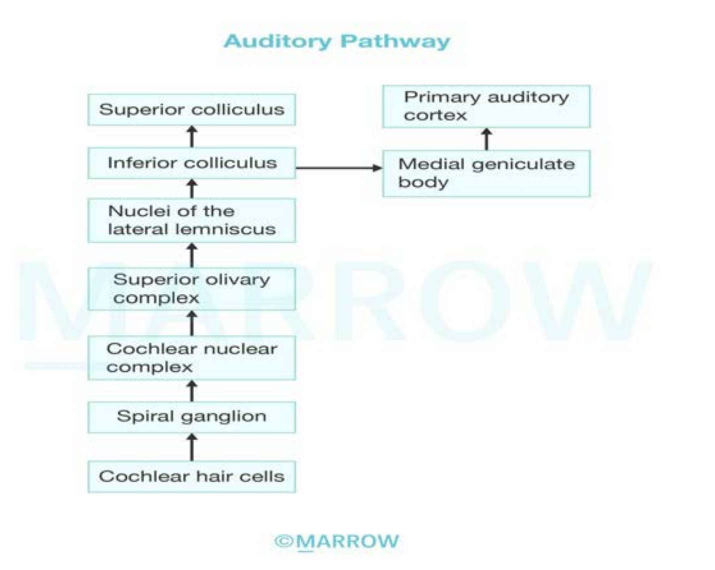
Option B: Midline lethal granuloma is a rare disease. It is characterized by necrotizing destruction and mutilation of the nose and other structures of respiratory passages. Generally, it is a diagnosis of exclusion. CT may be helpful in picking up the lesions. Negative c-ANCA and fungal stains also aid in the diagnosis.

Option C: NK/T cell lymphoma is a destructive lesion usually starting on one side of nose involving the upper lip, oral cavity, maxilla and sometimes even extending to orbit.Histologically polymorphic lymphoid tissue with angiocentric and angioinvasive features is seen. Immunochemical staining of biopsy material should be done for CD56+ marker for diagnosis. Nearly all cases of T/NK cel lymphomas are associated with EB virus. Localized T/NK-cell lymphoma is treated by radiation while a disseminated disease requires chemotherapy.

Solution to Question 2:

The correct sequence of the auditory pathway is spiral ganglion - cochlear nucleus - Olivary complex -Lateral leminiscus- Inferior colliculus-Medial geniculate body-Auditory Cotex

The spiral ganglion situated in Rosenthal's canal contains bipolar cells. The dendrites of these bipolar cells join to form the cochlear branch of the vestibulocochlear nerve which ends in the cochlear nuclei.



Solution to Question 3:

The given image shows a nasociliary nerve block.

The procedure is done by inserting the needle 1 cm superior and 1.5 cm posterolateral to the medial canthal ligament of the eye.

The nasociliary nerve is a branch of the ophthalmic nerve which in turn is a branch of the trigeminal nerve.

Branches:

• Long ciliary nerve- cornea, ciliary body, iris

• Ramus communicans to the ciliary ganglion

• Posterior ethmoidal nerve- sensory innervation to frontal sinus and posterior ethmoidal air cells

• Anterior ethmoidal nerve- sensory innervation to middle and anterior ethmoidal cells as well as lateral part of the nasal septum

• Infra-trochlear nerve (continuation of the nasociliary nerve)- provides innervation to the conjunctiva, lacrimal sac, caruncle, the medial part of the eyelids

Uses of nasociliary nerve block:

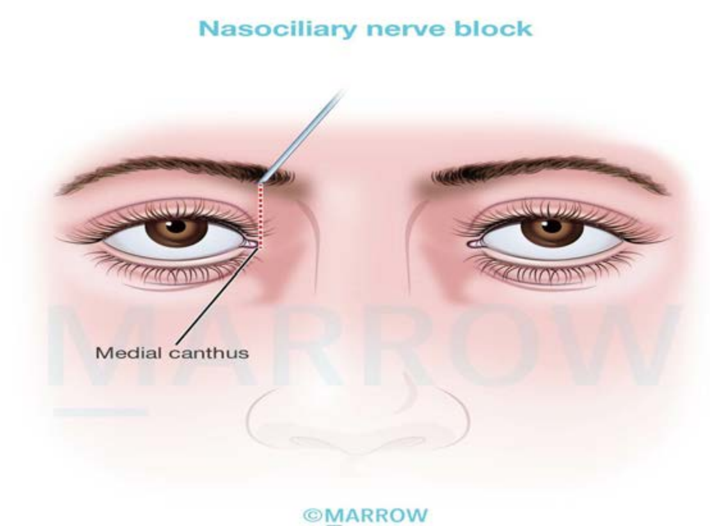
Used in

• Surgery of nasal fractures,

• Septoplasty,

• Rhinoplasty and

• Repair of skin laceration.



The above image represents an anterior ethmoidal nerve block.

Solution to Question 4:

The correct sequence of the auditory pathway is spiral ganglion (2) - cochlear nucleus (1) - superior olivary nucleus (3) - inferior colliculus (4) - medial geniculate body (5).

The spiral ganglion contains bipolar cells whose dendrites form the afferent supply for inner and outer hair cells of the organ of Corti and whose axons form the cochlear division of CN VIII that end in the cochlear nuclei.

The auditory pathway can be remembered with the mnemonic E.C.O.L.I.M.A

Eighth nerve

Cochlear nuclei

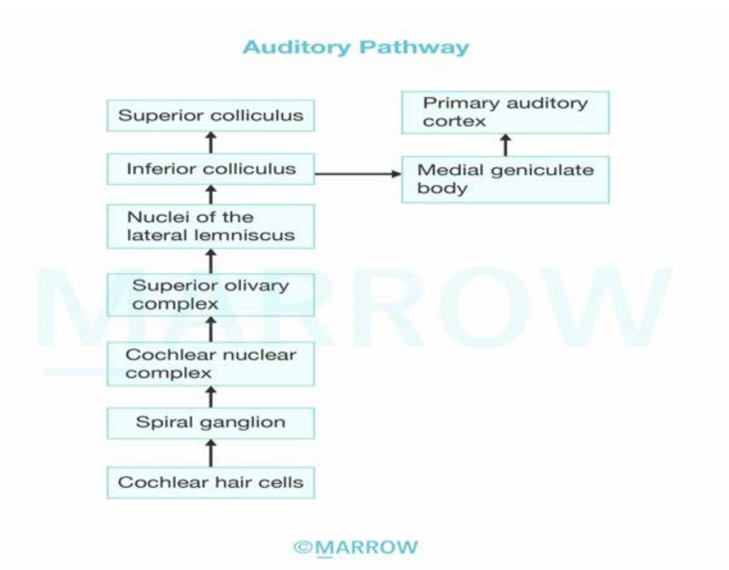
Olivary complex

Lateral lemniscus

Inferior colliculus

Medial geniculate body

Auditory cortex



Solution to Question 5:

In the clinical scenario, the history of a child presenting with complaints of inability to understand words with abnormal BERA and normal PTA(Pure Tone Audiometry) is suggestive of auditory neuropathy.

Auditory neuropathy/auditory dyssynchrony is a condition that affects the neural processing of auditory stimuli.

Pathophysiology

- Injury to the synaptic junction between inner hair cells of the cochlea and dendrites of spiral ganglion neurons
- Direct damage to the dendrites of the spiral ganglion neurons
- Direct injury to the spiral ganglion neurons
- Direct axonal damage to the auditory nerve

Auditory neuropathy should be suspected in any child with severe speech and language delay out of proportion with the presumed hearing loss.

All of the following must be present in newborns to diagnose auditory neuropathy:

- Absent or severely abnormal ABR test results at maximal stimulus (100 dBnHL)
- Normal outer hair cell function as determined by OAEs(Otoacoustic emission)
- Absent or elevated stapedial reflex thresholds

In addition, adults with mildly to moderately abnormal pure tone thresholds and speech discrimination scores out of proportion with suspected hearing loss should undergo additional evaluation. Some patients may have normal or slightly elevated audiogram pure tone thresholds

Management includes conventional hearing aids. If a child does not progress with hearing aid devices and shows limited speech discrimination abilities, cochlear implantation is the next viable option.

Other options

Option A:

Michel aplasia: There is a complete absence of bone and a membranous labyrinth. Even the petrous apex is absent but external and middle ears may be completely unaffected. The child presents with congenital sensorineural hearing loss. No hearing aids or cochlear implantation can be used.

Option B:

Malingering(Nonorganic hearing loss)

In this type of hearing loss, there is no organic lesion. Usually, there is a motive to claim some compensation for being exposed to industrial noises, head injury, or ototoxic medication. The patient may present with any of the three clinical situations: Total hearing loss in both ears, total loss in only one ear, or exaggerated loss in one or both ears.

Investigations

Inconsistent results on Repeat Pure Tone and Speech Audiometry Tests: Normally, the results of repeat tests are within +5 dB. A variation greater than 15 dB is diagnostic of nonorganic hearing loss (NOHL).

Absence of Shadow Curve: Normally, a shadow curve can be obtained while testing bone conduction, if the healthy ear is not masked. This is due to the transcranial transmission of sound to the healthy ear. The absence of this curve in a patient complaining of unilateral deafness is diagnostic of NOHL.

Inconsistency in PTA and SRT: Normally, the pure tone average (PTA) of three speech frequencies (500, 1000, and 2000 Hz) is within 10 dB of the speech reception threshold (SRT). An SRT is better than PTA by more than 10 dB points to NOHL.

Stenger Test: It helps in identifying malingering

Acoustic Reflex: Threshold Normally, the stapedial reflex is elicited at 70-100 dB sensation level. If the patient claims total deafness but the reflex can be elicited, it indicates nonorganic hearing loss (NOHL).

Option C:

Otosclerosis is the disease of the bony labyrinth in which a normal dense endochondral layer of the bony otic capsule is replaced by irregularly laid spongy bone. It has a positive family history with an autosomal dominant trait. It is more commonly seen in females between 20-30 years of age and the hearing loss is aggravated during pregnancy and menopause. It may be associated with osteogenesis imperfecta and a viral infection like measles.

Stapedial otosclerosis is the most common type of otosclerosis. Here the lesion starts just in front of the oval window in an area called fissula ante fenestram.

Patients present with a bilateral conductive type of hearing loss, tinnitus (commonly in cochlear type and active stage), paracusis Willisii (better hearing in noisy surroundings than in a quiet place), vertigo, and a monotonous, well-modulated soft speech.

The tympanic membrane is quite normal and mobile. Eustachian tube function is also normal. Pure tone audiometry shows a dip at 2000 Hz in the bone conduction curve known as the Carhart's notch. Tympanometry shows an As type of tympanogram in which there is reduced compliance at or near the ambient air pressure.

On examination, Schwartz sign/flamingo flush showing a red blush of the tympanic membrane due to vascularity can be seen.

Treatment includes medical, surgical management, and hearing aids. Medical management with sodium fluoride is used in the active stage to speed up the maturity of the active focus and halt further cochlear loss. The treatment of choice is surgical management which includes stapedotomy/stapedectomy with the placement of a prosthesis. Hearing aids are used in patients who refuse or are unfit for the surgery.

Solution to Question 6:

The clinical image showing a polyp and the histopathology showing several sporangia is suggestive of rhinosporidiosis.

Rhinosporidiosis is a chronic granulomatous disease caused by Rhinosporidium seeberi. It is most prevalent in rural districts, among persons bathing in public ponds or working in stagnant water, such as rice fields. Most common in persons in the age range of 15–40; males are more commonly affected than females.

The rhinoscopic examination will reveal papular or nodular, smooth-surfaced lesions that become pedunculated and acquire a papillomatous or proliferative appearance. The lesions are pink, red, or purple in color. Histopathology shows several sporangia, oval, or round in shape and filled with spores.

The treatment of choice is surgical excision of lesions, with or without cauterization. Not many drugs are effective against this disease. Dapsone has been tried with some success.

Solution to Question 7:

Statements i, ii, and iii are correct.

Statement i: There is 1 row of inner hair cells and 3 rows of outer hair cells is a correct statement.

Statement ii: Inner hair cells are responsible for transmitting auditory impulses is a correct statement. Inner hair cells are richly supplied by afferent cochlear fibres and are probably more important in the transmission of auditory impulses.

Statement iii: The majority of the afferent fibres of spiral ganglion supply the inner hair cells is a correct statement. Ninety-five percent of afferent fibres of spiral ganglion supply the inner hair cells while only five percent supply the outer hair cells. Efferent fibres to the hair cells come from the olivocochlear bundle.

Statement iv: Outer hair cells are important in the modulation of basilar membrane action is an incorrect statement. Outer hair cells mainly receive efferent innervation from the olivary complex and are concerned with modulating the function of inner hair cells.

Solution to Question 8:

In the given clinical scenario, a male patient with symptoms of an upper respiratory tract infection (URTI) exhibits opacity in the right ethmoidal sinus, as shown by the CT image, suggesting a probable diagnosis of sinusitis. CT scans show a typical lenticular rim-enhancing collection adjacent to the lamina papyracea with a fat plane between it and the displaced medial rectus muscle, suggesting a diagnosis of Sinusitis with subperiosteal abscess.

Orbital complications occur most commonly following ethmoid sinusitis as the ethmoid sinus is separated from the orbit by a thin papery bone known as lamina papyracea. Also, the venous supply of orbit and ethmoid sinuses is the same and additionally, these veins are valveless (superior and inferior ophthalmic veins). It can vary in degree and severity and is typically graded with the Chandler classification.

Contrast-enhanced computed tomography (CT) is advised as first-line imaging because of its superiority in demonstrating bony anatomy and pathology of the orbit and sinuses with its speed and ease of examination. MRI provides outstanding soft tissue detail without radiation exposure.

Chandler stages				
Stage 1	Stage 2	Stage 3	Stage 4	Stage 5
Preseptal cellulitis	Postseptal cellulitis or orbital cellulitis without abscess	Subperiosteal abscess	Orbital abscess	Cavernous sinus thrombosis/abscess

Chandler stages				
Inflammation does not extend beyond t he orbital septum (where the medial o rbital periosteal refl ection attaches to t he medial eyelid at the tarsal plate).	Inflammation exten ds into the tissues o f the orbit.	There is abscess for mation deep to the periosteum of the o rbital bones, typical ly at the lamina pap yracea from ethmoi d sinusitis.	There is abscess for mation within the o rbit which has brea ched the periosteu m.	The inflammatory p rocess has extended into the cavernous sinus which thromb oses and may progr ess to abscess form ation.

Solution to Question 9:

The malleo-incudal joint is a saddle type.

The type of joint between the middle ear ossicles is a synovial joint. The incudostapedial joint is a ball and socket type.

Note: The image in the question has been updated to include the arrow which was initially missing.

Solution to Question 10:

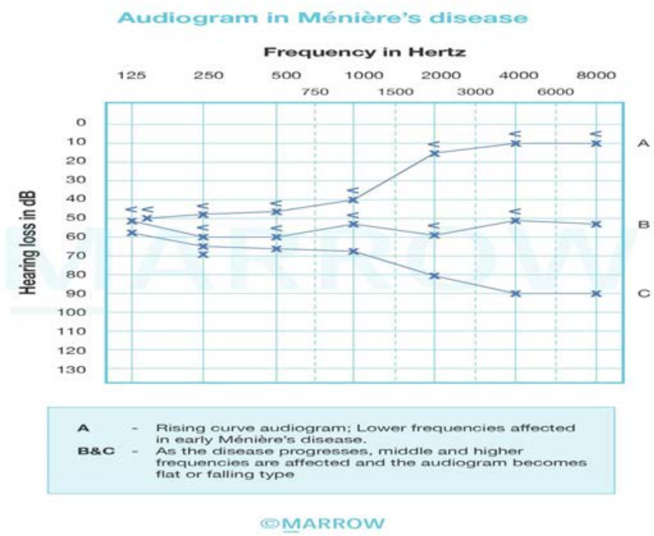
The given image shows an upsloping/ ascending audiogram in which lower frequencies are affected first like in the early stages of Meniere's disease also known as endolymphatic hydrops.

Meniere's disease (endolymphatic hydrops) is an inner ear disorder where the endolymph distends the endolymphatic system.

Meniere's disease is usually unilateral. It is characterized by episodic vertigo, fluctuating sensorineural hearing loss, tinnitus, and a sense of ear fullness.

Pure tone audiometry reveals a sensorineural type of hearing loss with lower frequencies affected first. It shows a rising type of curve. As the disease progresses, middle and higher frequencies get involved, and the audiogram becomes flat or downsloping type.

The illustration given below shows an audiogram seen in different stages of Meniere's disease.

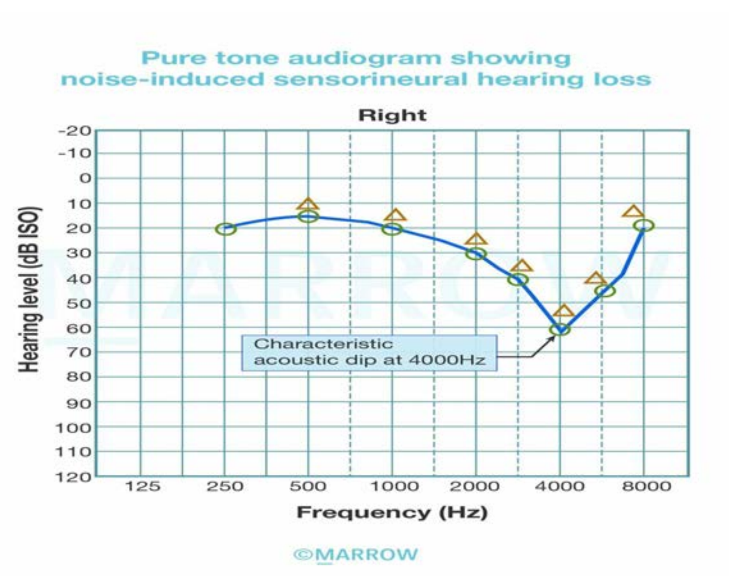


Other options:

Option A: Ototoxicity causes bilateral high-frequency sensorineural hearing loss. Audiometry shows a downsloping audiogram, typically involving frequencies above 8,000 Hz and progresses to lower speech frequencies. Common drugs causing the same include aminoglycoside antibiotics; loop diuretics (not thiazides); antimalarials; cisplatin; and NSAIDs like ibuprofen, salicylates, and indomethacin.

Option B: Presbycusis is a physiological sensorineural hearing loss occurring in individuals above 65 years of age. Audiometry shows a downsloping graph.

Option C: In noise-induced hearing loss, audiometry shows a typical notch at 4000 Hz. As the duration of noise exposure increases, the notch becomes deeper and wider, involving lower and higher frequencies.



Solution to Question 11:

In the above clinical scenario, a patient presenting with unilateral obstruction of the nasal cavity due to the presence of mass and nasal bleeding along with the histopathology showing several sporangia is most likely suggestive of the diagnosis of rhinosporidiosis.

Rhinosporidiosis is a chronic granulomatous disease caused by Rhinosporidium seeberi. It is most prevalent in rural districts, among persons bathing in public ponds or working in stagnant water, such as rice fields. Most common in persons in the age range of 15–40; males are more commonly affected than females.

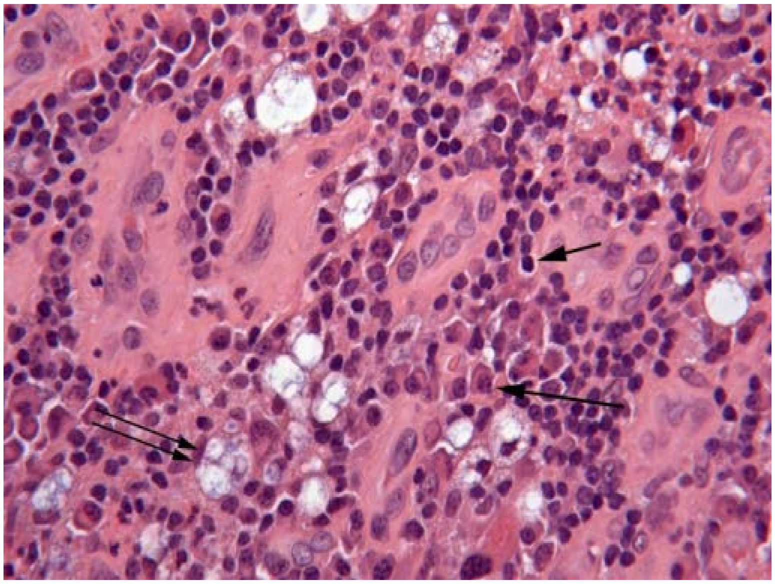
The rhinoscopic examination will reveal papular or nodular, smooth-surfaced lesions that become pedunculated and acquire a papillomatous or proliferative appearance. The lesions are pink, red, or purple in color. Histopathology shows several sporangia, oval, or round in shape and filled with spores.

The treatment of choice is surgical excision of lesions, with or without cauterization. Not many drugs are effective against this disease. Dapsone has been tried with some success.

A patient with rhinosporidiosis is shown below.

Other Options:

Option B: Rhinoscleroma is caused by Klebsiella rhinoscleromatis (Frisch bacillus). Histopathology shows large foam cells with a central nucleus and vacuolated cytoplasm containing the causative bacilli, which are Mikulicz cells. There are also homogenous eosinophilic inclusion bodies consisting of the immunoglobulins in the plasma cells called the Russell bodies.



Option C: Inverted papilloma (transitional cell papilloma) is a tumour originating from the nonolfactory mucosa of the nose (Schneiderian membrane) and paranasal sinuses. The unilateral growth of hyperplastic papillomatous tissue into the stroma characterizes this condition. It primarily affects men aged 40-70 and presents unilaterally with symptoms such as nasal obstruction, nasal discharge, and epistaxis. Invasive growth into the sinuses or orbit can lead to proptosis, diplopia, and lacrimation. A biopsy is essential for diagnosis. Histology shows a thin respiratory epithelium overlying the surface and an underlying stroma containing multiple nodules of thick neoplastic epithelium growing inward.

Option D: Juvenile nasopharyngeal angiofibroma is seen almost exclusively in males in the age group of 10–20 years and is very rarely seen in older people and females. Profuse, recurrent, and spontaneous epistaxis is the most common presentation. Diagnosis is mostly clinical. Contrast-enhanced CT scan of the head is the investigation of choice that reveals anterior bowing of the posterior wall of the maxillary sinus (Antral sign/Holman Miller sign) which is pathognomonic of angiofibroma.

Solution to Question 12:

Coughing is a type of forceful expiration. During forceful expiration, when pressure builds up, the glottis opens and then the air comes out. The cough reflex is a protective mechanism of the larynx to expel irritants and keep the airway clear.

Coughing begins with a deep inspiration followed by a forced expiration against a closed glottis. This increases the intrapleural pressure to 100 mm Hg or more. The glottis is then suddenly opened, producing an explosive outflow of air at velocities up to 965 km (600 ml) per hour.

Both chemical and mechanical stimuli can initiate the cough reflex. Chemical or mechanical irritation of the respiratory mucosa stimulates the afferent internal laryngeal nerve, which is the branch of the superior laryngeal nerve of the vagus nerve. These sensory fibres synapse in the nucleus tractus solitarius in the medulla oblongata. Complex neural networks here generate the conscious 'urge to cough' while also sending efferent signals to the various muscles involved in generating an effective cough which include the adductors of the vocal cord, the diaphragm, the external intercostals and the muscles of the anterior abdominal wall.

Other Options

Statement ii and iii: It is the forceful expiration, not just a normal expiration or deep inspiration which opens the glottis.

Solution to Question 13:

In the given clinical scenario, a female with progressive bilateral hearing loss, in the right ear more than the left, and a stapedotomy of the right ear was planned. Stapedotomy surgery is done for otosclerosis. Otosclerosis results in conductive hearing loss so, the tuning fork test results of the patient will be Bilateral Rinne's negative, Weber's lateralized to the right.

As there is conductive hearing loss in both ears, like that associated with otosclerosis (which may require a stapedotomy), the bilateral Rinne test will be negative. In Weber's test, the sound lateralizes to the affected ear, which, in this case, is the right ear with more significant hearing loss.

Tuning fork tests are performed commonly with tuning forks of frequencies such as 128, 256, 512, 1024, etc. The tuning fork of 512 Hz is considered ideal as forks of lower frequencies produce a sense of bone vibration, while those of higher frequencies have a shorter decay time.

Rinne’s test is done by placing a vibrating tuning fork on the patient’s mastoid and when he stops hearing, it is brought beside the external auditory meatus (EAC). If he still hears the sound, air conduction (AC) is better than bone conduction (BC) i.e. AC>BC. The inferences are as follows:

• Rinne's test is considered positive (normal) when the air conduction AC>BC is seen in normal people and people with sensorineural deafness.

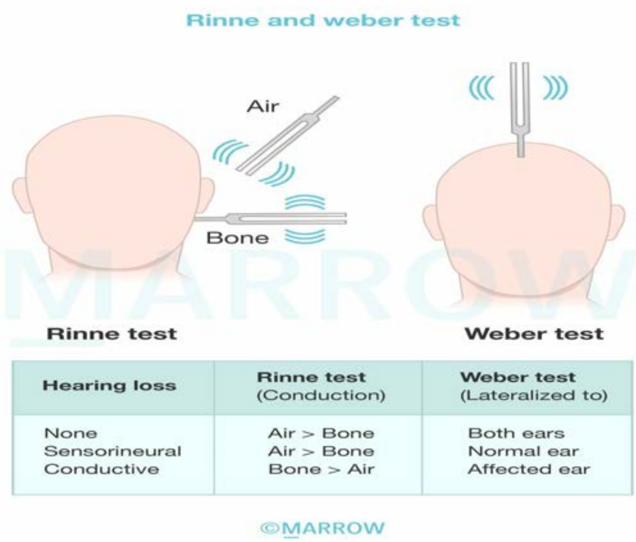
• It is considered negative when the BC>AC, and is seen in patients with conductive deafness.

Weber test is done by placing a vibrating tuning in the middle of the forehead or the vertex and the patient is asked in which ear the sound is heard. The inferences are as follows:

• Normal result is achieved when the sound heard equally in both ears

• In conductive deafness, it is lateralized to the worse ear

• In sensorineural deafness, it is lateralized to the better ear.



Solution to Question 14:

In the given clinical scenario, the patient can only hear when words are shouted, which is a severe hearing loss. The degree of his hearing impairment will be Severe hearing loss > 70%.

Our normal conversational speech is only up to 60 dB. If we have to shout, it means the patient is having a hearing loss greater than 60 dB.

WHO grades of hearing loss:

World Hearing Day 2023 observed on March 3, will highlight the importance of integrating ear and hearing care within primary care, as an essential component of universal health coverage. The theme for this year's campaign is "Ear and hearing care for all, let's make it a reality"

Grade of hearing loss range	Range of hearing loss	Performance	Recommendation
0: No impairment	25 dB or better	No or very slight hearing probl ems. Able to hear whispers	None
1: Slight impairment	26-40 dB	Able to hear and repeat words spoken in normal voice at 1 me ter	Counselling. Hearing aids may be needed
2: Moderate impairment	41-60 dB	Able to hear and repeat words using raised voice at 1 meter	Hearing aids are usually recom mended
3: Severe impairment	61-80 dB	Able to hear some words when shouted into better ear	Hearing aids needed. If no hea ring aids available, lip reading should be taught
4: Profound	81 dB or greater	Unable to hear and understand even a shouted voice whether in quiet or noise.	Hearing aids may help in unde rstanding words. Additional re habilitation is needed. Lip-read ing and sometimes signing ess ential

ENT INI-CET 2024

Question 1:

All of the following are features of otitis media with effusion EXCEPT?

a) Most cases have Ad type of curve on tympanometry and a conductive hearing loss

b) Resolves on its own after the symptoms subside

c) Usually occurs after an episode of common cold, but can occur de-novo

d) It is usually bilateral

Question 2:

Which of the following is typically not seen in allergic rhinitis?

a) Allergic shiners

b) Allergic salute

c) Otto Veraguth folds

d) Dennie-Morgan line

Question 3:

A 13 year old male patient presents with unilateral nasal obstruction and recurrent episodes of epistaxis for past 6 months. What is the most likely diagnosis?

a) Antrochoanal polyp

b) Juvenile nasopharyngeal angiofibroma

c) Coagulation disorder

d) Allergic rhinitis

Question 4:

Which of the following is not recommended in an 8 yr old child with bilateral sensorineural hearing loss?

a) Adenoidectomy and grommet insertion

b) Cochlear implant

c) Hearing aid

d) Sitting in front row of the class

Question 5:

A young male patient presents to ER with complaints of daytime somnolence and repeated awakening at night from sleep and his partner gives history of snoring in bed. All of the following are true except?

a) Apnea results in awakening

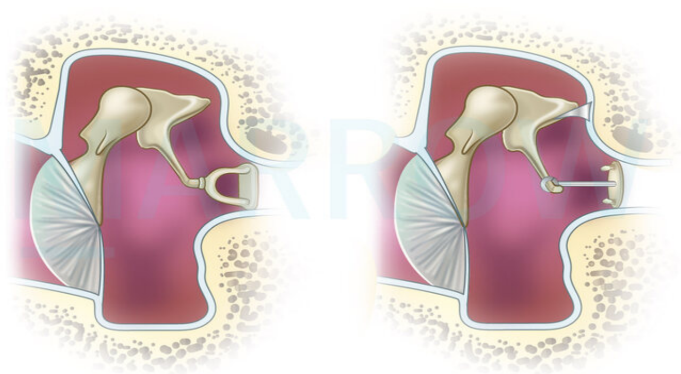
b) Apnea is due to fall in oxygen saturation

c) Low oxygen stimulates respiratory effort

d) Pharngeal muscle tone increases

Question 6:

Identify the procedure.



a) Stapedotomy

b) Myringoplasty

c) Tympanoplasty

d) Ossiculoplasty

Question 7:

What is the most commonly used test in screening for neonatal hearing?

a) Conduction audiometry

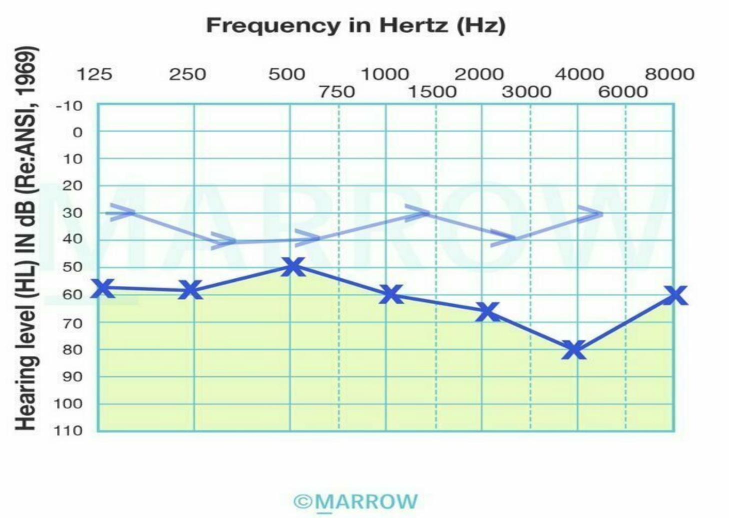
b) Otoacoustic emissions (OAE)

c) Brainstem evoked response audiometry

d) Clap and smile

Question 8:

Identify the type of hearing loss.



a) Mixed hearing loss

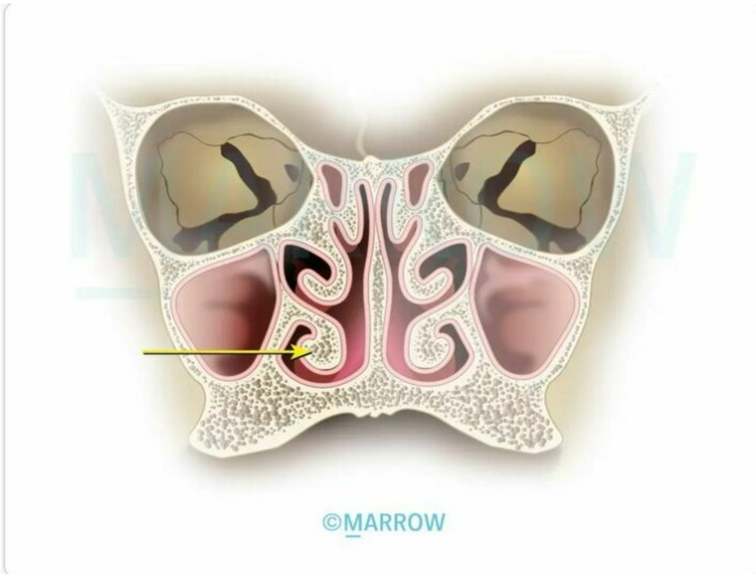
b) Conductive hearing loss

c) Sensory hearing loss

d) Normal

Question 9:

The posterior end of the marked structure is present at the level of which of the following?



a) Eustachian tube

b) Sphenopalatine foramen

c) Odontoid process

d) Sphenoidal sinus

Question 10:

Carhart's notch is seen in _______

a) Otosclerosis

b) Noise-induced hearing loss

c) Meniere's disease

d) Serous otitis media

Question 11:

Which of the following is the most dangerous lesion?

a) U/L Partial RLN lesion

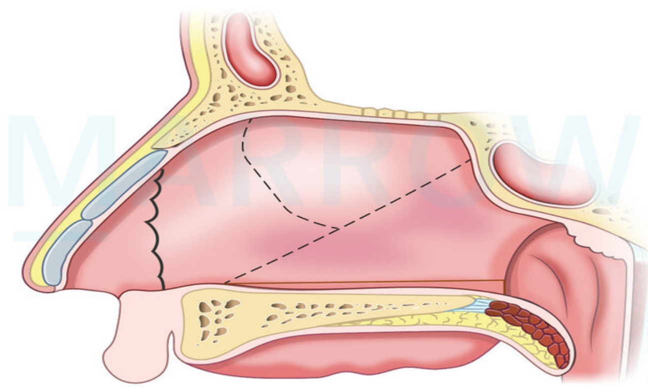
b) B/L Partial RLN lesion

c) U/L Complete RLN lesion

d) B/L Complete RLN lesion

Question 12:

Identify the nasal fracture.



a) Chevallet fracture

b) Jarjaway fracture

c) Simple fracture

d) Le fort fracture

Question 13:

What causes maximum hearing loss?

a) Malleus-incus discontinuity with intact tympanic membrane

b) Malleus-incus discontinuity with perforation of tympanic membrane

c) Partial fixation of the footplate of stapes

d) Otitis media with effusion

Answer Key

Question No.	Correct Option
1	a
2	c

3	b
4	a
5	d
6	a
7	b
8	a
9	a
10	a
11	d
12	a
13	a

Detailed Explanations

Solution to Question 1:

Type B tympanometry curve is obtained in otitis media with effusion.

Serous otitis media also known as secretory otitis media, mucoid otitis media, or glue ear is characterized by the accumulation of nonpurulent, serous effusion in the middle ear cleft.

Glue ear can occur either due to eustachian tube dysfunction (adenoid hyperplasia, chronic sinusitis or tonsillitis or rhinitis, cleft palate, tumors of nasopharynx) or due to increased secretory activity of the middle ear mucosa (allergy, unresolved otitis media, viral infections).

Glue ear is commonly seen in children aged 5-8 years and is associated with insidious onset of hearing loss (not exceeding 40 dB), mild ear aches, and a history of upper respiratory tract infections. Delayed or defective speech may also result from hearing loss.

Otoscopy shows:

• Dull and opaque tympanic membrane with loss of light reflex

• Thin leash of vessels along the handle of the malleus

• Fluid levels and air bubbles can be seen when the tympanic membrane is transparent.

The otoscopy image of serous otitis media is shown below:

Treatment is mainly aimed at the removal of fluid and preventing recurrence.

• Medical management includes decongestants, antihistamines, and antibiotics.

• Surgical management is done if the condition cannot be controlled with medications.
It includes myringotomy and aspiration of fluid with grommet insertion. Adenoidectomy and tonsillectomy may be required and are usually done at the time of myringotomy.

Hearing tests suggestive of conductive hearing loss as seen in serous otitis media:

• Rinne negative (BC>AC).

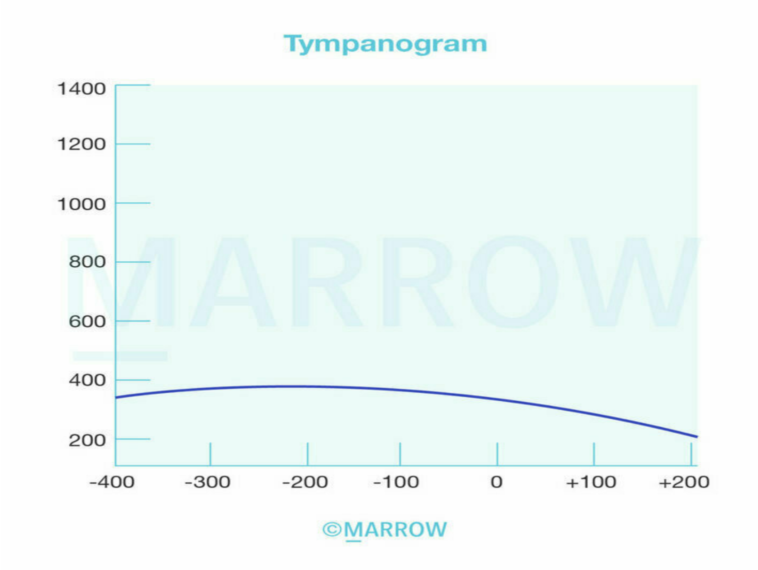
• Weber - lateralized towards the worse ear

• Audiogram - AC abnormal, BC normal, air-bone gap present

• Impedance audiometry - flat curve (reduced compliance, shift to the negative side)

B-type tympanogram is expected in a case of serous otitis media.

Here, compliance decreases. The fluid accumulation causes decreased mobility of the tympanic membrane leading to a flat or dome-shaped curve, this results in a B-type tympanogram, as shown below:



Note: Tympanograph is Flat when there is Fluid in the middle ear.

Solution to Question 2:

Otto Veraguth folds are not seen in allergic rhinitis.

Otto Veraguth fold is a physical sign of depression wherein there is a triangular fold in the nasal corner of upper eyelid.

Allergic rhinitis is an IgE-mediated immunological response of the nasal mucosa to allergens in the environment.

Patients typically present with recurrent bouts of sneezing, nasal obstruction, watery nasal discharge, and itching.

Clinical signs include:

• Allergic shiners: dark circles around the eyes.

• Allergic salute: black transverse nasal crease on the dorsum of the nose due to repeated upward rubbing.

• Dennie-Morgan lines - transverse skin creases in the infraorbital area.

Allergic rhinitis can be classified into intermittent and persistent, based on the duration of the disease:

• Intermittent- Symptoms are present for less than 4 days a week, or less than 4 weeks

• Persistent- Symptoms are present for more than 4 days a week, or more than 4 weeks

It can be classified as mild, moderate, or severe based on the severity of the disease:

• Mild- None of the following symptoms are present- sleep impairment, impairment of daily activities, school, or work, troublesome symptoms.

• Moderate- severe: If at least one of the symptoms mentioned above are present.

A step-care approach is recommended for allergic rhinitis and its impact on asthma (ARIA) for the treatment of allergic rhinitis includes the following:

• Oral antihistamines or intranasal cromolyn sodium are recommended for mild, intermittent disease. (option A).

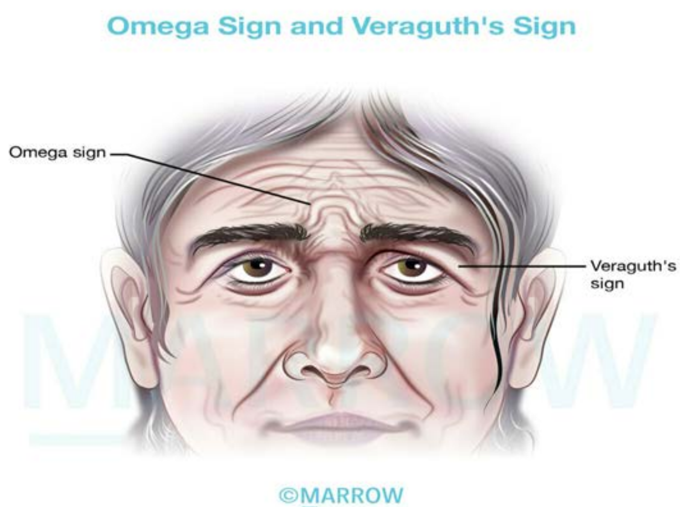
• Intranasal corticosteroids are used for moderate disease as monotherapy. (option B).

• Oral antihistamines and intranasal steroids can be used in combination to treat severe diseases. (option C).

• A short course of oral corticosteroids and immunotherapy can be used for severe, persistent disease in case the above intervention fails

In the given image, Otto Veraguth's sign or Veraguth's fold is shown, a diagonal palpebral fold running from the lateral corners of the eyes, medially upward to the medial end of the eyebrows. It is a physical sign of depression.

Another physical sign of depression is the Omega sign. In this, there is vertical wrinkling between the eyebrows joined at the top by a horizontal crease.



Solution to Question 3:

The most likely diagnosis is juvenile nasopharyngeal angiofibroma (JNA), as there is unilateral nasal obstruction and recurrent epistaxis.

Juvenile nasopharyngeal angiofibroma (JNA) is a highly vascular and locally aggressive tumor seen almost exclusively in adolescent males, associated with significant morbidity and occasional mortality.

Patients usually present with:

• Epistaxis (most common presentation): Profuse, recurrent, and spontaneous

• Mass: A mass may be present in either the postnasal space, resulting in progressive nasal obstruction, or nasopharynx, where it may obstruct both choanae. The mass is firm in consistency, but digital palpation of the same should not be done. If the mass obstructs the eustachian tube, conductive hearing loss and otitis media with effusion may be observed.

• Proptosis, swelling of the cheeks, broadening of nasal bridge, and involvement of 2nd, 3rd, 4th and 6th cranial nerves may occur

JNA can be diagnosed using CT, MRI, and carotid angiography.

Anterior bowing of the posterior wall of the maxillary sinus due to the presence of a mass in the pterygomaxillary space is a characteristic CT finding of JNA. This is known as the Holman-Miller sign or antral sign.

Recurrence has been reported up to 35%. Surgical excision is the treatment of choice. Recurrent angiofibroma is treated with revision surgery or radiotherapy when the tumor is inaccessible.

Solution to Question 4:

Adenoidectomy and grommet insertion is not recommended in bilateral sensorineural hearing loss (SNHL).

Adenoidectomy and grommet insertion is done in the management of serous otitis media.

Serous otitis media is also known as secretory otitis media or mucoid otitis media or glue ear. It is characterized by the accumulation of nonpurulent effusion in the middle ear cleft.

Treatment is mainly aimed at the removal of fluid and preventing recurrence.

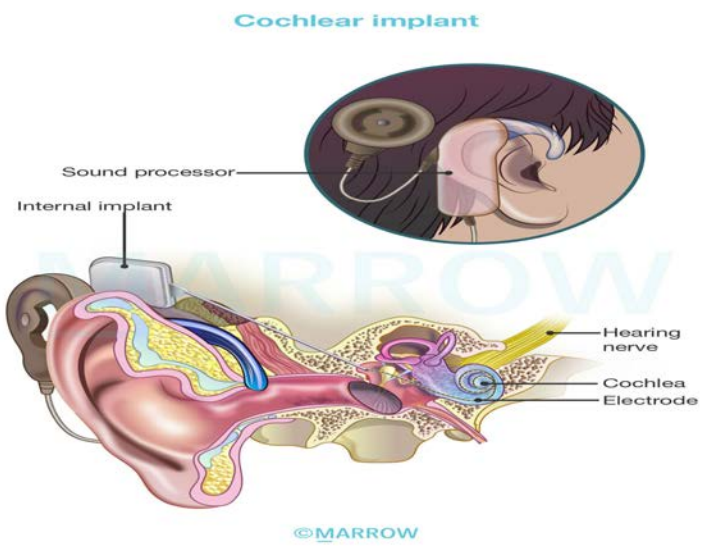
• Medical management includes decongestants, antihistamines, antibiotics, and middle ear aeration techniques.

• Surgical management includes myringotomy and aspiration of fluid, grommet insertion, tympanotomy, or cortical mastoidectomy. Adenoidectomy and tonsillectomy may be required and are usually done at the time of myringotomy.

Option B: Cochlear implant is an electronic device which helps in hearing for those with profound sensorineural hearing loss, who cannot benefit from hearing aids. It has two components :

• External component : speech processor and transmitter

• Internal component : receiver / stimulator and electrode array



Sound signals received by the processor is then conducted to the stimulator via the transmitter. It is then decoded and sent to the electrode array which is placed over the scala tympani. Scala tympani then transmits the signal to the spiral ganglion, which in turn stimulates the auditory nerve cells. Thus, sound is conducted to the brain.

Option C: Hearing aids are used in the management of SNHL. Hearing aids are of the following types

• Conventional hearing aids

• Bone-anchored hearing aids (BAHAs)

• Implantable hearing aids (vibrant soundbridge)

Option D: Sitting in the front row of class is recommended in SNHL to help the child hear the class better.

Solution to Question 5:

Pharyngeal muscle tone decreases during snoring.

Pharyngeal muscles are relaxed during sleep and cause partial obstruction. Breathing against obstruction causes vibrations of soft palate, tonsillar pillars and base of tongue producing sound.

Types of snoring:

• Primary snoring: without association with obstructive sleep apnea (OSA). Primary snoring is not associated with excessive daytime sleepiness and has apnea—hypnoea index of less than five.

• Complicated snoring: associated with OSA.

Obstructive sleep apnea-hypopnea syndrome (OSAHS) is diagnosed if the patient has either symptoms of nocturnal breathing disturbances or daytime sleepiness, and ≥5 episodes of obstructive apnea or hypopnea per hour of sleep. Each such episode should last for at least 10 seconds, resulting in ≥3% drop in oxygen saturation or a brain cortical arousal. OSA is diagnosed by overnight polysomnogram (PSG).

OSAHS is characterized by recurrent episodes of upper airway obstruction during sleep, leading to intermittent hypoxia and sleep fragmentation

Usually, there is a decline in neuromuscular output to pharyngeal dilator muscles during sleep. In predisposed individuals, this results in complete (apnea) or partial (hypopnea) pharyngeal collapse.

Solution to Question 6:

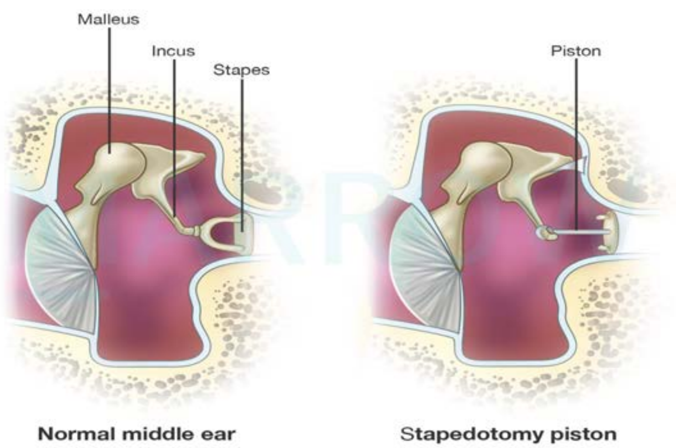
The above image shows a stapedotomy procedure performed in the middle ear.

It can be identified as a stapedotomy procedure by the following features in the image given:

• The footplate of the stapes is not removed

• The prosthesis is placed by creating a hole in the footplate of the stapes

Stapedotomy is the excision of the superstructure of stapes and drilling a hole in the footplate & placing a piston in the hole. It is done to bypass the immobile stapes using a piston which will take over the function of conducting sound.



Contraindications for Stapedotomy:

• Only hearing ear

• Positive Schwartze sign

• Otitis external

• Meniere’s disease: can damage the inner ear while drilling a hole in the footplate

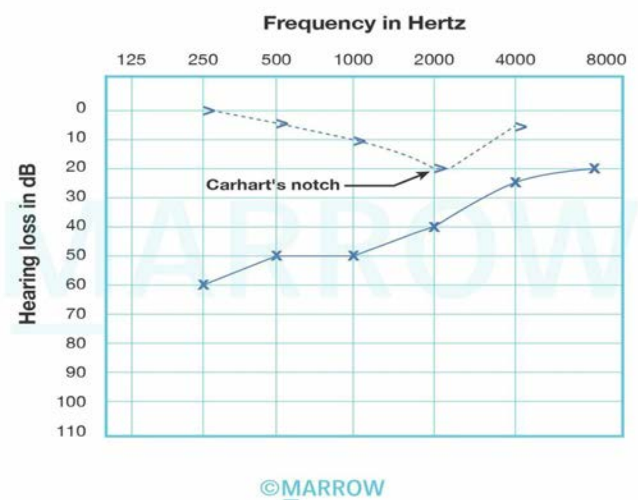
• Young children

• Professional athletes, pilots, construction workers, and divers: Sudden changes in pressure can displace the piston and cause vertigo.

• Those who work in noisy surroundings: Surgery makes them more vulnerable to noise-induced hearing loss.

Otosclerosis is a primary disease of the bony labyrinth where foci of irregularly placed spongy bone replace part of the normally dense enchondral layer of the bony otic capsule. Otosclerosis commonly involves the stapes, causing stapes fixation and conductive hearing loss. Clinically, patients present with painless progressive bilateral conductive hearing loss and paracusis willisii (patients hear better in noisy surroundings as other people raise their voices to be heard). Pure tone audiometry would reveal conductive hearing loss with a dip in bone conduction at 2000 Hz (Carhart’s notch). The definitive treatment for otosclerosis involves stapedectomy, in which the otosclerotic stapes is removed, and a prosthesis is placed in its place.

The image shown below shows Carhart's notch.



Other options

Option A: Myringoplasty is the term given for the closure of perforation of pars tensa of the tympanic membrane. It is done to check repeated infections from the external ear.

Option B: Tympanoplasty is the surgical reconstruction to correct damage in the middle ear and restore the integrity of the tympanic membrane. It involves ossiculoplasty (ossicular chain reconstruction) with or without myringoplasty.

Option C: Ossiculoplasty eradicates the disease from the middle ear and reconstruction of hearing.

Solution to Question 7:

Otoacoustic emissions (OAE) are used in the universal neonatal hearing screening protocol.

In India, a two-stage screening protocol is followed according to which infants are screened first with otoacoustic emissions (OAE). Infants who fail the OAE are screened with auditory brainstem response (ABR) or Brainstem Evoked Response Audiometry (BERA), which is also the definitive test. All babies kept in NICU must be screened by BERA test as per the protocol.

The first screening is usually conducted at birth or before discharge. Conducting the first screening before discharge is preferred to ensure better coverage and reduce loss to follow-up. A second screening is scheduled anytime between 1–3 weeks or during the next scheduled visit at 6 weeks.

All babies with abnormal BERA should undergo detailed ENT evaluation and hearing-aid fitting (cochlear implants) and auditory rehabilitation must be done before 6 months of age.

Other options

Option A: Conduction audiometry is used in older children and adults to measure hearing thresholds, requiring active cooperation, which is not feasible in neonates.

Option C: Brainstem evoked response audiometry (BERA): While BERA is more precise and can detect auditory pathway issues up to the brainstem, it is more time-consuming and costly. It is typically used for confirmation if OAE results are abnormal.

Option D: Clap and smile is an outdated, subjective test and is not reliable for neonatal hearing screening.

Solution to Question 8:

The given audiogram is suggestive of left-sided mixed hearing loss.

Interpretation of the audiogram:

• Blue colour indicates the left ear.

• > indicates bone conduction (BC), and the threshold is >25 dB. Hence, BC is abnormal, which implies that the abnormality is in the sensorineural pathway.

• × indicates air conduction (AC), and the threshold is >15 dB. Hence, AC is abnormal.

• Both AC and BC are abnormal. It indicates sensorineural hearing loss.

• In addition, the air-bone gap is present, which implies conductive hearing loss.

• Hence, the interpretation is left-mixed hearing loss.

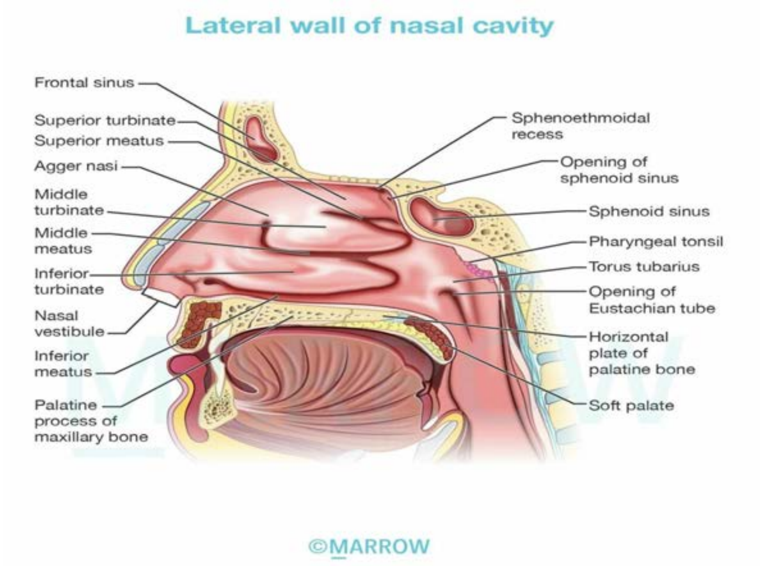
A mixed type of hearing loss presents with defects in both conductive and sensorineural pathways. This type of hearing loss is seen in chronic suppurative otitis media and otosclerosis.

Solution to Question 9:

The marked structure shown is the inferior turbinate. The posterior end of the inferior turbinate is present at the level of the opening of the eustachian tube.

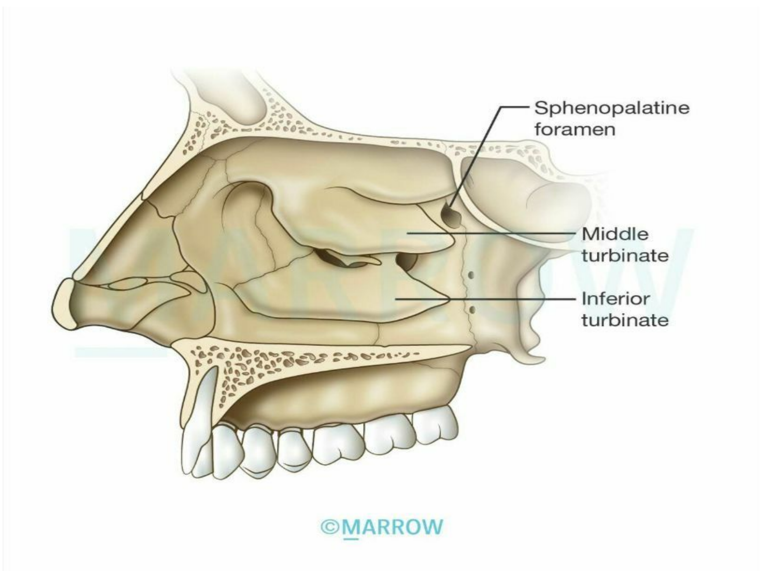
The eustachian tube connects the nasopharynx and the tympanic cavity in the middle ear. It is usually collapsed and opens during activities such as yawning or swallowing. Its function is to maintain the pressure in the middle ear cavity and drain any secretions from the middle ear. It opens approximately 1.25 cm behind the posterior end of the inferior turbinate, in the lateral wall of the nasopharynx.

The inferior turbinate is the largest turbinate located in the lower part of the lateral nasal wall. The Woodruff's plexus is located inferior to the posterior end of the inferior turbinate and is responsible for posterior epistaxis in adults.



Other options:

Option B: Sphenopalatine foramen is situated at the posterior end of the middle turbinate.

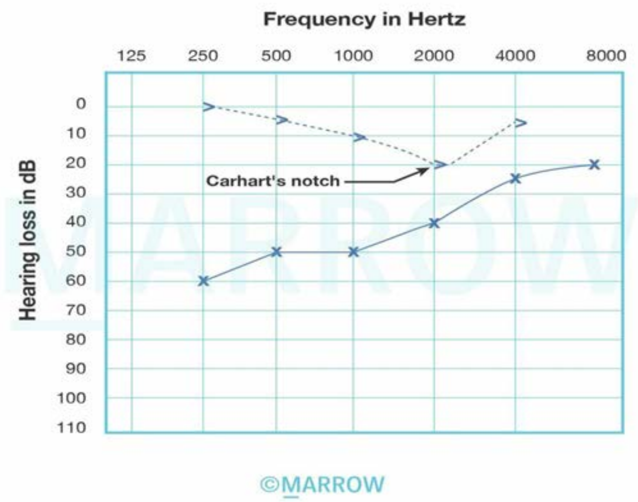


Option C: Odontoid process is a protuberance of the Axis (C2 vertebra). Part of the axis is located much lower in the neck and not associated with the nasal cavity.

Option D: Sphenoidal sinus opens into the nasal cavity at the superior meatus.

Solution to Question 10:

Carhart's notch is seen in otosclerosis. Carhart's notch is a dip at 2000 Hz in Bone Conduction, seen in pure tone audiometry.



Pure Tone Audiometry in Otosclerosis:

• Air conduction is poor & worse for lower frequencies.

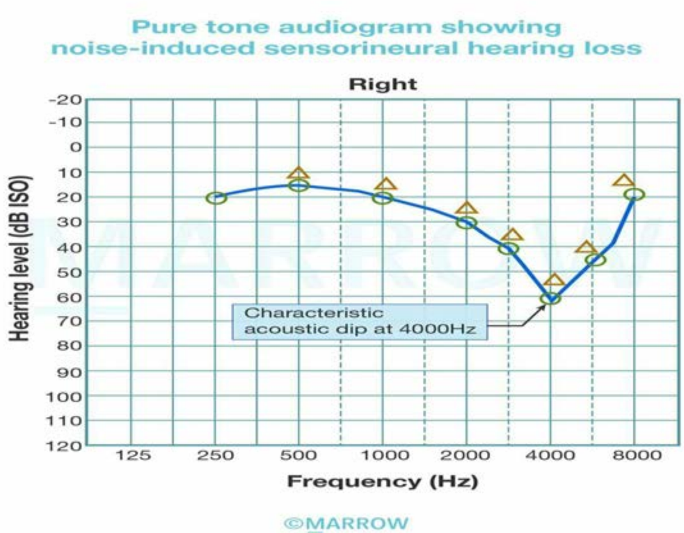
• Large A-B gap

• Bone conduction is normal.

• In some cases, there is a dip in the bone conduction curve.

• It is different at different frequencies but is maximum at 2000 Hz and is called Carhart’s notch.

Note: Boiler’s notch is a similar dip in the BC and AC curve but at 4000 Hz, as seen in noise-induced hearing loss.



Solution to Question 11:

Bilateral complete RLN (recurrent laryngeal nerve) lesion is the most dangerous lesion among the given options.

In bilateral RLN paralysis

• Both the cords lie in the median or paramedian position, the airway is inadequate causing dyspnea and stridor. This may lead to death if untreated.

• The voice is good.

• Dyspnea and stridor become worse on exertion or during an attack of acute laryngitis.

All laryngeal muscles are supplied by the ipsilateral recurrent laryngeal nerve (RLN), except cricothyroid which is supplied by the superior laryngeal nerve (SLN). Both SLN and RLN are branches of the vagus nerve.

The recurrent laryngeal nerves (RLN) on the right and the left side follow different courses-

• Right RLN takes origin in the root of the neck, winds around the first part of the subclavian artery, and then ascends in the tracheoesophageal groove.

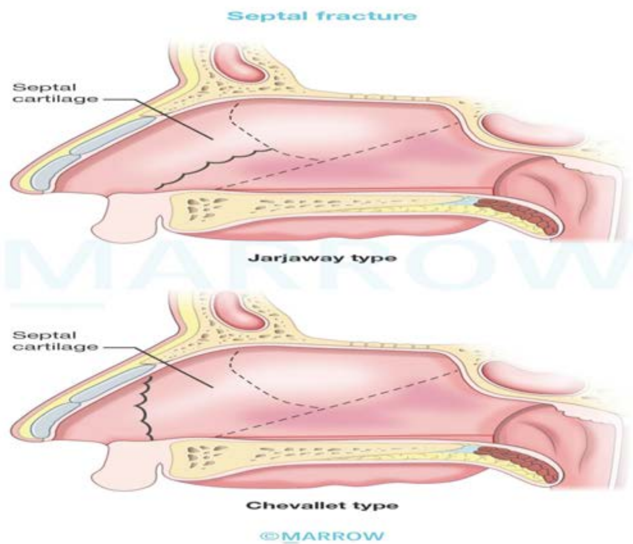
• Left RLN takes origin in the thorax, winds around the arch of the aorta, and then ascends in the tracheoesophageal groove. Since it has a longer course, it is more likely to be injured.

Solution to Question 12:

A Chevallet fracture usually results from a blow to the nose from below. This fracture line runs vertically. It extends from the anterior nasal spine upwards to the junction of the bony and cartilaginous dorsum of the nose.

Injuries to the nasal septum can result from trauma to the nose from the front, side, or below. The septum may get buckled on itself, fracture vertically and horizontally, or get crushed

A Jarjaway fracture (option B) results from a blow to the nose from the front. The fracture line runs horizontally backwards. It begins just above the anterior nasal spine and extends just above the junction of the septal cartilage with the vomer.



Early recognition and treatment of septal injuries are crucial, including drainage of hematomas.

Dislocated or fractured septal fragments should be repositioned and supported between mucoperichondrial flaps using mattress sutures and nasal packing. Nasal pyramid fractures often coincide with septal fractures and should be treated simultaneously.

Neglecting injuries to the septum can lead to deviations in the cartilaginous nose, as well as asymmetry in the nasal tip, columella, or nostrils.

Other options:

Option C: Simple fractures refer to the fractures of the nasal bone without displacement. Simple fracture describes the nature of the fracture.

Option D: Le Fort fractures are a group of fractures that affect the midface of the skull and collectively involve a partial or complete separation of the midface from the skull.

Solution to Question 13:

Malleus-incus discontinuity with an intact tympanic membrane causes maximum hearing loss.

This condition causes maximum conductive hearing loss, up to 60 dB, as there is no alternative pathway for sound transmission to the cochlea. Even though the tympanic membrane is intact, the disconnection of the ossicular chain severely disrupts sound transmission to the inner ear.

Hearing loss severity depends on the extent to which the normal transmission of sound waves through the middle ear is disrupted.

Average hearing losses seen in different lesions of conductive apparatus

- Complete obstruction of the ear canal: 30 dB
- Perforation of tympanic membrane: 10-40 dB
- Ossicular interruption with intact drum: 54 dB

• Ossicular interruption with perforation: 38 dB

• Malleus fixation: 10-25 dB

• Closure of oval window: 60 dB

Other options

Option B: Malleus-incus discontinuity with perforation of the tympanic membrane: Although this results in conductive hearing loss (38-40 dB), the perforated tympanic membrane allows some direct transmission of sound to the inner ear, slightly reducing the severity compared to an intact tympanic membrane with ossicular discontinuity.

Option C: Partial fixation of the footplate of stapes causes a milder form of conductive hearing loss compared to ossicular discontinuity, typically around 30–40 dB because some sound transmission still occurs through the fixed stapes.

Option D: Otitis media with effusion leads to mild to moderate conductive hearing loss (20–30 dB), as fluid in the middle ear dampens sound vibrations but does not fully block sound transmission.

ENT NEET 2024

Question 1:

Which of the following stain can be used for laryngeal mass seen during laryngoscopic examination?

- **a)** Congo red
- **b)** Silver nitrate
- **c)** H & E
- **d)** Supra vital

Question 2:

An 18-year-old male presents to the emergency department with epistaxis and a nasal mass. What is the investigation of choice to evaluate his condition?

- **a)** Plain CT
- **b)** CECT PNS and nose
- **c)** MRI
- **d)** X-ray

Question 3:

A female patient presents with nausea, vomiting, and a sensation of objects moving around her. She is diagnosed with left vestibular neuritis. Her head impulse test is positive. What is the correct finding?

a) Right saccade with right side movement

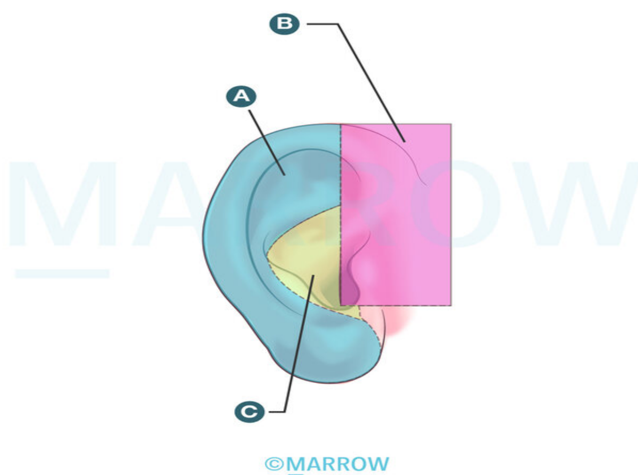
b) Left saccade with right side movement

c) Right saccade with left side movement

d) Left saccade with left side movement

Question 4:

Match the following with the nerve supply of external ear:



a) A - Auriculotemporal nerve; B - Greater auricular nerve; C - Facial and Vagus nerve

b) A - Greater auricular nerve; B - Auriculotemporal nerve; C - Facial and Vagus nerve

c) A - Greater auricular nerve; B - Facial and Vagus nerve; C - Auriculotemporal nerve

d) A - Facial and Vagus nerve; B - Greater auricular nerve; C - Auriculotemporal nerve

Question 5:

A female singer presents with hoarseness of voice and difficulty singing in a high pitch. On laryngoscopy, it was noted that one vocal cord is bowed, positioned lower than the other, and the associated muscle is shortened. Which muscle is likely affected in this scenario?

a) Posterior Cricoarytenoid

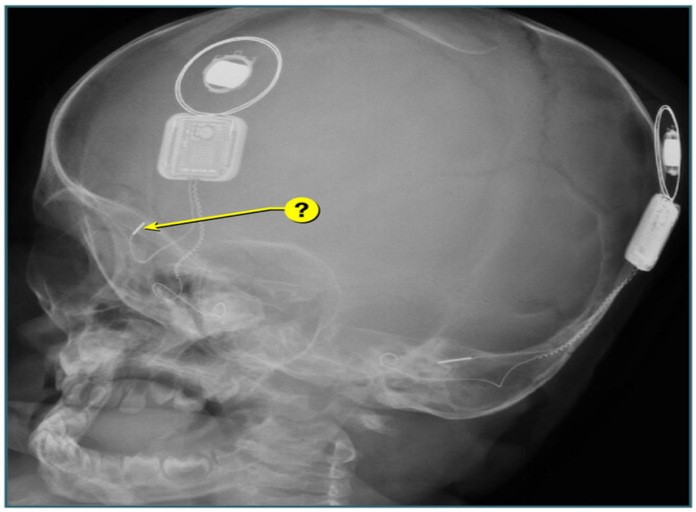
b) Lateral Cricoarytenoid

c) Interarytenoid

d) Cricothyroid

Question 6:

What is the marked structure of the cochlear implant shown below?



a) Ground electrode

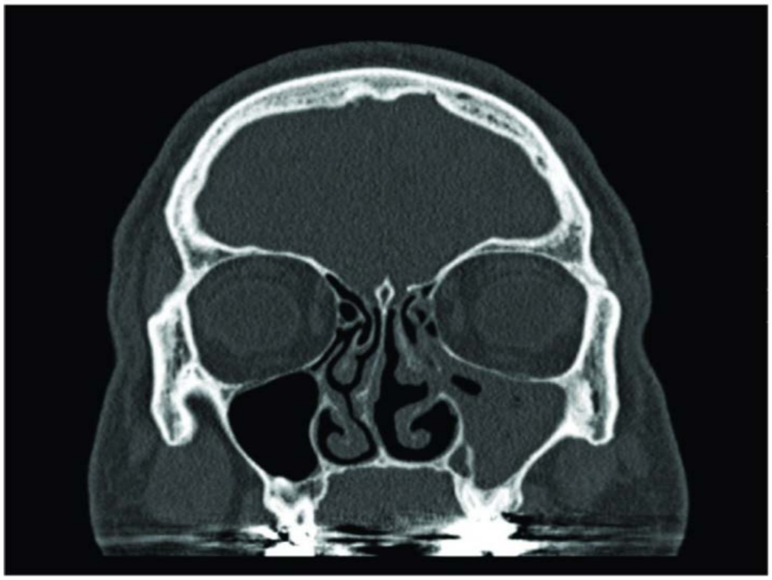
b) Internal magnet

c) Transmitter coil

d) Receiver stimulator antenna

Question 7:

A patient presented with nasal congestion and upon examination, nasal polyps were observed. A coronal CT scan of the patient was taken and is as shown below. Which of the following sinus drainage pathways is likely to be affected?



a) Ethmoid

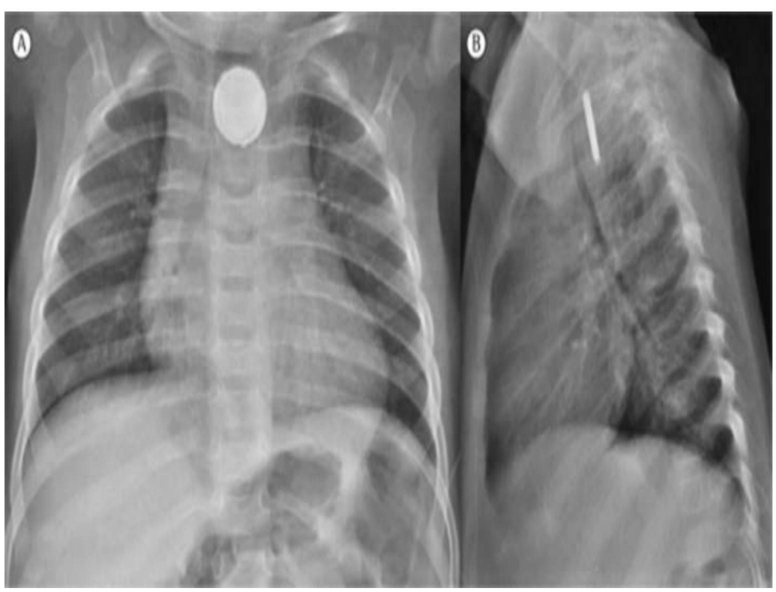
b) Frontal

c) Maxillary

d) Sphenoid

Question 8:

Which of the following statements is true regarding an esophageal foreign body?



a) Impaction can not cause mediastinitis

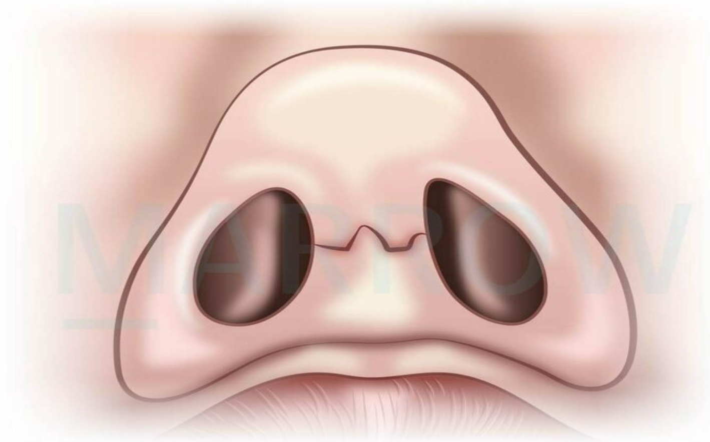
b) More commonly seen in adults

c) Most common site of impaction is at level of cricopharyngeus

d) One of the constrictions of the esopahgus is at the level of the right bronchus

Question 9:

In which procedure is the incision showed in the image used?



a) Caldwell Luc's procedure

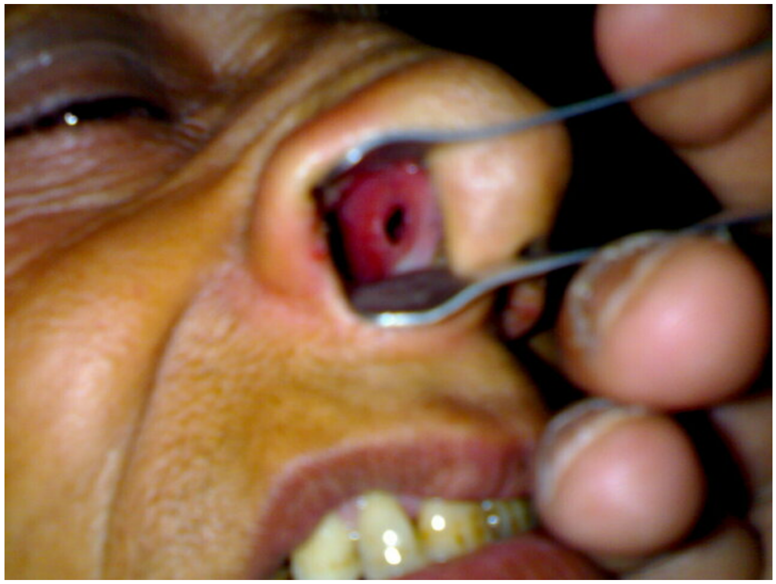
b) Modified Denker's

c) Open rhinoplasty

d) Midface degloving procedure

Question 10:

The patient complains of repeated episodes of crusting in the nose with occasional bleeding. She had undergone surgery for nasal obstruction four months back. On examination, the following finding is seen. Which surgery has the highest likelihood to result in this complication?



a) FESS

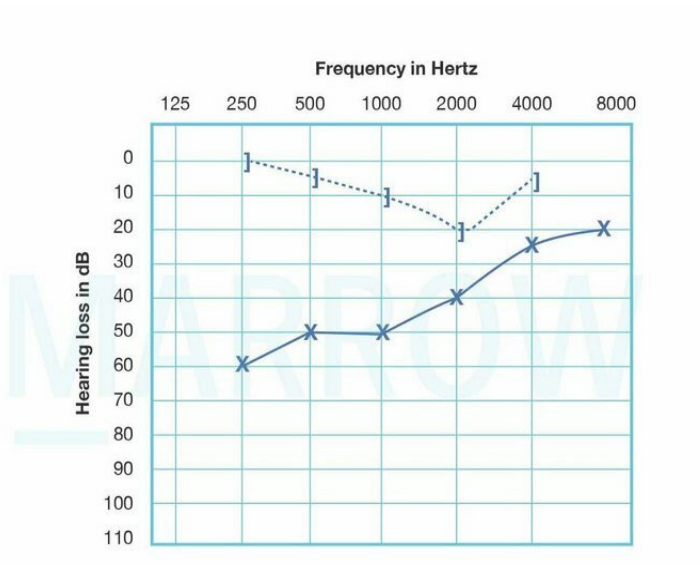
b) Submucosal resection

c) Denker's method

d) Inferior meatotomy

Question 11:

A 40-year-old woman presents with mild, progressive, bilateral hearing loss. A PTA showed the following findings. What is the most likely clinical finding in this patient?



a) Hennebert sign

b) Hitzelberger sign

c) Schwartze sign

d) Bryce's sign

Question 12:

A 70-year-old man presents with unilateral hearing loss. On examination, a small perforation is noted on his left tympanic membrane. What will be the left-sided Rinne's test finding in this patient?

a) Positive

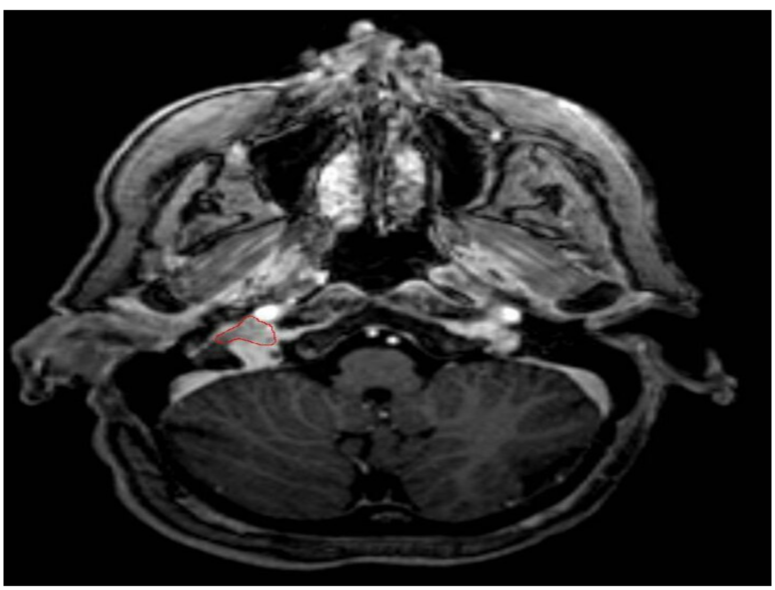
b) False postive

c) Negative

d) Intermediate

Question 13:

A 50-year-old woman presents with hearing loss in the left ear and episodic pulsatile tinnitus. On examination, there was a mass in the middle ear behind the tympanic membrane. A CT performed is as shown below. What is the probable diagnosis?



a) Glomus tympanicum

b) Reparative granuloma

c) Glomus jugulare

d) Acoustic neuroma

Answer Key

Question No.	Correct Option
1	d
2	b
3	c
4	b
5	d
6	a
7	c
8	c
9	c
10	b
11	c
12	c
13	c

Detailed Explanations

Solution to Question 1:

Laryngoscopic image shows growth over the vocal cords; supravital stain is used to detect laryngeal mass seen during laryngoscopic examination.

In the larynx, it is difficult to identify carcinoma in situ, early infiltrating carcinoma, and the exact margins of an established carcinoma. However, it has been found that the use of toluidine blue as a supravital stain during direct laryngoscopy is a reliable adjunct to the early and precise diagnosis of carcinoma of the larynx.

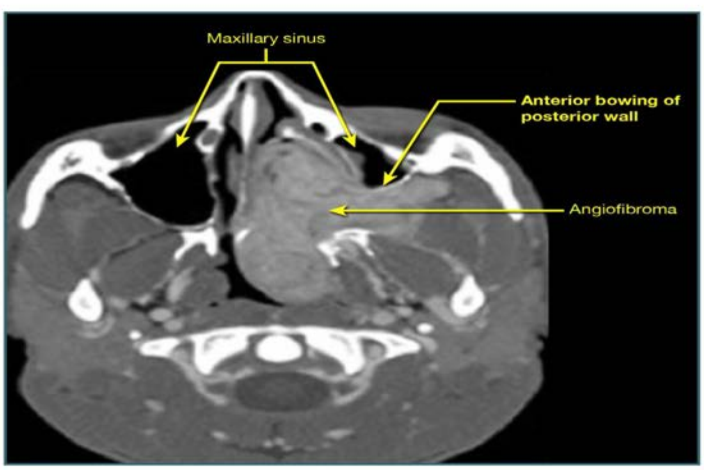
Also, toluidine blue staining is useful in improving the rate of negative resection margins of early glottic cancer (T1a–T2) treated by TLM (transoral oral microsurgery). This technique is called chromoendoscopy, where the interpretation is based on the colour: a dark blue (royal or navy) stain is considered positive, light blue staining is doubtful, and when no color is observed, it is interpreted as a negative stain, as shown below

Solution to Question 2:

History of nasal mass and associated epistaxis in an 18-year-old raises the suspicion of nasopharyngeal angiofibroma. CECT of the PNS and nose can be done to evaluate the patient.

CECT of the PNS and nose is particularly useful for revealing characteristic findings such as the anterior bowing of the posterior wall of the maxillary sinus, known as the Antral sign or Holman-Miller sign (shown in the image below), which is indicative of nasopharyngeal angiofibroma.

Antral sign / Holman-Miller sign



Nasopharyngeal angiofibroma is a benign but locally invasive tumor that predominantly affects adolescent males aged 10-20 years. It typically originates from the posterior part of the nasal cavity near the superior margin of the sphenopalatine foramen and is thought to be testosterone-dependent. Clinically, this condition presents with profuse, recurrent spontaneous epistaxis, progressive nasal obstruction, conductive hearing loss, and otitis media with effusion. It may also cause additional features such as proptosis, cheek swelling, and broadening of the nasal bridge.

Diagnosis is primarily based on clinical presentation, with CECT serving as the preferred imaging technique to confirm the diagnosis and assess the extent of the tumor. Carotid angiography may be performed to evaluate the vascularity of the tumor and its feeding vessels, typically arising from the external carotid artery, and potentially from the internal carotid system in cases of larger tumors with intracranial extension. Biopsy of the mass is generally avoided due to the risk of severe bleeding.

Staging of juvenile nasopharyngeal angiofibroma

Surgery is the treatment of choice. Embolization of the vessels can be done prior to surgery to reduce the risk of bleeding during the surgery. Radiotherapy, chemotherapy, and hormonal therapy can also be considered.

Stage	Spread
IA	Limited to the nose and/or nasopharynx
IB	Extension into ≥ one paranasal sinus
IIA	Minimal extension into the sphenopalatine foramen with minimal part of pterygopalatine fossa
IIB	Full occupation of pterygopalatine fossa with Holman-Miller sign, may with erosion of orbital bones
IIC	Extension into the cheek, temporal fossa, or posterior to pterygoids through the pterygomaxillary fossa
IIIA	Erosion of skull base with minimal intracranial
IIIB	Erosion of skull base with extensive intracranial with or without cavernous sinus

Solution to Question 3:

In the case of left vestibular neuritis, the correct finding on the head impulse test is a right saccade with left-side movement.

The head impulse test is to assess the vestibulo-ocular reflex (VOR). The abnormal head impulse test result is due to the VOR being impaired on the side of the vestibular lesion leading to saccades. Saccades are rapid, ballistic movements of the eyes that abruptly try to return to the point of fixation/ gaze.

The head impulse test is done by asking the patient to focus on a target. Then, the patient's head is quickly rotated to the left or right. If there is a deficiency in the VOR, the head movement will be followed by a catch-up saccade in the opposite direction. Thus, in left vestibular neuritis, a rightward saccade is seen following a left-side head rotation.

Vestibular neuritis is distinguished from central causes of vertigo by its characteristic findings: a positive head impulse test, absence of skew deviation, and spontaneous unidirectional nystagmus. Central vertigo, on the other hand, presents with a normal head impulse test, vertical skew deviation, bidirectional nystagmus, and additional neurological signs like diplopia, weakness, or dysarthria. In acute vestibular syndrome, especially in older patients with vascular risk factors, evaluation for a possible stroke is warranted.

Vestibular neuritis resolves spontaneously, but glucocorticoids can improve outcomes if administered within 3 days of symptom onset. Antiviral medications are only suggested for Herpes zoster oticus (Ramsay Hunt syndrome). Vestibular suppressant medications can reduce acute symptoms. Patients should be encouraged to resume normal activity as soon as possible. Vestibular rehabilitation therapy can also be tried.

Causes of vertigo

Peripheral causes of vertigo	Central causes of vertigo (Brainstem & central lesions)
• Meniere's disease • Benign paroxysmal positional vertigo • Vestibular neuronitis • Labyrinthitis • Vestibulotoxic drugs • Acoustic neuroma • Perilymph fistula • Head trauma • Syphilis	• Vertebrobasilar insufficiency • Posterior inferior cerebellar artery syndrome (Wallenberg syndrome) • Basilar migraine • Cerebellar disease • Brainstem tumors • Multiple sclerosis • Cervical vertigo • Epilepsy

Solution to Question 4:

The correctly matched option regarding the nerve supply of the pinna is option B: A - Greater auricular nerve; B - Auriculotemporal nerve; C - Facial and Vagus nerve.

The nerve supply of the pinna is provided by the following:

• Greater auricular nerve (A) supplies most of the medial surface of the pinna and the posterior part of the lateral surface.

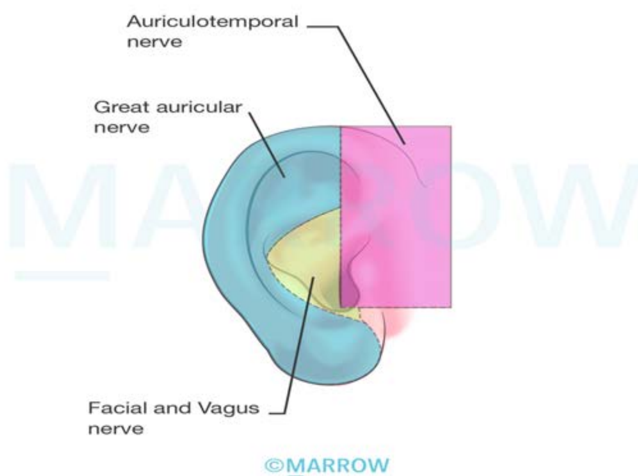
• Lesser occipital nerve supplies the upper part of the medial surface.

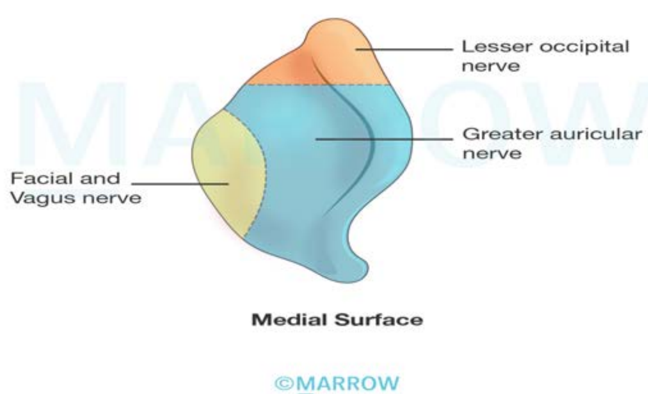
• Auriculotemporal nerve (B) supplies the tragal area and crus helix.

• Arnold's nerve (C) i.e. the auricular branch of the vagus nerve (CN X) supplies the concha and the corresponding area on the medial surface.

• Facial nerve (CN VII) (C) which is distributed along with Arnold's nerve also supplies the concha and the retro auricular groove.

The nerve supply has been demonstrated in the image below.



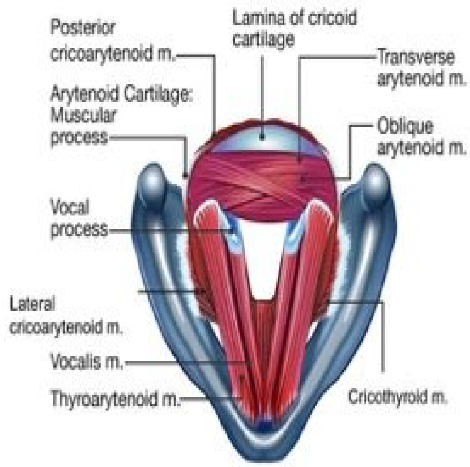


Solution to Question 5:

The affected muscle is likely the cricothyroid muscle, one of the paired intrinsic muscles of the larynx, which is responsible for elongating and tensing the vocal cords to produce higher pitches.

The cricothyroid muscle and the vocalis muscle(the internal part of the thyroarytenoid) are crucial for adjusting the tension of the vocal cords to produce higher pitches. The cricothyroid muscle, in particular, plays a unique role by elongating and tensing the vocal cords, which is essential for achieving high-pitched phonation. This muscle's function is directly influenced by the external branch of the superior laryngeal nerve, which supplies it and facilitates the adjustment of pitch.

The external branch of the superior laryngeal nerve descends along the larynx, situated beneath the sternothyroid muscle, to innervate the cricothyroid muscle. By activating this muscle, the external branch enables the tension of the vocal cords to be increased, thereby contributing to the production of higher-pitched sounds. This intricate nerve-muscle interaction highlights the specialized role of the cricothyroid muscle in pitch modulation and vocal quality.



Solution to Question 6:

The marked structure in the radiology image of the cochlear implant is the ground electrode.

A cochlear implant is a medical device that uses electricity to stimulate the spiral ganglion cells of the auditory nerve to restore sensorineural hearing loss. It can be used in a patient with bilateral severe sensorineural hearing loss.

It converts sound to an electrical signal (transduction) and delivers this to the hearing nerve, which bypasses the damaged hearing apparatus. It cannot be used in patients with cochlear aplasia or without the VIIIth cranial nerve.

Cochlear implant:

• An electronic device for severe to profound SNHL and who cannot benefit from a hearing aid.

• Principle- surgery to place the electrode array within the scala tympani of the cochlea.

• Indication- bilateral severe to profound deafness

• Prerequisites- intact 8th nerve and higher auditory pathway

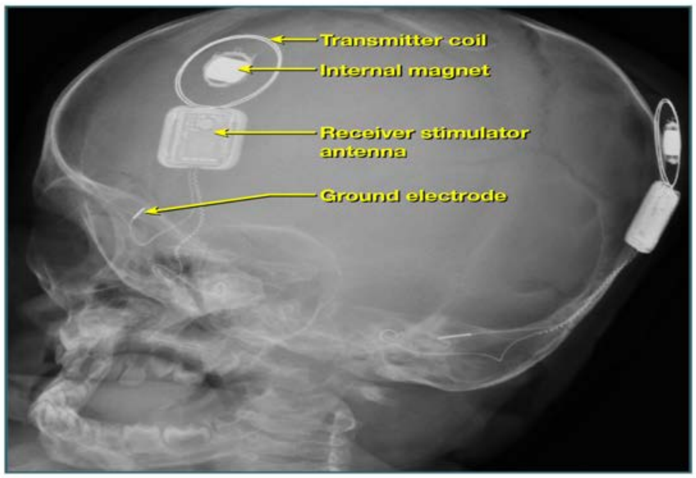
• Surgical approach- facial recess (posterior tympanotomy) approach

• The electrode of a cochlear implant is placed in the scala tympani. (Note: In the auditory brainstem implant, the site of the implant is the lateral recess of the 4th ventricle.)

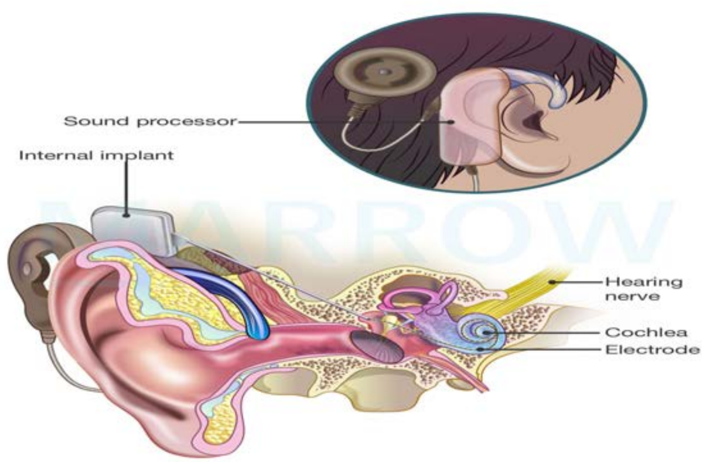
The receiver-stimulator implanted during surgery (under the skin) and the electrode array (implanted in the scala tympani of the cochlea) are part of the internal component of the cochlear implant. The microphone, speech processor and transmitter are part of the external component of the cochlear implant, which remains outside the body.

Cochlear implants replace the non-functional transducer system of hair cells of the cochlea and convert the sound signal to electrical impulses, which directly stimulate the fibres of cranial nerve VIII.

X-ray of bilateral cochlear implant



Cochlear implant



Solution to Question 7:

In this patient with rhinosinusitis and nasal polyps, the CT scan indicates that the maxillary sinus is likely affected, as evidenced by the presence of a polyp in this sinus.

Antrochoanal polyp is a type of non-neoplastic nasal polypoidal masses of edematous nasal or sinus mucosa. It arises from the mucosa of the maxillary antrum near its accessory ostium, comes out of it, and grows in the choana and nasal cavity. It is usually single and unilateral which are more commonly seen in children and young adults.

It has 3 parts namely

• Thin antral stalk,

• Round and globular choanal part, and

• Flat nasal part.

The clinical features include unilateral nasal obstruction, mucoid nasal discharge, and a hyponasal voice.

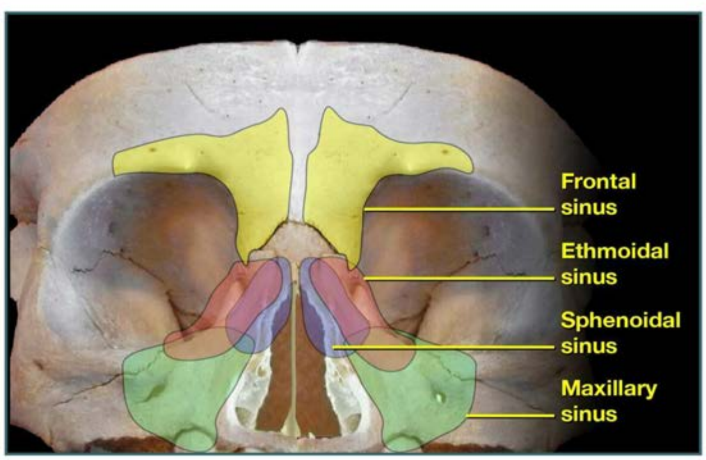
Anterior rhinoscopy may miss the antrochoanal polyp when it starts to grow posteriorly. A large polyp may protrude from the nasal cavity and may appear pink and congested. It is soft and can be moved with a probe. Posterior rhinoscopy may reveal a globular mass filling the choana and hanging down behind the soft palate.

X-ray (lateral view) of soft tissue nasopharynx, reveals a globular swelling in the postnasal space, it is differentiated from angiofibroma by the presence of a column of air behind the polyp known as Dodd sign or crescent sign.

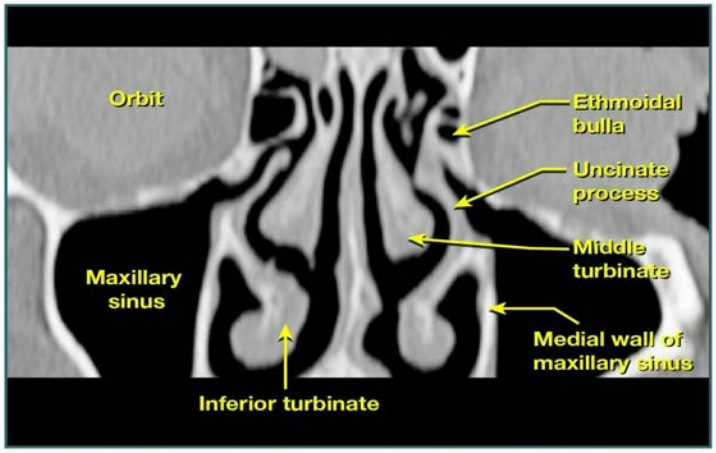
Diagnosis is made by radiograph of the paranasal sinuses reveals opacity of the involved antrum. Non-contrast CT scan reveals the extent of the polyp. Endoscopic examination reveals the antrochoanal polyp hidden posteriorly in the nasal cavity.

Endoscopic sinus surgery is the treatment of choice. Recurrence is uncommon after the complete removal of the polyp.

Paranasal sinuses



CT scan of ostiomeatal complex



Solution to Question 8:

The correct statement with regard to the oesophageal foreign body in the image is that the most common site of impaction is at the level of the cricopharyngeus.

Foreign bodies commonly ingested by children are coins, disk batteries, and small toy items. In adults, it is usually a food bolus (mostly meat) that gets lodged.

The flat surface of a coin is seen on the anteroposterior view (AP view) and the edge is seen on the lateral view. This indicates that it is oriented in the coronal plane, which is seen when it is lodged in the esophagus (and not in the trachea).

Initial symptoms are choking, gagging, and coughing followed by excessive salivation, dysphagia, food refusal, vomiting, or pain in the neck and throat. Respiratory symptoms such as stridor, wheezing, cyanosis, or dyspnea are seen if the esophageal foreign body impinges on the larynx or tracheal wall. Imaging with chest and abdominal X-rays should be performed in AP and lateral positions. Chest CT should always be performed if oesophageal perforation is suspected.

Endoscopic removal of the esophageal foreign body is the procedure of choice. It should be performed

• Immediately in esophageal obstruction, disk batteries, and sharp pointed objects.

• Within 12-24 hours in non-pointed objects, pointed objects in the stomach, and coins in the esophagus.

• Non-urgent removal in objects in stomach >2.5cm in diameter, and disk battery in the stomach (up to 48h)

Other options:

Option A: If a foreign body becomes lodged in the esophagus and causes perforation, it can lead to mediastinitis, a potentially life-threatening infection of the mediastinum.

Option C: Esophageal foreign bodies are more commonly seen in children, although they do occur in adults, particularly in those with underlying esophageal abnormalities (e.g., strictures or motility disorders).

Option D: The esophagus has several natural constrictions, including at the cricopharyngeus (C6 level), at the aortic arch, at the level of the left main bronchus and at the diaphragmatic hiatus.

Solution to Question 9:

The incision shown is used in open rhinoplasty.

This procedure involves a transcolumellar incision, an incision made across the columella to lift the nasal skin, allowing full access to the underlying nasal structures. This approach is used for cosmetic or reconstructive surgery. The open technique allows better visibility of the nasal framework compared to closed rhinoplasty.

Other options:

Option A: The Caldwell-Luc operation, also known as anterior antrostomy, involves opening the maxillary sinus via the canine fossa through a sublabial approach to address various pathologies within the sinus. It's indicated for conditions like chronic maxillary sinusitis, dental cysts, oroantral fistula, neoplasms, and certain fractures. Surgery is contraindicated for patients below 17 years of age.

Option B: The modified Denker's procedure is similar to the Caldwell-Luc procedure but involves more extensive exposure of the maxillary sinus and nasal cavity. It's used for the removal of larger sinus tumors.

Option D: Midface degloving approach is a facial incision-less approach used for accessing the midface, including the nasal cavity and sinuses.

It's commonly used in removing large nasal and sinus tumors, and it involves incisions within the oral cavity and nasal vestibule, avoiding external scars. No incision on the nose itself is involved.

Solution to Question 10:

The given clinical vignette of a patient with complaints of repeated episodes of crusting in the nose with occasional bleeding and a history of nasal surgery for nasal obstruction is likely to have undergone correction of her DNS(Deviated Nasal Septum). The image is that of septal perforation; SMR (Submucosal resection) has a higher likelihood of causing the same.

Septal perforation can occur after nasal surgery. Features depend on the size of the perforation. Smaller perforations can present with whistling while the larger perforations will present with epistaxis and crust formation. Therefore, septoplasty has become a more favored treatment of DNS than SMR due to its conservative approach.

Septoplasty is a conservative technique in the surgical correction of septal deformity. The key advantage is that it can be carried out as a daycare procedure, most of the bone and cartilage are retained, and complications are less. The deformed septum may corrected by various techniques like scoring the concave side, shaving, and wedge excision. Sometimes realignment by joining straight pieces of the septum outside the nose and replacing them can be done which is called extra-corporeal septoplasty.

The indications for an SMR are:

• DNS causing symptoms of nasal obstruction and recurrent headaches.

• DNS causing obstruction of paranasal sinuses and middle ear, resulting in recurrent sinusitis and otitis media.

• Recurrent epistaxis from a septal spur.

• As a part of septorhinoplasty, for cosmetic correction of external nasal deformities.

• As a preliminary step in hypophysectomy or vidian neurectomy.

It is contraindicated in children below 17 years of age, during acute respiratory infection, bleeding diathesis, untreated diabetes, or hypertension.

In this procedure, mucoperichondrial and mucoperiosteal flaps are raised on both sides of the septum. Bony and cartilaginous parts of the septum are excised. This is followed by extensive dissection of the septum removing all deformed bony and cartilaginous parts of the septum and preserving only a caudal and a dorsal strut of cartilage.

There is a greater chance of complications associated with SMR that include bleeding, septal hematoma, septal abscess, perforation, depression of the nasal bridge, retraction of the columella, and persistence of the deviation. The flapping of the nasal septum and toxic shock syndrome following SMR can also occur.

Other options:

Option A: FESS (Functional Endoscopic Sinus Surgery) is a minimally invasive procedure to enlarge nasal drainage pathways and improve sinus ventilation; rarely causes septal perforation.

Option C: Denker Approach is a surgical technique to access the maxillary sinus, pterygopalatine, and infratemporal fossae, used for large sinonasal tumors and extensive sinus pathology.

Option D: Endoscopic Surgical Treatment of Odontogenic Maxillary Sinusitis involves middle or inferior meatotomy; inferior meatotomy is preferred for foreign bodies or sinus floor cysts.

Submucous resection	Septoplasty
Non-conservative procedure with extensive dissection of cartilage	Conservative procedure
Not done in < 18 yrs	Can be done for adults and persons < 18 yrs if bilateral and severe
Killian's incision is used	Freer's incision is used
Mucoperichondrial/periosteal flaps raised on both sides of the septum	Flaps raised on only one side
More Complications	Less complications
Reoperation is difficult	Reoperation is easier

Solution to Question 11:

The clinical scenario describes a 40-year-old female patient with mild, progressive, bilateral hearing loss, and a PTA showing Carhart's notch (dip in bone conduction at 2000 Hz) are all suggestive of Otosclerosis and the classical sign seen here is Schwartze sign.

The presence of a reddish hue on the promontory through the tympanic membrane, known as the Schwartze sign, suggests an active focus.

Otosclerosis is caused by bony overgrowth involving the enchondral layer of the bony otic capsule. It presents with bilateral progressive conductive hearing loss and tinnitus. It has an autosomal dominant inheritance and can worsen with stress and hormonal changes like during pregnancy and menopause.

Stapedial otosclerosis is the most common type of otosclerosis. Here the lesion starts just in front of the oval window in an area called fistula ante fenestrae.

PTA shows loss of air conduction with air-bone gap, more for lower frequencies and Carhart's notch may be seen. Tympanometry shows an As type of tympanogram in which there is reduced compliance at or near the ambient air pressure.

Medical management with sodium fluoride is used in the active stage. Definitive treatment is stapedectomy/stapedotomy with the placement of a prosthesis. Carhart notch usually disappears after a successful stapedectomy.

Other options:

Option A: Hennebert sign is pressure-induced dizziness seen in patients with Superior semicircular canal dehiscence syndrome, congenital syphilis, and Méniére's disease.

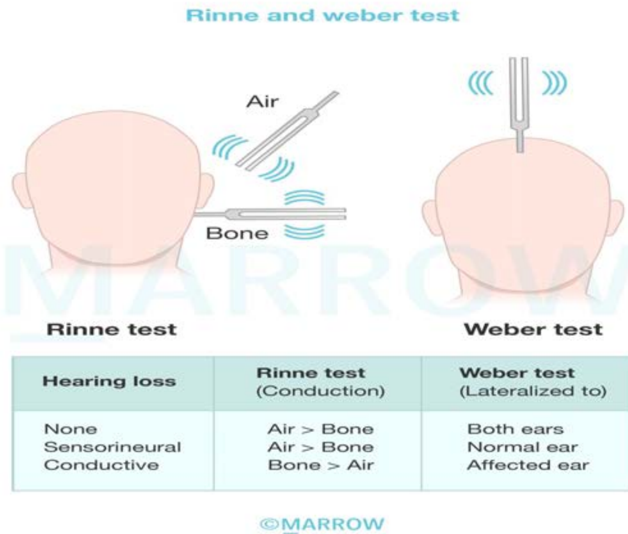
Option B: Compression of the facial nerve due to acoustic neuroma will result in hypoesthesia of the posterior wall of the external auditory canal. This is called the Hitzelberger sign.

Option D: The gurgling sound produced on pressing the swelling in the laryngocele is known as Bryce's sign.

Solution to Question 12:

This patient with unilateral hearing loss and a small perforation of his tympanic membrane is likely to have conductive hearing loss in the left ear, which will show a negative Rinne's test in the affected ear.

Tuning fork tests are performed commonly with tuning forks of frequencies such as 128, 256, 512, 1024, etc. The tuning fork of 512 Hz is considered ideal as forks of lower frequencies produce a sense of bone vibration while those of higher frequencies have a shorter decay time. Some examples of commonly done tuning fork tests are as follows:



Hearing loss	Rinne test (Conduction)	Weber test (Lateralized to)
None	Air > Bone	Both ears
Sensorineural	Air > Bone	Normal ear
Conductive	Bone > Air	Affected ear

Rinne's test is done by placing a vibrating tuning fork on the patient's mastoid and when he stops hearing, it is brought beside the external auditory meatus (EAC). If he still hears the sound, air conduction (AC) is better than bone conduction (BC). The inferences are as follows:

• Rinne's test is considered positive (normal) when the air conduction AC>BC is seen in normal people and people with sensorineural deafness.

• It is considered negative when the BC>AC, and is seen in patients with conductive deafness. It suggests a minimum hearing loss of 15-20 dB.

Weber test is done by placing a vibrating tuning in the middle of the forehead or the vertex and the patient is asked in which ear the sound is heard. The inferences are as follows:

• A normal result is achieved when the sound heard equally in both ears

• In conductive deafness, it is lateralized to the worse ear

• In sensorineural deafness, it is lateralized to the better ear

Other options:

Option A: Rinne's test is positive in normal subjects.

Option B: Rinne's test is falsely negative in patients with sudden severe unilateral sensorineural hearing loss.

Option B: There is no such interpretation as intermediate in Rinne's test.

Solution to Question 13:

The given clinical vignette of a woman with hearing loss and pulsatile tinnitus and examination suggesting a mass behind the tympanic membrane and CT image showing a mass in the region of jugular foramen makes Glomus Jugulare the likely diagnosis.

Glomus tumors are the most common benign tumors of the middle ear. The tumors comprise paraganglionic cells and arise from glomus bodies which resemble carotid bodies. Glomus bodies are present in the dome of the jugular bulb and on the promontory along the course of the tympanic branch of the IX cranial nerve.

They are most commonly seen in females between the ages of 40-50 years. They are very slow-growing, thus it may take years before changes in symptoms are appreciated. It follows the rule of 10s - 10% are familial, 10% are multicentric, and 10% are functional (secrete catecholamines).

Depending on the area from which it arises, glomus tumors are classified into two types:

• Glomus jugulare, arising from the dome of the jugular bulb

• Glomus tympanicum, arising from the promontory of the middle ear

The presentation of glomus tumors is as follows:

• Intratympanic presentation

• Progressive hearing loss

• Pulsatile tinnitus is synchronous with pulse and can be stopped by carotid pressure

• Otoscopy: red reflex through the intact tympanic membrane

• Rising sign when a tumor arises from the floor of the middle ear

• Pulsation or Brown sign: when the ear canal pressure is raised with Siegel's speculum the tumor pulsates and blanches. This reverses when pressure is released.

• Polyp presentation

• Profuse bleeding

• Otorrhea if there is a secondary infection

• Red vascular polyp filling the meatus that bleeds easily

• Cranial nerve palsies

• Dysphagia and hoarseness with unilateral paralysis of the soft palate, and pharynx due to IX and X

• Vocal cord paralysis due to X

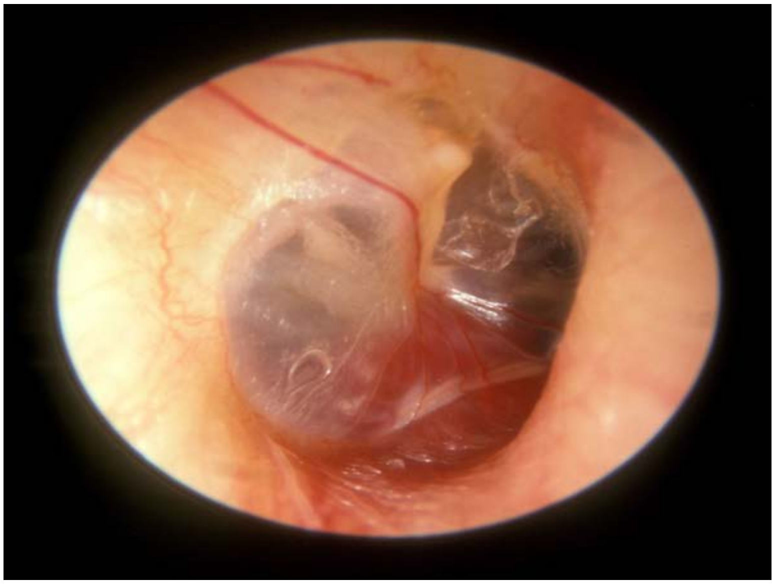
• Weakness of the trapezius and sternomastoid muscles XI

• Atrophy of half of the tongue XII

• Audible bruit over the mastoid

They can be diagnosed by CT scan, MRI, four-vessel angiography, brain perfusion and flow studies, embolization, and biopsy. It is treated by surgery, radiation, embolization, or a combination of them. Phlep sign is seen in a CT scan of the glomus jugulare tumor. It is erosion of the normal crest of bone between the carotid canal and jugular foramen.

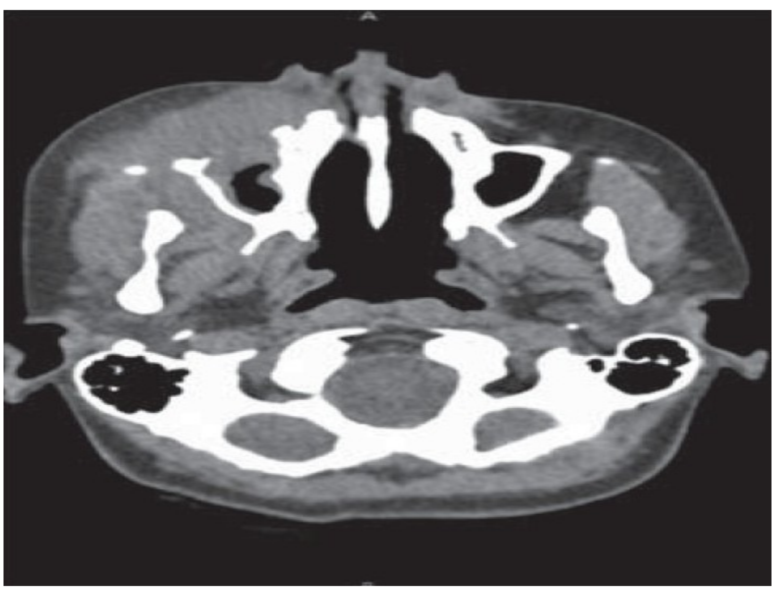
The following image shows the rising sun sign in the case of glomus jugulare.



Other options:

Option A: Glomus tympanicum is a close differential to the jugular tumor, however, it commonly arises from the promontory of the middle ear rather than the jugular bulb and does not show the Phlep sign and it is also found in relation to the temporal bone on CT.

Option B: Giant cell reparative granuloma accounts for 1–7% of all benign lesions of the jaw. It often arises in the maxilla followed by the mandible and affects children and young adults. It is usually a slow-growing lesion. The fast-growing lesions are rare and despite the innocent histological appearance, have an aggressive behavior mimicking a malignant lesion. The following image shows the CT of the reparative granuloma of the maxilla showing the extension of tumor mass over the pterygoid plates and infratemporal fossa.



Option D: Acoustic neuromas are noncancerous, usually slow-growing tumors that form along the vestibulocochlear nerve. They arise from Schwann cells. Schwannomas can occur on any cranial or peripheral nerve in the body, but in the brain, acoustic neuromas are the most common schwannomas. The following image depicts the classical ice cream cone appearance of acoustic neuroma on MRI.